SIMATS ENGINEERING
Department of Information Technology
ITA05 – Computer Vision
ASSIGNMENT 1 – CO1

Assignment Title: Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Textile Fabric Defect Inspection System

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

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Department/ Section	Information Technology	**AcademicYear/ Semester**	2025–2026 / AIML
Faculty Name	Dr Murali	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	30.10.2026	**Date of Evaluation**	07.11.2026

Industrial Scenario

Sri Lakshmi Textile Industries, Tiruppur, plans to develop a computer vision-based inspection system for its fabric production line. The system must inspect moving cotton and cotton-polyester fabric, measure fabric width, and identify holes, broken threads, oil stains, weaving irregularities and colour variations. It must estimate the location of each defect and trigger a marking mechanism so that the defective portion can be identified for rework or rejection.

Given Industrial Data and Constraints

Parameter	Specification
Fabric width	1,000 mm
Camera field of view	1,200 × 400 mm
Conveyor speed	0.6 m/s
Smallest defect to be detected	1.5 mm
Oil-stain contrast	5–8 grey levels

Required fabric-width measurement accuracy	±2 mm
Maximum decision time	0.5 s
Distance between camera and marking mechanism	0.3 m
Lighting variation	500–3,000 lux
Camera mounting height	700 mm above the fabric
Available computing system	4-core CPU, 16 GB RAM, no GPU
Software environment	Python, OpenCV and NumPy
Network interface	Gigabit Ethernet
Storage capacity	2 TB
Maximum project budget	□1,50,000
Image retention period	7 days
Maximum acceptable missed-defect rate	2%

General Instructions

1. Answer all ten tasks.
2. For every task, three candidate techniques or approaches are provided. Analyse all three alternatives, compare their suitability for the given industrial scenario, select the most appropriate option and justify your selection.
3. Support your answers with calculations, diagrams, tables, observations, screenshots or experimental results wherever applicable.
4. Clearly state assumptions and show the steps used in numerical calculations.
5. Your justification must consider the given operating conditions, accuracy, processing time, hardware limitations and budget.
6. Do not provide only definitions. Relate every answer to the textile fabric inspection application.
7. Where implementation is required, use Python, OpenCV and NumPy. Include relevant code snippets and sample output.

TASK 1: Overview and Features of Computer Vision – Compare Grading/Inspection Approaches

Marks: 4

Compare the following three approaches for inspecting fabric defects:

Candidate techniques / approaches:

- Manual inspection by trained workers
- Image-processing-assisted inspection, in which an operator reviews computer-generated results

 Fully automated computer vision inspection

Student tasks:

8. Explain the working principle of each approach.
9. Compare their accuracy, speed, consistency, labour requirement and suitability for continuous fabric production.
10. Select the most suitable approach for Sri Lakshmi Textile Industries and justify your decision usingthe given production conditions.
11. Identify at least two limitations of your selected approach and suggest how they can be reduced.

TASK 2: Levels of Computer Vision – Pipeline Design and Operation

Marks: 5

Compare the following system designs:

Candidate techniques / approaches:

 Low-level pipeline: image acquisition, preprocessing and enhancement
 Three-level pipeline: low-level processing, intermediate-level feature extraction and high-level defect decision
 Deep-learning-based pipeline: image acquisition, neural-network inference and defect classification

Student tasks:

12. Draw a block diagram for each pipeline.
13. Explain the input and output of each processing stage.
14. Identify the stages required to detect holes, broken threads, stains and weaving irregularities.
15. Compare computational complexity, explainability, hardware requirements and response time.
16. Recommend a pipeline that can operate within the available 4-core CPU system and the 0.5-seconddecision constraint.

TASK 3: Image Representation Analysis

Marks: 4

Compare the following image representations for fabric inspection:

Candidate techniques / approaches:

 RGB colour representation
 HSV colour representation
 Grayscale representation

Student tasks:

17. Explain how each representation stores image information.
18. Discuss the suitability of each representation for identifying colour variations, oil stains, holes andbroken threads.

19. Compare memory requirements and processing complexity.
20. Choose a suitable representation or combination of representations and justify your choice withexamples from the scenario.

TASK 4: Connectivity Analysis

Marks: 4

Compare the following pixel-connectivity approaches for identifying defect regions:

Candidate techniques / approaches:

- 4-connectivity
- 8-connectivity
- m-connectivity (mixed connectivity)

Student tasks:

21. Define each connectivity approach with a suitable pixel-neighbour diagram.
22. For a small binary image containing diagonally and horizontally connected defect pixels, identifythe connected components using all three methods.
23. Explain how the choice of connectivity may affect defect-region size and component counting.
24. Recommend the most appropriate connectivity method for detecting small holes and irregularfabric defects.

TASK 5: Image Acquisition Analysis

Marks: 5

Compare the following camera acquisition options for the moving fabric line:

Candidate techniques / approaches:

- Rolling-shutter camera
- Global-shutter camera
- Line-scan camera

Student tasks:

25. Explain the image acquisition principle of each camera type.
26. Analyse the effect of fabric movement at 0.6 m/s on image distortion and defect visibility.
27. Compare image quality, suitability for continuous web inspection, installation complexity and cost.
28. Select a camera type for the given fabric width and production process.
29. Justify the selection with reference to the 1.5 mm minimum defect size and 0.5-second decisionrequirement.

TASK 6: Sampling Analysis

Marks: 5

Evaluate the following image sampling resolutions:

Candidate techniques / approaches:

- Low resolution: 1280 × 1024 pixels
- Medium resolution: 2448 × 2048 pixels
- High resolution: 4096 × 3000 pixels

Student tasks:

30. Calculate the total number of pixels for each resolution.
31. Estimate the approximate spatial sampling interval across the 1,200 mm field of view, assumingthe image width corresponds to the full field-of-view width.
32. Determine whether a 1.5 mm defect can be represented adequately at each resolution.
33. Compare image storage, processing time and defect-detection capability.
34. Select a resolution that balances defect visibility and the available computing resources. Show allcalculations.

TASK 7: Quantization Analysis

Marks: 4

Compare the following intensity quantization options:

Candidate techniques / approaches:

- 8-bit quantization
- 12-bit quantization Reduced

4–6 bit quantization

Student tasks:

35. Calculate the number of intensity levels available in each option.
36. Explain how quantization affects the representation of low-contrast oil stains having only 5–8 greylevel differences.
37. Compare image storage requirements and processing implications.
38. Recommend a quantization level for reliable defect detection under the stated lighting conditions. Justify your answer.

TASK 8: Image Formation Analysis

Marks: 5

Compare the following image formation and calibration approaches:

Candidate techniques / approaches:

- Standard perspective imaging without calibration
- Calibrated imaging with geometric correction
- Controlled-illumination imaging with suitable calibration

Student tasks:

39. Explain the image formation process for a camera mounted 700 mm above the fabric.
40. Discuss perspective distortion and its effect on fabric-width measurement.
41. Explain how camera calibration can improve measurement accuracy.
42. Analyse how lighting variation from 500 to 3,000 lux can affect defect visibility.
43. Recommend an image formation and illumination arrangement that can achieve the required widthaccuracy of ±2 mm.

TASK 9: Implementation and Performance Evaluation

Marks: 6

Design and compare three complete implementation configurations:

Candidate techniques / approaches:

- Budget configuration: minimum-cost camera, basic illumination and CPU-based processing
- Balanced configuration: suitable camera, controlled illumination and optimised CPU-based processing
- Premium configuration: high-resolution industrial camera, advanced illumination and higherperformance processing hardware

Student tasks:

44. Prepare a block diagram showing camera, illumination, processing unit, storage, network andmarking mechanism.
45. Prepare a comparison table covering estimated cost, resolution, processing time, storage andexpected inspection capability.
46. Explain how the system will detect and record holes, broken threads, stains and weavingirregularities.
47. Describe how the system will communicate the defect location to the marking mechanism located 0.3 m downstream.
48. Evaluate whether each configuration can meet the 0.5-second decision time, 7-day image retentionand 2% maximum missed-defect rate.
49. Select one configuration that fits the ₹1,50,000 budget and justify your selection. Clearly state any assumptions used in cost estimates.

TASK 10: Front-End Adaptation Challenge

Marks: 8

Assume that the textile company receives a new export order with the following changed requirements: the system must detect defects as small as 0.4 mm, identify stains visible under ultraviolet (UV) illumination, operate at a conveyor speed of 1.2 m/s, inspect twice the current production volume and produce a decision within 0.3 seconds. The existing PC and project budget must remain unchanged.

Candidate techniques / approaches:

- Retain the existing system with only software changes
- Modify the existing system by upgrading selected components
- Redesign the inspection system using a different acquisition and processing architecture

Student tasks:

50. Identify which existing system components will become inadequate under the new requirements.
51. Analyse the effect of the increased conveyor speed and smaller defect size on image sampling andacquisition.
52. Explain how UV illumination or an additional imaging channel could help identify the new staintype.
53. Compare the three adaptation strategies in terms of feasibility, accuracy, processing time, cost andimplementation risk.
54. Recommend the most practical strategy under the unchanged PC and budget constraints.
55. Prepare an updated system block diagram and list the changes required.
56. Clearly identify any requirements that cannot realistically be achieved without additional resources,and justify your conclusion.

Mark Distribution

Task	Topic	Marks
1	Overview and Features of Computer Vision	4
2	Levels of Computer Vision	5
3	Image Representation	4
4	Connectivity Analysis	4
5	Image Acquisition	5
6	Sampling Analysis	5
7	Quantization Analysis	4
8	Image Formation	5
9	Implementation and Performance Evaluation	6
10	Front-End Adaptation Challenge	8
	TOTAL	50

Submission Requirements

- Submit a neatly formatted assignment report with task-wise headings.
- Include hand-drawn or computer-generated diagrams wherever required.

 Show all numerical calculations with formulae, substitution and final units.
 Include comparison tables and evidence supporting the selected techniques.
 For implementation tasks, attach source code, sample input images and output screenshots. Mention references, tools and software libraries used.

Industrial Scenario and Requirements

Sri Lakshmi Textile Industries, Tiruppur, plans to inspect moving cotton and cotton–polyester fabric using computer vision. The system must measure fabric width, detect holes, broken threads, oil stains, weaving irregularities and colour variations, estimate defect locations, and trigger a marking mechanism so that defective fabric can be reworked or rejected.

Given operating data: fabric width 1,000 mm; field of view 1,200 × 400 mm; web speed 0.6 m/s; smallest defect 1.5 mm; oil-stain contrast 5–8 grey levels; width accuracy ±2 mm; decision time at most 0.5 s; marker distance 0.3 m; lighting 500–3,000 lux; camera height 700 mm; 4-core CPU, 16 GB RAM, no GPU; 2 TB storage; image retention 7 days; budget □1,50,000; maximum missed-defect rate 2%.

The design must satisfy the whole system requirement. A good classifier cannot recover a defect erased by motion blur or insufficient sampling. A good camera cannot guarantee reliable marking unless coordinate mapping, communication and actuator timing are also tested.

Derived quantities from the given data. The fabric is 1,000 mm wide and the field of view is 1,200 mm wide, so a centred web leaves about 100 mm of background on each side. This margin is needed to see both fabric edges for the width measurement and to tolerate lateral wandering of the web. The 400 mm field height corresponds to 400/600 = 0.667 s of fabric travel at 0.6 m/s. The marker is 0.3 m downstream, so a defect needs 0.3/0.6 = 0.5 s to reach it, which equals the maximum permitted decision time. The design therefore has no spare time if processing is allowed to run up to the limit.

Given data	Value	Design consequence
Smallest defect	1.5 mm	Needs several pixels across it; about 0.3 mm/pixel for five pixels
Oil-stain contrast	5–8 grey levels	Needs low-noise sensor, stable light and no coarse quantization
Width accuracy	±2 mm	Needs calibration, edge detection and a pixel pitch well below 1 mm
Decision time / marker distance	0.5 s / 0.3 m	Latency budget must be below the 0.5 s travel time; use encoder position
Lighting	500–3,000 lux	Enclosure and controlled LED lighting; flat-field correction
Computer	4-core CPU, 16 GB, no GPU	Efficient classical pipeline; compact models only after benchmarking
Storage / retention	2 TB / 7 days	Store events, crops and metadata; compress; check capacity daily
Missed-defect limit	2 %	Measure per-class sensitivity on a hold-out set

The scenario therefore couples optics, illumination, computing and mechanical control. A requirement can only be declared satisfied when the complete chain from exposure to physical mark has been measured on real fabric, which is why every later task states its acceptance evidence.

Expected defect behaviour on the line. The five defect classes differ in geometry and contrast, and the requirements must be read class by class. Holes and thread breaks are mainly structural, so they depend on sampling and sharp edges. Oil stains and colour variation are mainly photometric, so they depend on bit depth, uniform light and colour calibration. Weaving irregularities are textural and depend on comparing a local patch with its surroundings. A single threshold on a single grey image would not serve all five classes.

Deliverables expected from the design. The report must justify the choice of inspection approach, pipeline depth, image representation, connectivity rule, camera type, sampling density, bit depth and imaging geometry; show a complete implementation with timing and storage; and then show how the front end must change when the customer's export order tightens the specification. These deliverables are answered in Tasks 1–10 in that order.

General Method and Assumptions

The report compares the three approaches named in each task, then recommends one for the stated textile application. Calculations use the given dimensions and SI units. Where a value such as frame rate, exposure or compression ratio is not specified, it is treated as a design assumption that must be confirmed during procurement and commissioning.

Unless explicitly stated as measured, numerical examples are engineering calculations or planning estimates. The assignment source does not provide a real labelled textile dataset or measured camera benchmark. Therefore the report does not invent experimental accuracy values; it specifies how those values should be measured on actual fabric.

Testing should include multiple weave structures, cotton and cotton-polyester blends, light and dark colours, low-contrast oil stains, small holes, thread breaks, uneven weaving, clean regions and the full lighting range. A hold-out test set should be separated by roll or production batch to avoid near-duplicate leakage.

Design assumptions to be confirmed. Because the assignment does not supply a camera data sheet or a labelled textile dataset, several values are treated as explicit assumptions. They are listed below so that a reader can distinguish given data, calculated values and quantities that still require procurement or measurement.

Quantity	Status	How it is treated in this report
Fabric width, FOV, speed, defect size	Given	Used directly in all calculations
Pixel pitch, defect span, blur, data rate	Calculated	Derived from given data using stated formulas
Exposure time, frame/line rate	Design assumption	Chosen to satisfy blur and coverage; confirm with vendor and bench test
Compression ratio, retention policy	Design assumption	Verify against 2 TB capacity on real image statistics

Sensitivity, false-alarm rate, latency	To be measured	Reported only after tests on actual fabric
Camera and lighting price	Planning estimate	Replace with written vendor quotations

Verification principle. Each task follows the same pattern: define the approach, show a calculation where numbers are available, compare alternatives under the same conditions, recommend one, and state the limit of the conclusion. Where a figure cannot be derived from the given data, the report states the measurement that would produce it instead of inventing a value. Hold-out data are split by roll or production batch so that neighbouring crops of the same fabric are never shared between development and testing.

A final point of method is that every result is traceable to evidence: the image, the mask or overlay, the configuration used, the timestamp and the operator or software version. This allows any disputed rejection or missed defect to be reproduced and analysed later.

Standard test conditions. When two options are compared, they should be tested with the same fabric samples, the same speed, the same lens and lighting, and the same labelled defects. Otherwise a difference in result cannot be attributed to the option being compared. A full test matrix would cross the defect classes (five), fabric types (cotton and cotton–polyester), colours (light and dark), and ambient light conditions (500 lux, 1,500 lux and 3,000 lux), with clean fabric included in every cell as a false-alarm check.

Section summary	Statement
Given	Given data are used directly; other values are labelled assumptions.
Finding	Claims about accuracy require measurements on actual fabric.
Decision	Report estimates, simulations and measurements separately.

Report Map and Evaluation Metrics

Tasks 1–10 address inspection approaches, pipeline depth, image representation, connectivity, camera acquisition, sampling, quantization, image formation, implementation/performance, and adaptation to the new export order. Each task includes application-specific explanation, comparisons, calculations where relevant, a recommendation and limits.

Core metrics: sensitivity = TP/(TP+FN); missed-defect rate = FN/(TP+FN); precision = TP/(TP+FP); width error = measured width minus reference width; latency = time from acquisition to decision; marker error = actual mark position minus expected position. Per-class results must be reported rather than only one aggregate accuracy.

Evidence for final acceptance should include source images, ground-truth labels, masks or overlays, configuration files, timestamps, processing logs, calibration records, confusion matrices and a list of failures. Estimates and simulated results must be clearly distinguished from measurements.

Task	Question addressed	Key evidence in this report
1	Compare manual, assisted and automated inspection	Capacity, traceability decision-time and comparison

2	Compare pipeline depth (low level, three level, learned)	Stage input/output, timing and failure analysis
3	Choose image representation (RGB, HSV, grey)	Memory calculation and low-contrast stain example
4	Choose pixel connectivity (4, 8, mixed)	Binary-pattern example and regionmeasurement effects
5	Choose acquisition (rolling, global, line scan)	Motion-blur and line-rate calculations
6	Choose sampling resolution	mm/pixel, defect span, data rate, width margin
7	Choose quantization depth	Level counts, step size, experiment plan
8	Choose image-formation model and calibration	Focal-length estimate, width error budget
9	Implement and evaluate configurations	Architecture, timing, storage, acceptance test
10	Adapt the front end to the export order	0.4 mm, UV, 1.2 m/s, 0.3 s feasibility

Worked metric example. Suppose a validation run contains 200 holes, of which 196 are detected and 4 are missed, with 12 false alarms among all alarms. Sensitivity = 196/200 = 98 %, missed-defect rate = 4/200 = 2 %, and precision = 196/(196+12) = 94.2 %. The missed-defect rate then sits exactly at the 2 % limit, so the point estimate alone is not convincing; the confidence interval and the per-class results must also be reported. This example is illustrative arithmetic, not a measured result from the plant.

Per-class reporting. Aggregate accuracy is misleading because clean fabric dominates production. If only 1 section in 200 contains a defect, a program that always says “clean” scores 99.5 % accuracy while missing every defect. For that reason sensitivity, precision and false alarms per metre are reported for each class, and the confusion matrix and a gallery of misses accompany the acceptance table. Width error and marker-position error are reported in millimetres, and latency as median, 95th percentile and maximum.

Acceptance gate. The commissioning gate is the conjunction of the following: per-class missed-defect rate within 2 %, width error within ±2 mm, worst-case latency below the marker travel time with margin, marker-position error within the agreed tolerance, and seven-day retention within the 2 TB disk. A system that meets only some of these is reported as partially compliant, with the unmet requirement named.

Reproducibility. Every figure in the report can be reproduced from stored inputs: the given operating data, the formulas in the Key Formula Sheet and the stated assumptions. Where a student or engineer repeats a calculation with a different assumption, for instance a different exposure time or sensor size, the same formulas apply and only the result changes. This also lets the reader check the report arithmetic independently.

TASK 1 – Overview and Features of Computer Vision

1. Inspection problem and operating requirements

Manual inspection is the traditional method of checking fabric quality. In this method, trained workers visually examine the fabric to identify defects such as holes, stains, tears and colour differences. The workers use their experience and eyesight to find defects. The accuracy of this method depends on the skill and attention of the workers.

Sri Lakshmi Textile Industries in Tiruppur needs continuous inspection of cotton and cotton–polyester fabric. The system must measure fabric width and identify holes, broken threads, oil stains, weaving irregularities and colour variation. It must also record defect coordinates and send a marking command to a mechanism located downstream. The design should be judged against the actual operating limits rather than against a generic laboratory image.

The fabric width is 1,000 mm and the camera field of view is 1,200 × 400 mm. The web moves at 0.6 m/s, the smallest required defect is 1.5 mm, and oil-stain contrast is only 5–8 grey levels. Width measurement must be within ±2 mm. A decision is required within 0.5 s, the marker is 0.3 m downstream, lighting can vary from 500 to 3,000 lux, and the available PC has a 4-core CPU, 16 GB RAM and no GPU.

These constraints imply that acquisition quality is as important as the classifier. A defect that is blurred, undersampled or hidden by glare cannot reliably be recovered by later processing. The system should therefore use controlled illumination, a calibrated view, sufficient sampling, and an inspection pipeline whose worst-case latency is measured. The 2% maximum missed-defect rate is a validation target, not an assumed result.

What the system must output. For every defect the system must return the roll identifier, defect class, centroid and bounding box in millimetres, along-web position from the encoder, severity and the action requested from the marker. The width result is a separate measurement returned for every inspected section. Treating coordinates and width as first-class outputs, rather than as by-products of classification, is what makes the inspection useful for rework or rejection.

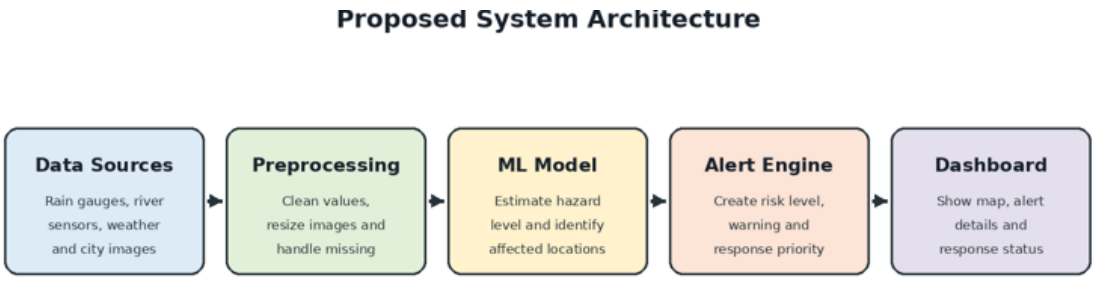
Requirement	Consequence for the design
1.5 mm smallest defect	Pixel pitch near 0.3 mm or finer so that the defect covers about five pixels
5–8 grey-level stain contrast	Low-noise sensor, stable light, at least 8-bit data, no coarse quantization
Width within ±2 mm	Calibrated lens and edge detection with margin for optical and fabricflutter error
Decision within 0.5 s	Measured worst-case latency, encoder-based marker control, deadline check
500–3,000 lux ambient	Enclosed camera station with controlled diffuse LED lighting
2 % missed-defect limit	Per-class sensitivity measured on a roll-separated hold-out set

The table links each given number to a concrete engineering response. If any row cannot be satisfied by the chosen hardware, the other rows cannot compensate for it, because the requirements are joint conditions on the whole system rather than separate goals.

Boundary of the problem. The system does not repair or classify fabric grade by itself; it observes, measures, decides and marks. The inputs are light reflected from moving fabric, encoder pulses that give web position, and the product recipe (fabric colour and weave). The outputs are defect records, a width log and marker commands. Keeping this boundary clear makes it possible to test each interface separately: optics to image, image to decision, decision to marker.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.



2. Approach A: manual inspection

In image-processing-assisted inspection, cameras capture images of the fabric and image-processing software analyses them to identify possible defects. The software highlights the defective areas, and the operator checks the results and confirms the defects. This method reduces manual effort and helps workers inspect fabric more easily.

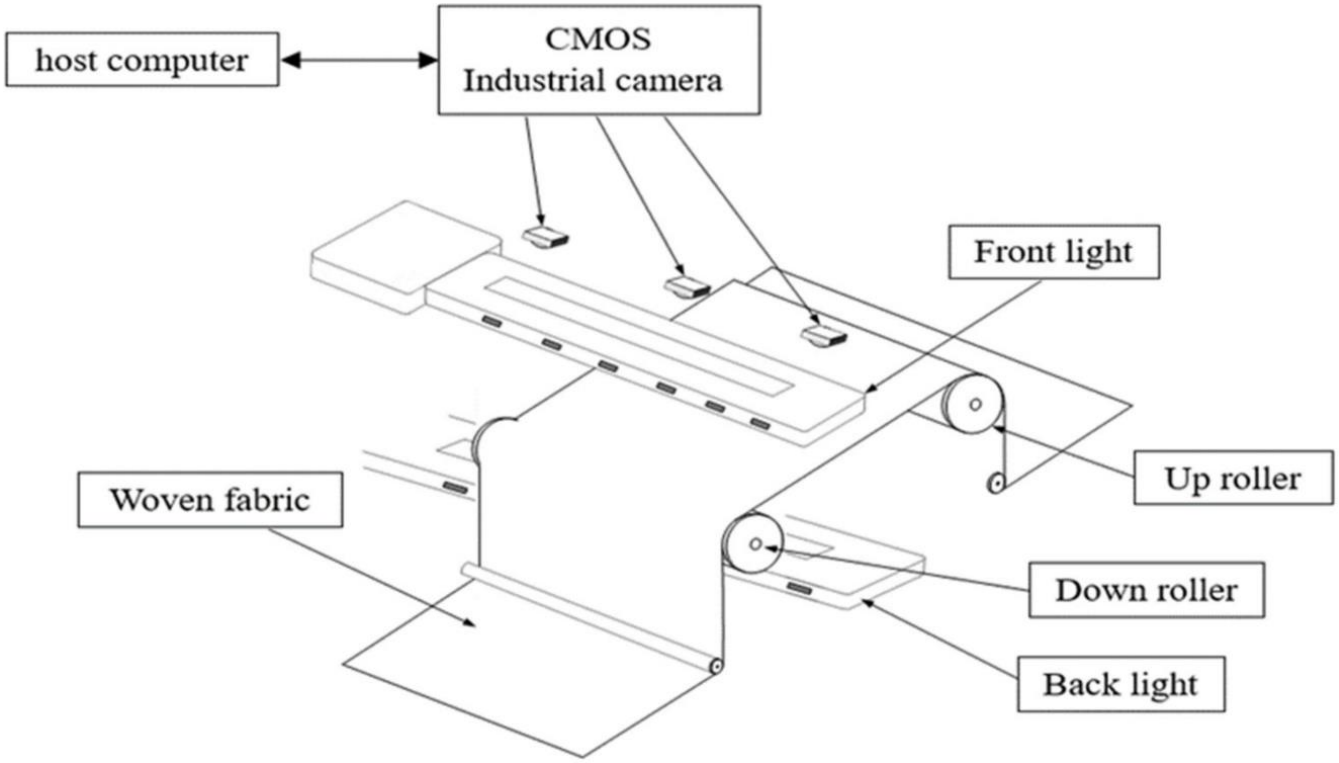
Manual inspection relies on trained workers who look at the moving fabric and identify visible defects. Its strengths are flexible judgement, experience with unfamiliar patterns and low initial equipment cost. A worker can sometimes recognize an unusual defect without the company having to collect examples or write a new rule. Manual inspection can also provide useful labels during the first stage of building a dataset.

The disadvantages are fatigue, inconsistent judgement, dependence on operator training, difficulty measuring exact width and defect position, and weak traceability unless every decision is recorded. A small 1.5 mm hole or low-contrast oil mark may pass unnoticed at production speed. The worker cannot reliably stop and inspect each region when the web is continuously moving at 0.6 m/s.

For this plant, manual inspection is best retained as a quality-audit and exception-review activity. It is not the strongest primary method because the company needs repeatable coordinates, a sub-second decision, and evidence that the missed-defect rate is below 2%. Any human comparison in an experiment should

use actual operators and recorded decisions; simulated human accuracy must be labelled as simulated rather than measured.

Why speed is the limiting factor. At 0.6 m/s, a 1.5 mm defect passes a fixed point in 1.5/600 = 2.5 ms. A typical human visual reaction time is of the order of 0.2–0.25 s, during which the fabric advances about 600 × 0.25 = 150 mm. An operator therefore cannot stop the line and inspect each small defect; the worker sees it only after it has already moved well past the viewing position, and the exact location is lost unless a mark is made immediately. These figures are physical estimates from the given speed, not measured operator data.



Manual-inspection factor	Typical issue	Practical control
Fatigue and attention	Missed small or low-contrast defects late in a shift	Short rotations and scheduled breaks
Viewing conditions	Glare and shadow at the inspection table	Standard diffuse lighting and fixed viewing angle

Recording	Defect position written by hand, often imprecise	Roll ID and metre-count log, with photograph for rejects
Training	Different inspectors classify differently	Reference defect album and periodic calibration of inspectors

Manual inspection remains valuable in a sampling role. A fixed proportion of rolls can be re-examined by an experienced inspector, and the results compared with the automatic decisions. This provides an independent ground truth and gives early warning if the automatic system drifts.

Simple cost view. Manual inspection has no capital cost for cameras, but it needs trained inspectors on every shift and offers no digital evidence. If a customer later disputes a delivery, the only record is a handwritten log. The cost of one escaped defect, such as a rejected export lot, can exceed the annual cost of an automated system, so the comparison should include the cost of missed defects as well as equipment and labour. These cost statements are qualitative and should be replaced by the plant's own figures.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

3. Approach B: image-processing-assisted inspection

Fully automated computer vision inspection uses cameras and computer software to examine fabric automatically. The system captures fabric images and analyses them to detect defects such as holes, stains and tears. It can identify defects and alert workers without requiring manual checking for every fabric piece. This method is suitable for modern textile industries.

In assisted inspection, a camera captures the web, software enhances the image and highlights suspected regions, and an operator confirms or rejects the alerts. Processing may include grayscale conversion, contrast enhancement, noise reduction, thresholding, edge detection and connected-component labelling. This approach helps an operator focus on suspicious areas and allows the company to save images and annotations.

Assistance can improve visibility when a stain has only a small intensity difference from the surrounding fabric. A display can show a zoomed region and report the measured width or defect coordinates. However, the operator remains in the decision loop, so the overall response time includes image transfer, processing, display and human reaction. An alert-heavy system may also create operator fatigue.

This approach is a useful transitional solution for collecting verified examples and refining thresholds. It is less suitable as the final production control if every defect must be marked within 0.5 s. A practical design can use the assisted mode during commissioning and keep a human-review queue for lowconfidence regions after automatic inspection is deployed.

Operator workload example. Assume, for illustration, that the software raises 20 alerts per minute and the operator needs 3 s to look at the zoomed region and confirm or reject each one. The review demand is 20 × 3 = 60 s per minute, i.e. 100 % of the operator's time, with no allowance for movement or breaks. The queue would grow, so a human-confirmed decision cannot meet a 0.5 s limit. Reducing false alerts, or accepting only low-confidence cases for review, is essential.

Use of assisted mode	Benefit	Limitation
Commissioning	Engineers see candidate regions and tune thresholds on real fabric	Operator time is consumed during tuning
Collecting labelled data	Confirmed defects and clean regions build the validation set	Labels depend on operator consistency
Low-confidence review queue	Ambiguous regions are not silently accepted	Throughput depends on how many are queued
Training and audit	Shows inspectors what the system detects	Does not replace automatic marking

The assisted approach is therefore best treated as a stage in the project life cycle rather than the final control method. The overlays and annotations it produces are exactly the evidence needed later to validate the fully automatic detector.

Typical assisted workflow. A camera captures a frame, the software applies flat-field correction and local contrast enhancement, candidate regions are outlined on a monitor, and the operator accepts or rejects each outline. Accepted regions are saved with the image, roll number and position. The saved set grows into a labelled collection that includes both defects and false alarms, which is precisely the material needed to select thresholds and to test any later automatic classifier.

Practical limits. Assisted inspection still depends on illumination and viewing geometry. A display that shows contrast-enhanced regions may exaggerate noise, leading the operator to confirm false alerts. Therefore the enhancement parameters should be fixed, documented and applied identically to all rolls, and the alert rate per metre should be monitored. If it rises suddenly, the cause is usually dirt, a lamp change or a new fabric lot, not a sudden increase in defects.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

4. Approach C: automated computer vision

An automated system acquires images, corrects illumination and geometry, extracts candidate regions, classifies defects, calculates their positions and sends a timed marking signal. Low-level processing operates on pixels, intermediate processing creates regions and features, and high-level logic assigns defect labels and severity. The decision and image metadata can be stored with a timestamp and fabric roll identifier.

Automation gives consistent processing rules and can operate continuously, but only for defect types represented by the selected rules or training data. It can fail when the illumination changes, fabric texture changes, defects overlap, or a new defect type appears. The system therefore needs reject/review logic for uncertain regions and a routine for adding newly confirmed defects to its validation set.

For this scenario, automation is the preferred primary approach, provided that it is paired with controlled lighting, sufficient spatial resolution, calibration, and a real-fabric acceptance test. The system must not claim a 2% missed-defect rate merely because it performs well on synthetic images. The factory should measure sensitivity on a labelled hold-out set covering every defect type, fabric colour, weave and operating light condition.

Latency budget. The 0.5 s decision limit must be shared among all stages. The following is a planning allocation, not a measurement, and each entry must be replaced by timestamps from the real system.

Stage	Planning allocation	Notes
Exposure, readout and transfer	≤ 50 ms	Depends on camera interface and strobe timing
Calibration and preprocessing	≤ 80 ms	Flat-field correction, denoising, colour conversion
Candidate extraction and classification	≤ 150 ms	Region-based processing on the 4-core CPU
Coordinate mapping and marker message	≤ 30 ms	Pixel to millimetre conversion, encoder position, communication
Safety margin	≥ 190 ms	Absorbs worst-case frames, disk activity and jitter
Total	500 ms	Equal to the marker travel time at 0.6 m/s

The margin is deliberately large because worst-case frames, with many candidate regions or heavy background storage activity, take longer than average frames. Reporting the 95th percentile and maximum latency, not just the mean, is therefore part of the acceptance test.

Operating states. The automatic system should have defined states rather than a simple pass/fail output: inspecting, defect detected, review required, image quality poor, deadline missed and system fault. For example, if lighting fails and the frame is too dark, the correct state is “image quality poor”, and the section should be flagged for manual inspection. A system with only “defect” and “no defect” would report a dark frame as clean, which is the most dangerous failure mode.

Section summary	Statement
Given	0.5 s decision limit; 0.6 m/s web; 1.5 mm defect.

Finding	Only an automated chain can give repeatable coordinates and timing.
Decision	Adopt automation, with review bin and audit.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

5. Feature comparison and decision criteria

The main comparison dimensions are speed, repeatability, objectivity, measurement precision, continuity, traceability, adaptability and lifecycle cost. Manual inspection has low initial capital cost but recurring labour cost. Assisted inspection adds a camera and processing station while retaining human decision time. Automated inspection has the highest integration requirement but creates a consistent digital record for every inspected section.

A useful acceptance table should report the number of known defects, true positives, false negatives, false positives, sensitivity, specificity, processing time per frame, and end-to-end marking delay. Sensitivity is TP/(TP+FN); the missed-defect rate is FN/(TP+FN). These quantities should be reported separately for holes, broken threads, stains, weaving irregularities and colour variation because an aggregate score can hide a weak defect class.

The final decision is automated vision for routine inspection, with assisted review for uncertain cases and periodic manual audits. This is not a claim that automation is infallible; it is a system-level decision based on the need for speed, location reporting and repeatability. The design must be revised if field tests fail the missed-defect target.

Criterion	Manual	Assisted	Automated CV
Decision speed	Slow, variable	Limited by human review	Fast, deterministic
Repeatability	Low	Medium	High
Width measurement	Tape or rough estimate	Software, calibrated	Software, calibrated
Defect coordinates	Approximate	Recorded	Recorded in mm with encoder
Traceability	Weak	Good	Good, with stored evidence
Initial cost	Low	Medium	Highest

Running cost	Labour-intensive	Labour plus maintenance	Maintenance and recalibration

How many defects are needed to demonstrate 2 %? If the system misses none of n independent defects, the approximate 95 % upper bound on the true miss rate is 3/n (the "rule of three"). To support a claim of at most 2 % misses, n should be about 150 defects per class with zero misses. If a few misses are observed, more samples are needed. This shows why a handful of test images cannot validate the 2 % target, and why rare defect classes need deliberate collection of examples.

The qualitative ratings above are design expectations. Final selection still depends on measured per-class results, because a system with a good overall score may be weak on the one defect class that matters most to the customer.

Weighted decision example. A weighted scoring matrix can document the choice. Assign weights to the criteria, for instance decision speed 25 %, repeatability 20 %, measurement of width and position 20 %, traceability 15 %, initial cost 10 % and running cost 10 %, then score each approach from 1 to 5. These weights are an example and should be agreed with the plant manager. Whatever weights are used, the approach must still pass the hard requirements (0.5 s decision time and 2 % missed defects), because a weighted score must not allow a strong cost rating to compensate for a failed requirement.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

6. Image processing, computer vision, machine vision and graphics

Image processing transforms one image into another image. Examples include denoising, contrast adjustment, colour conversion and geometric correction. Computer vision interprets image data to estimate properties such as defect type, size and position. Machine vision combines the camera, optics, lighting, processing, communication and actuator into an industrial inspection system. Computer graphics creates synthetic images or diagrams and can help build test cases.

For the textile line, a grayscale image with reduced noise is an image-processing output. A detected oil stain with coordinates and a confidence score is a computer-vision output. A camera connected to the encoder, PC, storage and marking mechanism is the complete machine-vision system. A synthetic fabric image containing a known hole is a graphics-generated test input, not proof of real-world accuracy.

Keeping these terms separate helps avoid an important mistake: enhancement does not itself identify a defect, and a classifier alone is not a complete industrial system. The end-to-end design must connect the pixels to a reliable physical mark and preserve the evidence needed to investigate missed or falsely reported defects.

Field	Input → output	Textile example

Image processing	Image → image	Denoise, flat-field correct, convert to grey
Computer vision	Image → description	Oil stain, 4 mm², centred at (312 mm, 8.4 m), confidence 0.86
Machine vision	Scene → physical action	Camera, light, PC and encoder trigger a marker
Computer graphics	Description → image	Synthetic fabric with a rendered hole for software testing

Why the distinction matters in practice. A vendor who supplies only filtering software has delivered image processing, not an inspection system. A classifier that labels images in a laboratory has delivered computer vision, but it does not yet satisfy the requirement to mark the fabric 0.3 m downstream. The assignment is a machine-vision problem: it includes the lens, illumination, trigger, communication and actuator, and each of these has to be specified and tested.

Computer graphics has a supporting role. Synthetic holes, thread breaks and stains can be rendered on real fabric backgrounds to test coordinate conversion and software robustness. They must not be used to report detection accuracy because rendered defects do not reproduce the texture, noise and optical blur of real defects.

Where Tiruppur's project sits. The mill needs all four fields in different roles. Image processing prepares the picture, computer vision interprets it, machine vision delivers a working inspection station, and graphics supports testing and operator displays. Confusing the roles leads to wrong purchases, for example buying a general photo-editing library when a calibrated industrial camera and an encoder interface are what the mill actually requires.

Procurement impact. When specifying the system, each field becomes a separate line in the purchase and test documents: image-processing functions (flat-field, filtering), computer-vision functions (detection, measurement), machine-vision hardware (camera, lens, light, encoder, marker) and test tools (synthetic images). Asking a supplier to confirm all four avoids the situation where a delivered package contains good algorithms but no way of triggering or marking.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability
Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

7. Capacity, risks and corrective controls

Manual inspection capacity depends on the number of workers, fabric speed, viewing width and break schedule. Because the web is continuous, a worker cannot inspect each small region with unlimited attention. The line should not rely on a single person to notice every 1.5 mm defect while also maintaining

throughput. Automated inspection can examine each acquired region, but the acquisition rate must be matched to motion and the PC must finish each required decision before the marker arrives.

A new defect type may be invisible to a fixed threshold or absent from a trained model. Controls include an anomaly score, a low-confidence review bin, regular audits, saving representative images for seven days, and versioning the detection rules or model. A defect taxonomy should define each class, minimum size, labelling rule and severity before performance is measured.

The recommended approach is accepted only after a controlled trial on actual fabric. The trial should include clean fabric, each known defect, low-contrast stains, different weave structures, changing light, splices and fabric motion. Any failure should be recorded with the image, configuration and cause so that improvements can be tested reproducibly.

Risk	Likely cause	Corrective control
Missed small defect	Undersampling or motion blur	Verify pixel pitch and exposure with a moving test target
Missed faint stain	Contrast 5–8 grey levels lost in noise or quantization	Keep high bit depth; local background compensation; stable lighting
False alarm burst	Shadow, glare, texture change at fabric lot change	Recalibrate, retune thresholds, review queue
Marker misses target	Speed change or encoder slip	Encoder-based scheduling and mark-position test
Deadline exceeded	Worst-case frame or background disk activity	Separate real-time and archive paths; deadline fault state
Unknown defect type	Not in rules or training data	Anomaly score, review bin, periodic audit

Capacity of a single station. One camera station covers the full 1,000 mm width, so capacity is set by the line speed rather than by the number of inspectors. At 0.6 m/s the system inspects 36 m of fabric per minute. A 1,200 mm field and 400 mm height cover 0.48 m² per frame, so the pipeline must process fabric at 0.6 m² per second, i.e. 0.6 m/s × 1.0 m width. This rate is sustained for the entire shift, which is the reason the worst-case latency must be measured under continuous load.

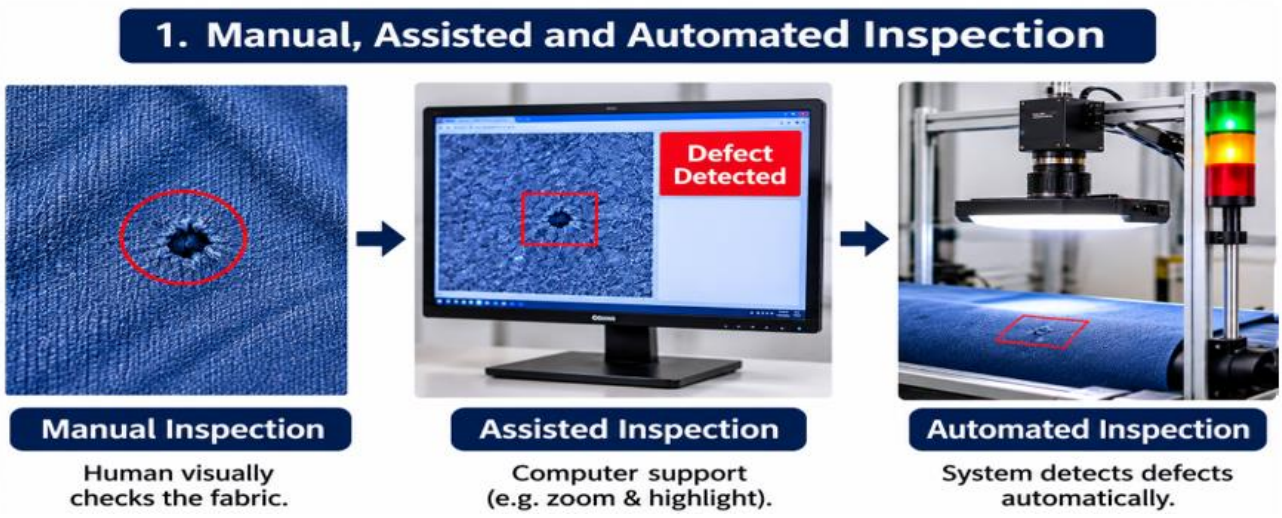
The controls in the table also define the maintenance routine: periodic recalibration, cleaning of the lens and lamp cover, review of the reject log, and checking disk capacity. Without these, performance measured at commissioning will gradually degrade.

Evidence kept for seven days. For each defect event the system should keep a full-resolution crop, a downsampled context view, the mask, the class and confidence, the coordinates, the timestamp, the roll ID and the software version. For ordinary clean frames it is enough to keep metadata and, if the buyer requires it, periodic overview images. This keeps the 2 TB disk usable while preserving what is needed to investigate an escape or a false rejection.

Approach	Strength	Main limitation
Manual	Flexible judgement	Fatigue, weak traceability

Assisted	Highlights candidates	Human decision time remains
Automated CV	Consistent and traceable	Needs validation and maintenance

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.



TASK 2 – Levels of Computer Vision: Pipeline Design and Operation

1. Low-level pipeline

The low-level pipeline performs pixel operations without explicitly reasoning about defect meaning. A typical sequence is image acquisition, denoising, flat-field correction, grayscale or colour conversion, local contrast enhancement and thresholding. The output remains an image or binary mask. This is computationally inexpensive and easy to explain, but a dark weave pattern, shadow or oil stain can produce similar pixel values.

For textile inspection, a low-level-only detector could flag every dark region. This can detect a clear hole or stain, but may also report ordinary dark yarns, texture variation and illumination gradients. It does not adequately distinguish a broken thread from the normal periodic pattern of the weave. Fixed thresholds are especially fragile when the lighting changes from 500 to 3,000 lux.

This pipeline is useful as a preprocessing stage or as a simple baseline. It should not be the sole decision maker because the task requires several defect categories and coordinate-level marking. A low-level-only result must be evaluated on the same labelled images as the more complete alternatives.

Low-level operation	**Purpose**	**Typical risk**
Flat-field correction	Remove illumination gradient across the field	Needs fresh white/dark reference
Gaussian or median filter	Suppress sensor noise	Can blur a 1.5 mm defect if the kernel is too large
Contrast enhancement	Make 5–8 level differences easier to threshold	Amplifies noise and banding
Thresholding	Create a binary candidate mask	Texture and shadow exceed the threshold
Morphology	Remove isolated pixels, close small gaps	May erase thin thread breaks

Computation. These operations are linear in the number of pixels N. For a 2448 × 2048 frame, N = 5.01 × 10^6, and a 5 × 5 filter needs about 25 multiply-add operations per pixel, i.e. about 125 million per frame. This is comfortably within the capability of a 4-core CPU if the code is vectorised, but the real time must be measured on the target computer, not inferred from operation counts.

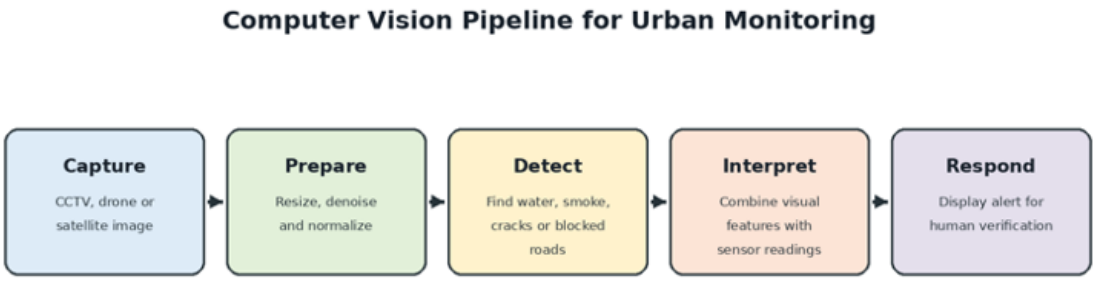
Illustration of fixed-threshold failure. Suppose the fabric background is 150 and a stain is 144. A threshold at 147 separates them under nominal lighting. If the lamp output drifts so that the local background becomes 140, the same threshold labels the whole region as defective; if it becomes 160, the stain is missed. A fixed global threshold therefore cannot cover a six-fold illumination range without local normalisation.

When low-level processing is enough. A low-level pipeline can be the right solution for a single clear task, such as finding the two fabric edges for width measurement. Edges produce a strong, consistent intensity step against the background, so a threshold or gradient method followed by line fitting is simple and fast. It is not enough for stains, weave irregularities and colour variation, where meaning depends on context rather than on a single pixel value.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy thespecification.



2. Three-level pipeline

The three-level design separates pixel preparation, feature extraction and final interpretation. Low level: correct illumination, suppress noise and create candidate masks. Intermediate level: calculate connected components, contour geometry, local texture, edge continuity, colour difference and region contrast. High level: combine these features to label a region as hole, broken thread, oil stain, weaving irregularity, colour variation or background.

This design is explainable because each decision can be linked to a measured feature. A hole may have a sharp boundary and background-like interior; a broken thread may appear as a thin elongated discontinuity; an oil stain may have a local colour or intensity shift; a weave irregularity may change periodic texture. Rules can be tuned for the expected fabric types and run on a CPU.

Its limitation is that handcrafted features can be sensitive to new materials, unusual patterns and defects that do not match the assumed geometry. The design should retain an unknown/uncertain category and record the intermediate masks so engineers can understand why a region was accepted or rejected.

Defect class	Intermediate features used	Typical evidence
Hole	Area, compactness, edge strength, interior similar to background	Sharp closed boundary
Broken thread	Elongation, orientation, length, discontinuity of yarn direction	Thin linear region crossing the weave
Oil stain	Local intensity and saturation difference, soft boundary	Small shift from nearby fabric
Weaving irregularity	Texture energy, local periodicity, yarn spacing	Pattern break over a patch
Colour variation	Mean colour difference in a calibrated space	Large smooth region with different hue

Example of an interpretable rule. A candidate whose bounding-box aspect ratio is above 5 and whose width is below about 3 pixels may be flagged as a thread-like region, then confirmed if its orientation is aligned with the warp or weft and it interrupts the periodic yarn pattern. Such rules are illustrative; actual thresholds must be tuned on labelled fabric, and every rule should be recorded so that a reject can be explained to the customer.

Because each stage produces an inspectable output (mask, region list, feature vector, label), an error can be localised. If a defect was lost, engineers can check whether it was already absent from the corrected image, removed by morphology, merged by labelling or rejected by a rule.

Stage-wise validation. The intermediate level is the main source of engineering control. Region lists can be compared with a labelled mask using overlap measures, so a candidate that overlaps a ground-truth defect is counted as found even before classification. This separates detection (was a region produced?) from classification (was it labelled correctly?), and shows which of the two limits the missed-defect rate.

Section summary	Statement
Given	Five defect classes with different geometry and contrast.
Finding	Explicit features allow class-specific rules and traceable decisions.
Decision	Use the three-level CPU pipeline as the baseline.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

3. Deep-learning pipeline

A learned pipeline can use a convolutional neural network or a vision transformer to map image patches to defect labels, or to generate a pixel-level segmentation mask. A typical flow is image capture, normalization, model inference, confidence thresholding, coordinate conversion and marker communication. Training requires labelled images with examples of each defect and representative defectfree fabric.

Deep learning can learn texture features that are difficult to express as hand-written rules. It may be more robust to weave patterns if the training set covers the variation. However, the available computer has no GPU, so model size and inference time must be tested on the actual 4-core CPU. A high-accuracy model that misses the 0.5-second limit is not a valid solution for this plant.

A practical progression is to begin with a three-level OpenCV pipeline, collect audited images, and benchmark a compact segmentation model later. The model should be adopted only if it improves perclass recall without breaking latency, storage or budget constraints. Training and validation data must be separated by fabric roll or production batch to reduce leakage.

Model option	Output	Suitability on a 4-core CPU without GPU
Small patch classifier	Class per image patch	Feasible if patches are gated to candidate regions
Lightweight segmentation network	Pixel-level defect mask	Needs benchmarking; may require reduced input size
Autoencoder / anomaly model	Anomaly map of clean texture	Useful for unknown defects; needs defect-free training data
Large transformer model	Rich features	Unlikely to meet 0.5 s on this hardware

Why gating is needed. If a 2448 × 2048 frame were cut into non-overlapping 224 × 224 patches, there would be about 11 × 9 = 99 patches. Assuming, for illustration, 5 ms per patch on the CPU, the model alone would take about 0.5 s, which uses the entire decision time. Running the model only on the few candidate regions found by the fast detector reduces the number of patches sharply. The 5 ms value is an assumption that must be replaced by a benchmark.

Data requirements are also demanding. Each defect class needs enough labelled examples across fabric colours, weaves and lighting, and defect-free textures must be well represented. Splitting by roll or production batch prevents the model from memorising fabric-specific texture and reporting an optimistic accuracy.

Hybrid approach. A hybrid design is attractive for this plant. Fast rules or statistical background models find candidate regions on the CPU, and a compact learned model labels only those regions. This keeps latency predictable, uses learning where handcrafted features are weak, and permits a safe fall-back to the rule-based decision if the model is unavailable. The learned component should be adopted only after it improves per-class recall on the hold-out set without breaking the timing budget.

Section summary	Statement
Given	4-core CPU, no GPU; 0.5 s limit.
Finding	Learned models help only if gated to candidate regions.

Decision	Benchmark a compact model after the baseline works.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

4. Block diagrams and stage I/O

Pipeline A: Camera → image → denoise/threshold → binary mask → simple defect/no-defect rule. Pipeline B: Camera → corrected image → candidate regions → feature vector → rule or classifier → defect label and coordinates. Pipeline C: Camera → normalized image/patches → neural inference → segmentation/class labels → confidence filter → coordinate mapping → marker command.

The input to acquisition is reflected light from the fabric; the output is a digital image. Preprocessing receives the image and outputs a corrected image. Segmentation receives the corrected image and outputs a mask or region proposals. Feature extraction receives regions and outputs numeric descriptors. Classification receives descriptors or image patches and outputs labels and confidence. Coordinate mapping converts pixel coordinates to millimetres, and the control stage converts those coordinates to a marker event.

A robust implementation stores the raw or minimally processed frame as well as the corrected view used for analysis. This allows a later audit to distinguish a sensor/acquisition failure from a processing failure. The raw image should not be overwritten by enhancement because enhancement can alter apparent contrast.

Stage	Input	Output	Check performed
Acquisition	Reflected light	Raw frame, timestamp, encoder count	Frame present; not saturated; trigger valid
Preprocessing	Raw frame	Corrected image	Brightness and focus metrics
Segmentation	Corrected image	Mask or region proposals	Region count within expected range
Feature extraction	Regions	Feature vectors	Features finite and in range
Classification	Features or patches	Label and confidence	Low confidence goes to review
Coordinate mapping	Pixel position and encoder count	Position in mm	Calibration residual within limit
Control	Defect event	Marker command	Deadline met; event sent once

Every record passed between stages should carry the frame number, timestamp and encoder count. This allows the later stages to join a decision with the raw image, and it prevents a late result from being applied to the wrong section of fabric if frames are processed out of order.

Raw versus processed images. The raw frame should be stored or at least retained in memory until the decision is complete, because enhancement can change apparent contrast and may remove evidence. If the audit store keeps only the processed image, it becomes impossible to tell whether a missed stain was invisible to the camera or removed by software. This is why the diagram keeps the acquisition buffer separate from the corrected image.

Section summary	Statement
Given	Each stage has defined inputs and outputs.
Finding	Raw frames and intermediate masks enable fault localisation.
Decision	Store timestamps, encoder counts and raw frames.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

5. Required stages for each defect

Holes: acquire with enough spatial detail; normalize illumination; detect a region whose interior and edge differ from the local fabric; measure area and shape; reject likely lint or gaps using context. Broken threads: emphasize line continuity and directional texture; identify a local interruption in expected yarn structure; measure orientation and length. Oil stains: use colour and intensity features with local background compensation, because contrast may be only 5–8 grey levels.

Weaving irregularities: compare local texture statistics or periodic structure with neighbouring regions; detect repeated spacing changes or missing yarns. Colour variation: estimate colour in a calibrated colour space and compare it with a reference or local neighbourhood. The same preprocessing can be shared, but class-specific features should be retained because one universal threshold will not perform equally on all defect types.

The pipeline should include a confidence and severity stage. If two defect classes overlap or the image quality is poor, the system should mark the region for review rather than silently classify it as defect-free. A failure-to-acquire state should also be explicit: a missing frame must not be treated as a clean fabric region.

Defect	Dominant cue	Preferred representation	Special risk
Hole	Edge and interior difference	Grayscale and local contrast	Lint or a gap between yarns mistaken for a hole

Broken thread	Line interruption	Grayscale, directional filters	Weave pattern resembles a thread
Oil stain	Faint local change	Local contrast plus HSV/colour	Contrast of only 5–8 levels
Weaving irregularity	Texture change	Texture features on patches	Natural periodic variation
Colour variation	Hue or lightness difference	Calibrated colour space	Illumination change looks like colour change

Severity grading. After detection, a severity stage can combine class, size and count per metre to decide between marking for rework, marking for rejection or only logging. The grading rules are a commercial agreement with the customer, not a computer-vision result, and must therefore be written into the specification before testing so that the detector is evaluated against the correct target.

The dedicated stages also make it possible to set a different operating point for each class. A hole may justify a conservative threshold with few false alarms, whereas a faint stain may deserve a more sensitive setting with human review of uncertain cases.

Shared and class-specific stages. Acquisition, flat-field correction, coordinate mapping and marker control are shared by all classes. Feature extraction and decision thresholds are class-specific. The software should therefore be organised as a common front end followed by parallel class detectors, whose candidate outputs are merged and de-duplicated before severity grading. De-duplication matters because one physical defect, such as an oil-stained hole, can trigger two detectors, and the marker should be fired once.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

6. Complexity, explainability and timing

Low-level operations such as Gaussian filtering and thresholding are usually linear in the number of pixels. Connected-component labelling and contour analysis are also practical for CPU use. Deep models have costs depending on architecture, image size and number of regions. For a full-width fabric image, reducing unnecessary pixels and using region-of-interest processing can improve response time, but the region must not exclude possible defects.

The relevant latency is not just inference time. It includes sensor exposure/readout, transfer, preprocessing, detection, coordinate conversion, communication and marker scheduling. The marking mechanism is 0.3 m downstream, so at 0.6 m/s the nominal travel time is $0.3/0.6 = 0.5$ s. The system must issue the command before the relevant fabric section reaches the marker and should allow a safety margin.

The 0.5-second decision requirement is a hard system constraint. Benchmarks should use the worst representative frame, not only an average. The PC should be tested while image saving and network communication are active, since background tasks can affect latency.

Operation	Typical cost	Comment
Flat-field and filtering	Linear in pixel count	Vectorised; reuse buffers
Thresholding, morphology	Linear in pixel count	Very fast on the CPU
Connected components	Close to linear	Practical for full frames
Texture features	Per-patch computation	Run on candidate regions only
Neural inference	Depends on model and patch count	Must be benchmarked; gate by candidates

Timing check. At 0.6 m/s a section of fabric reaches the marker after 0.5 s. If the measured pipeline needs 0.31 s at the 95th percentile and the marker command needs 0.03 s, the remaining margin is $0.5 - 0.34 = 0.16$ s. This arithmetic is only an example of how the budget is checked; the numbers must come from measurement. If the margin is negative for even a few frames, the system must raise a fault or use a safe response rather than mark in the wrong position.

Explainability supports maintenance as well as acceptance. A rule-based detector can be tuned by a technician who sees the overlay, whereas a black-box score offers little guidance when a new fabric lot produces false alarms.

Worst-case planning. The slowest frame is not the average frame. A frame containing a large stained area or a crowd of candidate regions may need many times the usual feature computation. Two controls are useful: a cap on the number of regions analysed in detail per frame, with the remainder marked “review” rather than discarded, and a watchdog timer that raises a fault when the per-frame time exceeds the allocation. Both prevent a rare heavy frame from silently delaying the marker.

Documentation. For every pipeline stage the report to the plant should list its algorithm, parameter values, expected time and measured time. This table lets maintenance staff find the stage responsible when latency increases and supports re-validation after a software update. Any change to a parameter should be logged with date and author, since an untracked threshold change is a common cause of unexplained shifts in the false-alarm rate.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

7. Failure analysis and mitigation

Low-level-only false positives arise when shadows, yarn texture and oil stains have similar brightness. In a three-level system, inadequate features can lead the classifier to receive a poor representation: if the segmentation mask merges two defects, the high-level stage cannot reliably recover the true region boundaries. This illustrates why each stage must be validated separately.

Mitigations include controlled diffuse lighting, a reference white/dark calibration, local rather than global thresholds, morphology tuned to the expected defect scale, texture features, and an unknown-defect route. A diagnostic overlay should show candidate regions and class labels. During commissioning, operators should be able to review the raw frame and the intermediate mask.

The system should report per-class confusion matrices and false-negative examples. A single overall accuracy figure is inadequate because a system can achieve high accuracy by classifying most pixels or images as defect-free while missing rare but important defects.

Failure	Effect	How it is detected	Mitigation
Shadow treated as a stain	False alarm	Review of overlays	Diffuse light, local thresholds
Two defects merged by the mask	Wrong area and marker position	Region geometry check	Distance and orientation split logic
Defect removed by morphology	Missed defect	Compare pre- and postmorphology masks	Tune kernel in millimetres
Missing or late frame	Uninspected fabric	Frame counter and timestamps	Treat as fault, never as clean
Model fails on a new fabric lot	Drop in recall	Monitor review-queue rate	Collect samples, retrain, revalidate

Per-stage validation. Each stage should be tested with its own reference data: the preprocessing stage with a uniform grey target, the segmentation stage with masks drawn by an expert, the classifier with a labelled confusion matrix, and the control stage with a marker-position test. A stage that is validated only through the final result can hide compensating errors.

A useful reporting format is a confusion matrix for each defect class and a gallery of false negatives, because the false negatives are the cases that determine whether the 2 % missed-defect limit is satisfied.

Monitoring in operation. After commissioning, simple statistics reveal drift: the number of candidate regions per metre, the fraction sent to review, the average frame brightness and focus score, and the number of dropped frames. A sudden change in any of these is an early warning of dirty optics, a failing lamp, a new fabric lot or a software fault, and should trigger an inspection before defects are missed in volume.

Section summary	Statement
Given	Shadows, merges, lost frames and new fabric lots are credible failures.
Finding	Each failure has a detection method and a control.

Decision	Validate every stage and report per-class results.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Decision rule: accept only when measured defect recall, width error, latency and marker position satisfy the specification.

Pipeline Flow:

Fabric Image Acquisition

↓

Low-Level Processing

↓

Mid-Level Feature Extraction

↓

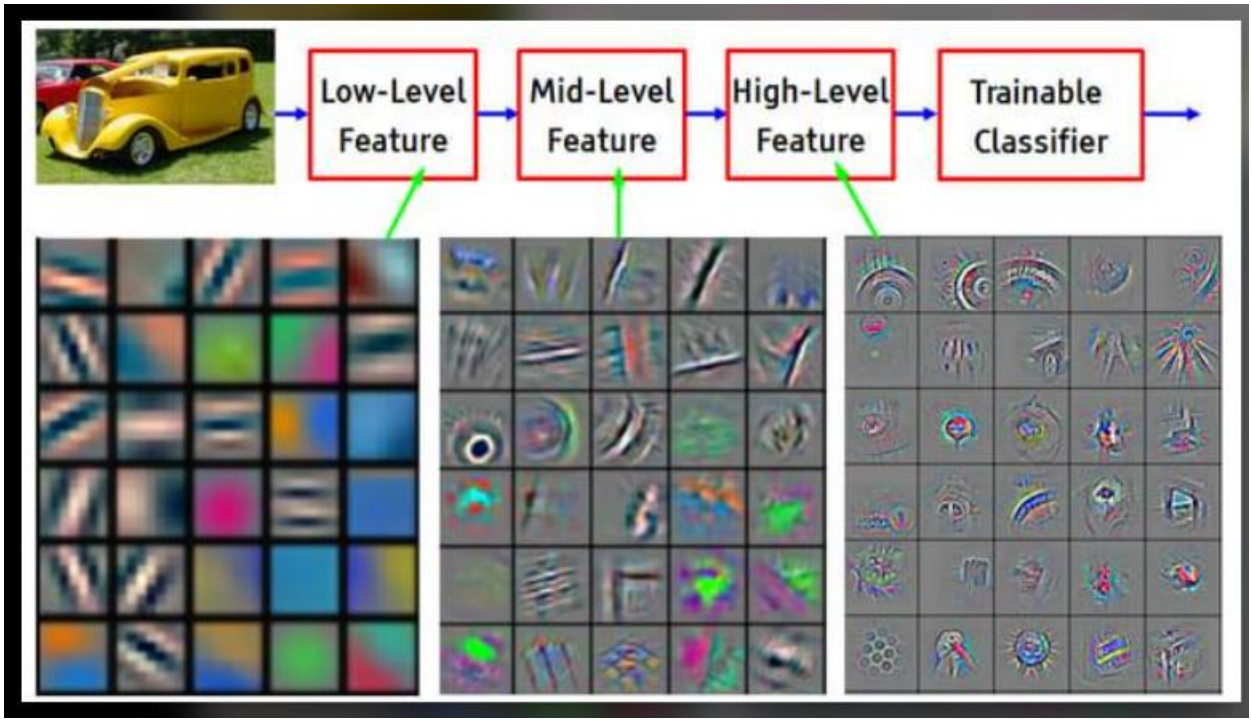
High-Level Defect Classification

↓

Defect Detection and Grading

↓

Final Result / Operator Alert



TASK 3 – Image Representation Analysis

1. RGB representation

RGB stores each pixel as red, green and blue channel values. With 8 bits per channel, each channel has 256 possible values and a three-channel image uses 24 bits per pixel before compression. RGB preserves colour information useful for colour variation and some oil stains. It also provides a familiar input for many pretrained vision models.

The main weakness is that brightness and colour are mixed across the channels. A change in illumination can change all three values even when the material is unchanged. A global threshold in RGB is therefore unreliable in a shed where light varies from 500 to 3,000 lux. For this application, RGB should be calibrated and paired with local normalization rather than used as raw fixed thresholds.

RGB is appropriate as the retained source representation and for colour inspection, but not necessarily as the only feature space. Keeping the original RGB image supports audit and later model development.

Intensity-invariant colour example. A fabric pixel measured as (R,G,B) = (180,170,160) under strong light might read (90,85,80) under half the exposure. The raw values differ by a factor of two, yet the normalised chromaticities are identical: r = R/(R+G+B) = 180/510 = 0.353 and 90/255 = 0.353; g = 0.333 in both cases; b = 0.314 in both cases. Working with channel ratios, rather than raw channel values, removes much of the effect of overall brightness and is a cheap first step towards illumination-robust colour comparison.

Aspect of RGB	Advantage	Limitation for this plant
Bit cost	24 bits per pixel, 15.04 MB for a 2448 × 2048 frame	Three times the data of a grey image
Colour fidelity	Keeps the full appearance for audit	Needs white-balance and colour calibration
Model input	Matches pretrained network input	Not needed for rule-based edge detection
Illumination	Direct and simple	Brightness change affects all three channels together

Role in the final design. RGB is stored as the master record and used for colour measurements after white-balance calibration with a reference tile. It is not the best space for applying a single intensity threshold, because a lamp that fades by 10 % changes R, G and B at once. For defect search, a derived grey or local-contrast image is more convenient, while RGB remains available when an engineer needs to review what the camera actually saw.

A practical white-balance routine images a neutral grey or white reference at the start of every shift and stores the gain for each channel. Without it, a slow yellowing of the lamp would appear as a colour variation defect across the entire roll.

Requirement	Recommended check	Acceptance evidence

Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

2. HSV representation

HSV represents colour using hue, saturation and value. Hue describes the dominant colour family, saturation indicates colourfulness, and value approximates brightness. It can help separate colour changes from brightness variation, especially when the illumination is controlled. A suspected oil stain may differ in saturation or hue even when its grayscale contrast is small.

HSV is not perfectly invariant to illumination. Hue becomes unstable in low-saturation or very dark pixels, and specular reflections can distort the values. The conversion also introduces computation and should be tested on the exact fabric colours. It is useful as one channel family rather than as a universal replacement for RGB.

For colour variation, HSV can be informative; for holes and broken threads, geometry and grayscale texture may be more useful. The recommended pipeline can compute HSV for colour features and grayscale/local-contrast features for structure.

Worked conversion. Take a clean pixel (R,G,B) = (180,170,160) and a slightly stained pixel (172,160,148). For the clean pixel V = max = 180, S = (180 − 160)/180 = 0.111 and H = 60 × (G − B)/(max − min) = 60 × 10/20 = 30°. For the stained pixel V = 172, S = (172 − 148)/172 = 0.140 and H = 60 × 12/24 = 30°. The hue is unchanged, brightness falls by 8 levels, and saturation rises by about 26 % in relative terms. The saturation change gives a second cue in addition to the small grey-level change. In OpenCV's 8-bit convention, hue is stored as H/2, so both pixels have H = 15, and S is scaled to 0–255 (28 versus 36).

HSV channel	What it measures	Use in textile inspection	Caution
Hue (H)	Colour family	Detect dye or shade drift	Unstable when saturation is very low
Saturation (S)	Colour purity	Oil can change surface and reflectance colourfulness	Specular highlights lower S
Value (V)	Brightness	Compare with local background	Strongly affected by lamp level

Limitation to note. On white or pale fabrics, saturation is near zero and the hue angle is mathematically defined but physically meaningless, so hue-based rules must be disabled below a saturation threshold. On dark fabrics, V is small and noise dominates the hue. HSV is therefore enabled only for pixels whose S and V lie in a reliable range, and the unreliable pixels are handled by grey-level or texture features instead.

Practical tip. The conversion should be applied after flat-field correction, not before. If HSV is computed from an uncorrected image, the lamp gradient appears as a slow change in value and saturation across the field and is mistaken for a colour drift. A short test is to image a uniform grey card and check that H, S

and V are almost constant across the frame; large variations indicate that illumination correction is still insufficient for colour work.

Section summary	Statement
Given	Hue, saturation and value separate colour from brightness.
Finding	Saturation gives a second cue for faint stains; hue is unreliable for dark or pale pixels.
Decision	Use HSV for colour features within valid ranges.

Requirement	Recommended check	Acceptance evidence
Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

3. Grayscale representation

A grayscale image stores one intensity value per pixel. At 8 bits it uses 8 bits per pixel, approximately one third of an 8-bit RGB image before compression. Grayscale simplifies edge detection, thresholding, local contrast measurements and texture analysis. Holes and broken threads can often be detected from structural intensity differences.

Grayscale discards chromatic information. Two regions with similar brightness but different colours can become indistinguishable, so colour variation and some oil stains may be missed. Low-contrast stains with only 5–8 grey levels require careful noise control and stable illumination; a contrast difference of a few levels can be overwhelmed by sensor noise or uneven lighting.

Grayscale is efficient for structural defects but should be combined with colour channels when the task includes colour variation and oil stains. It should not be the only stored source image.

Conversion and information loss. A common luminance formula is $Y = 0.299\,R + 0.587\,G + 0.114\,B$. The colours (200,100,50), an orange, and (100,150,50), a green, give $Y = 59.8 + 58.7 + 5.7 \approx 124.2$ and $Y = 29.9 + 88.1 + 5.7 \approx 123.7$. Their grey values are almost identical although the colours are clearly different. This is the reason a grey-only detector cannot reliably find colour variation, and why grey should be one of several parallel representations rather than the only input.

Defect	Value of grayscale	Comment
Hole	High	Strong edge and interior contrast with the surrounding yarn
Broken thread	High	Line interruption visible in intensity
Oil stain	Medium	Useful only if the 5–8 level change exceeds noise

Weaving irregularity	High	Texture statistics computed on intensity
Colour variation	Low	May be invisible in grey

Noise consideration. Averaging three colour channels into one grey value reduces random sensor noise by a factor of up to $\sqrt{3} \approx 1.7$ if the noise in each channel is independent, which can slightly help faint stains. However, if the stain differs mainly in one colour channel, averaging dilutes that difference. A better approach is sometimes to choose the channel or channel combination with the highest contrast for the fabric and stain under investigation, determined on calibration samples.

Choice of conversion. Different grey conversions are available: a luminance weighting, a plain channel average, or a single channel such as green. On a fabric whose stain absorbs mainly blue light, the blue channel may show more contrast than the weighted grey. The conversion should be selected using calibration samples, comparing the contrast-to-noise ratio of known defects for each option, and the selected formula must be fixed in the configuration to keep results comparable between rolls.

Section summary	**Statement**
Given	One channel, 8 bits, one third of RGB data.
Finding	Strong for structure; blind to some colour differences.
Decision	Use grey for holes, threads and texture; do not use it alone.

Requirement	**Recommended check**	**Acceptance evidence**
Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

4. Memory and computation

For an image of W × H pixels, an 8-bit grayscale image requires W×H bytes; an 8-bit RGB image requires 3×W×H bytes. A 12-bit sensor often stores samples in 16-bit words in common processing pipelines, so the practical memory requirement may be 2×W×H bytes per single channel. Compression changes disk usage but not the uncompressed working-buffer estimate.

At 2448 × 2048, the image has 5,013,504 pixels. One 8-bit grayscale frame requires about 5.01 MB decimal, or about 4.78 MiB; RGB requires about 15.04 MB decimal. If the frame rate is 10 frames/s, the raw RGB stream is about 150 MB/s before protocol overhead. These values illustrate why acquisition rate and storage policy must be designed together.

For CPU efficiency, perform the minimum necessary conversions and reuse buffers. Do not repeatedly convert RGB to HSV and grayscale inside every defect loop. Compute shared representations once per frame, then pass them to the class-specific detectors.

Representation	Bytes per pixel	Frame size (2448 × 2048)	Rate at 10 frames/s
Grayscale 8-bit	1	5.01 MB	50 MB/s
Grayscale 12-bit in 16-bit word	2	10.03 MB	100 MB/s
RGB 8-bit	3	15.04 MB	150 MB/s
HSV 8-bit (derived)	3	15.04 MB	150 MB/s
RGB + HSV + grey kept together	7	35.09 MB	351 MB/s

Interpretation of the table. The pixel count is 2448 × 2048 = 5,013,504. Keeping RGB, HSV and grey simultaneously needs 7 bytes per pixel, about 35 MB per frame, which is trivial for 16 GB of RAM but would be a heavy memory-bandwidth load if every operation made a full copy. The values in the last column are working-memory throughputs, not disk rates. For a 2 TB disk, writing even the 150 MB/s RGB stream continuously would produce about 13 TB per day, so only events and compressed crops can be archived.

Efficient implementation. Compute each representation once per frame, in place where possible, and share the buffers among the class detectors. If the line-scan camera supplies grey and colour only for the central region, or if the HSV conversion is needed only inside candidate regions, the conversion can be limited to those regions. This reduces CPU time without removing any information the detectors need.

Buffer management. In a continuous system, frames pass through a ring buffer of a few preallocated arrays, avoiding repeated memory allocation that causes unpredictable delay. For 5.01 MB grey frames, a ring of eight buffers uses only about 40 MB. The number of buffers must be large enough that a slow frame does not force the camera driver to overwrite one that is still being processed, and the software should count and report any such overwriting as a dropped frame.

Section summary	Statement
Given	2448 × 2048 = 5,013,504 pixels.
Finding	Grey 5.01 MB, RGB 15.04 MB, all three maps 35.09 MB per frame.
Decision	Compute each map once and reuse buffers.

Requirement	Recommended check	Acceptance evidence
Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

5. Representation suitability matrix

The following qualitative matrix summarizes the role of each representation. RGB: strong for broad colour differences, moderate for stains, moderate for holes and thread breaks when edges are visible. HSV: strong

for hue/saturation changes, moderate for stains, moderate for holes and thread breaks. Grayscale: weak for pure colour changes, useful for stain contrast, strong for edges and structural defects.

These ratings are design expectations, not experimental measurements. A real evaluation should use a labelled dataset and report sensitivity and false-positive rate separately for each defect type. The same images, split and thresholds should be used when comparing representations so that the result is not biased by different test conditions.

The representation choice should be based on complementary information. RGB preserves the original appearance, HSV supports colour features, and grayscale supports structural features. This multirepresentation approach costs additional processing but is appropriate when defects differ in both colour and geometry.

Defect	**RGB**	**HSV**	**Grayscale**	**Best combination**
Hole	Moderate	Moderate	Strong	Grayscale edges + RGB for audit
Broken thread	Moderate	Moderate	Strong	Grayscale directional filters
Oil stain	Moderate	Moderate	Useful	Local contrast + saturation
Weaving irregularity	Moderate	Weak	Strong	Grayscale texture statistics
Colour variation	Strong	Strong	Weak	Calibrated colour space (RGB/HSV)

How to turn the ratings into evidence. Each rating is a hypothesis. To test it, run the same detector on the same labelled images using only RGB, only HSV, only grayscale and the combination, and record sensitivity and false alarms for each defect class. A representation earns its place if it raises sensitivity of at least one class by an amount that justifies its extra computing, for instance by reducing missed oil stains while keeping the false-alarm rate constant.

Statistical caution. Differences between representations must be judged with their uncertainty. If one representation detects 48 of 50 stains and another 46 of 50, the difference of two stains is within what could arise by chance for such small samples. Larger test sets, or paired comparisons on the same images, are needed before the simpler or cheaper representation is discarded.

Defect interactions. Real defects often combine classes: an oil stain may soften the surrounding weave, or a hole may be surrounded by discolouration from heat. A detector that checks only one representation may report only one of the effects. The final decision stage should therefore look at the combination of features, merge overlapping candidates, and report the most severe class while keeping the others in the record for later analysis.

Section summary	**Statement**	
Given	Ratings are design expectations, not measurements.	
Finding	Different defect classes favour different representations.	
Decision	Test each representation on identical labelled images.	
Requirement	**Recommended check**	**Acceptance evidence**
Defect recall	Per-class test set	TP/FN counts

Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

6. Low-contrast oil-stain example

Suppose healthy fabric intensity is 150 and an oil stain is 144, giving a difference of 6 levels. A fixed threshold might work in a stable region but fail if illumination causes the local background to shift by 10 levels. Local background estimation compares the candidate region with nearby fabric and can reduce the effect of broad illumination gradients.

For RGB, the stain may also produce a subtle channel ratio change. HSV saturation may change if the oil changes the surface reflectance. These effects must be verified with real oil-stained samples; no representation guarantees detection solely from the numeric contrast value. Polarized or directional illumination can help reduce glare, while a dark-field arrangement may emphasize surface irregularities.

The detector should combine local intensity difference, colour difference, shape and texture context. Lowconfidence stains should be logged for human review during commissioning. This reduces the risk that a stain is missed because a single threshold is poorly matched to a new fabric lot.

Contrast-to-noise calculation. Using the numbers in the example, the contrast is $150 - 144 = 6$ grey levels. If the sensor noise has a standard deviation of 2 levels (an assumed value for illustration), a single pixel has a contrast-to-noise ratio of $6/2 = 3$, which is not reliable. Averaging a region of 16 pixels reduces the noise by $\sqrt{16} = 4$ to 0.5 levels, giving a ratio of 12. This is the reason oil stains should be detected as regions, using local means, rather than pixel by pixel.

Step	Operation	Purpose
1	Estimate local background with a large Gaussian or median filter	Remove smooth illumination gradients
2	Subtract background from the image	Leave only local departures from normal fabric
3	Average over a region of a few millimetres	Increase contrast-to-noise ratio
4	Apply a threshold chosen from clean-fabric statistics	Control false alarms
5	Check saturation and shape cues	Confirm that the region resembles a stain

Choosing the threshold. A threshold of k standard deviations of the background-subtracted clean fabric controls the false-alarm rate. With $k = 3$, about 0.13 % of Gaussian noise pixels exceed the threshold on the high side, but a region test that requires several adjacent pixels reduces false alarms much further. The value of k must be tuned on real clean rolls, because fabric texture is not Gaussian noise, and then confirmed on a separate set of oil-stained samples.

Validation sample. To validate the stain detector, prepare test patches by applying controlled drops of the actual oils used in the mill onto clean fabric, photograph them immediately and again after drying, and

measure the grey-level and colour difference for each. Stains age and spread, so contrast changes with time. Testing at both stages shows whether the system detects fresh and dried stains, and gives a realistic spread of contrast values around the 5–8 level case.

Section summary	Statement
Given	150 versus 144 grey levels; 6-level difference.
Finding	Region averaging raises contrast-to-noise ratio from 3 to 12 in the example.
Decision	Use local background subtraction and region-level tests.

Requirement	Recommended check	Acceptance evidence
Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

7. Recommended combined representation

The recommended approach is to retain RGB as the source frame and compute HSV plus grayscale or a local-contrast map for analysis. Colour variation is evaluated using calibrated colour features. Holes and broken threads use grayscale edges and texture discontinuities. Oil stains use local contrast plus colour/saturation cues. Weaving irregularities use texture descriptors over local patches.

The pipeline should normalize illumination before comparing colour or intensity. A flat-field reference can estimate systematic brightness variation across the field of view. The system should also mask saturated highlights because a clipped reflection contains little reliable colour information. If the fabric has different colours or finishes, calibration profiles may be required for each material recipe.

Final choice: combined RGB/HSV/grayscale processing. It preserves the source data and gives the detector distinct features for colour and structure. The choice should be revised only if measured results show that the added representation does not improve defect recall enough to justify its CPU and memory cost.

Processing stage	Representation	Output
Master record	RGB (raw, calibrated)	Stored frame for audit and retraining
Structure analysis	Grayscale or local-contrast map	Holes, thread breaks, texture features
Stain analysis	Background-subtracted grey + saturation	Oil-stain candidates
Colour analysis	Calibrated colour (RGB or HSV or Lab)	Colour-variation regions
Decision	Feature vector from all maps	Class, confidence, coordinates

Calibration profiles. Each fabric recipe (colour, weave and finish) should have a stored profile containing the white reference, the flat-field image, the thresholds and the colour tolerances. A change of recipe loads

a different profile, so the detector never applies the thresholds of a light cotton to a dark cotton–polyester blend. The profile identifier is saved with every defect record so that results can be reproduced.

Cost-benefit rule. The additional representations are retained only if their measured benefit exceeds their cost. A simple test is to remove one representation at a time and observe the change in per-class sensitivity and in the processing time. If removing HSV leaves all sensitivities unchanged within the measurement uncertainty, the HSV stage can be dropped to save computing time.

Implementation summary. In software the recommended design is a single function that accepts the raw frame and the stored profile and returns three maps (grey or local contrast, saturation or colour difference, texture energy) plus the original. These maps are generated once and reused. Because all detectors read from the same maps, the pipeline is easy to test: replacing a map with a synthetic one allows each detector to be checked independently of the camera.

Section summary	Statement
Given	Defects differ in colour and geometry.
Finding	RGB master plus grey and saturation maps covers all five classes.
Decision	Adopt the combined representation; revalidate by ablation.

Requirement	Recommended check	Acceptance evidence
Defect recall	Per-class test set	TP/FN counts
Latency	Timestamp full pipeline	Median and worst-case
Coordinate accuracy	Calibrated target	Error in mm

OpenCV example: hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV); gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

TASK 4 – Connectivity Analysis

1. Pixel neighbourhood definitions

A binary mask labels each pixel as foreground or background. Four-connectivity considers only the horizontal and vertical neighbours: north, south, east and west. Eight-connectivity also includes the four diagonal neighbours. Mixed or m-connectivity uses 4-neighbours and selected diagonal neighbours when the diagonal connection does not create ambiguous adjacency through a shared 4-neighbour.

The choice affects whether diagonally touching defect pixels are considered one component or separate components. For fabric, a thin diagonal broken thread may form a connected line under 8-connectivity, while 4-connectivity can split it into multiple small pieces. Conversely, 8-connectivity may merge nearby diagonal noise into a larger region.

The connectivity definition must be consistent between component labelling, area measurement and downstream coordinate calculation. The selected method should be validated on masks representing real holes, thread breaks, stain boundaries and weave patterns.

Connectivity	**Neighbours of pixel P**	**Distance to neighbours**	**Typical use**
4connectivity	N, S, E, W (4 pixels)	1 pixel	Conservative separation of nearby objects
8connectivity	All 8 surrounding pixels	1 pixel (edge) or √2 pixels (corner)	Continuous diagonal lines and irregular outlines
mconnectivity	4-neighbours, plus a diagonal only when both shared 4-neighbours are background	1 or √2 pixels	Avoid ambiguous double paths in digital topology

Foreground and background must be paired. If the foreground uses 8-connectivity, the background should use 4-connectivity, and vice versa. Otherwise the same image can seem to have contradictory properties; a one-pixel-wide diagonal line would both connect the foreground across itself and, by 8connectivity of the background, also connect the two sides of the line. The pairing is the digital version of the Jordan curve theorem and is relevant to hole detection: a hole in a fabric mask is a background component enclosed by foreground.

Which choice is the default here? For defect candidates in this plant, 8-connectivity is used first because thread breaks and hole edges are often oblique and one pixel wide. The number of components is then cleaned by area and shape rules. The choice is documented in the configuration file and validated on the labelled masks, so a different connectivity can be justified if the measured fragmentation or merging proves worse.

Neighbourhood sketch (1 = foreground pixel):

4-neighbour:

```
        N
    W   P   E
        S
```

8-neighbour: NW N NE
W SW S P E
SE

OpenCV example: n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)

2. Illustrative binary pattern

Consider the binary matrix: 1 0 0 / 0 1 0 / 0 0 1. Under 4-connectivity, each foreground pixel is isolated because none shares an edge with another foreground pixel. Under 8-connectivity, all three pixels belong to one component through diagonal adjacency. Under m-connectivity, diagonal adjacency is permitted only when the shared 4-neighbour conditions do not make the connection ambiguous.

For a horizontal pattern such as 1 1 1 / 0 0 0 / 0 0 0, all three methods identify one connected component because each foreground pixel touches another horizontally. This simple example shows why diagonal patterns are important when testing connectivity.

The number of components is not itself a defect count. A single physical defect may fragment into several components because of noise, while several defects may merge because of thresholding. Connectivity should therefore be combined with morphology and geometric constraints.

Binary pattern (rows)	**4connectivity**	**8-connectivity**	**m-connectivity**
100 / 010 / 001 (diagonal)	3 components	1 component	1 component
010 / 101 / 010 (four pixels around a gap)	4 components	1 component	1 component
110 / 011 / 000 (stair-step)	1 component	1 component	1 component
11 / 01 (L shape of three pixels)	1 component	1 component, with a redundant diagonal link	1 component, redundant link removed

Reading the table. In the second pattern, four foreground pixels surround a background centre. They touch only at corners, so 4-connectivity sees four separate dots, whereas 8-connectivity sees a ring of one object. In a fabric mask this could be the broken outline of a small hole, so 8-connectivity returns one defect with a hole in its middle, while 4-connectivity returns four tiny objects that could all fall below the minimum-area filter. In the last pattern the diagonal link between the two end pixels is redundant, and mconnectivity removes it because the pixels are already connected through the shared 4-neighbour.

Counting caution. The counts in the table are pattern properties, not defect counts. After labelling, the software still has to decide how many physical defects each component represents, for example by

checking size, elongation and the distance between neighbouring components. A label count used directly as a defect count will be wrong whenever noise splits or merges real defects.

Neighbourhood sketch (1 = foreground pixel):

```
4-neighbour:      N
                W P E
                  S

8-neighbour:    NW N NE
                W P E
                SW S SE
```

OpenCV example: `n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)`

3. 4-connectivity in fabric masks

Four-connectivity is conservative. It avoids joining diagonal pixels that only meet at a corner and can reduce accidental merging between nearby defects. It can be helpful when component separation is more important than continuity, for example when small isolated spots must be counted separately.

Its weakness is that a real thread break or narrow diagonal hole can be fragmented. A fragmented region may fall below the minimum area threshold and be discarded as noise. If the detector uses 4-connectivity, a later linking stage may be needed to join fragments that lie on the same orientation and within a permitted distance.

A 4-connected mask should be tested on diagonal threads, jagged hole edges and oblique fabric defects. If fragmentation is common, it is not a suitable default for the full inspection pipeline.

Fragmentation example. Suppose a broken thread is imaged as a diagonal line of six pixels. Under 4connectivity it becomes six isolated components, each of area one pixel. If the noise filter removes components smaller than three pixels, all six are discarded and the defect disappears. Under 8-connectivity the same line is one component of six pixels. At 0.49 mm per pixel, a diagonal run of six pixels is about $6 \times 0.49 \times \sqrt{2} \approx 4.2$ mm long, i.e. well above the 1.5 mm minimum defect size, so losing it would be a genuine miss.

Situation	Effect of 4-connectivity	Remedy
Diagonal thread break	Fragments into single pixels	Use 8-connectivity or link fragments their along orientation
Jagged hole boundary	Outline broken into pieces	Close small gaps before labelling
Two nearby spots	Kept separate	Benefit: counts and positions remain distinct
Noise speckles	Cannot chain diagonally	Benefit: fewer false bridges

Linking stage. If 4-connectivity is retained for counting spots, fragments can be re-joined by a linking step: two fragments are merged when the gap between them is less than about 2 pixels (about 1 mm at

0.49 mm per pixel) and the line joining them is within a few degrees of their orientation. This adds code and tuning but keeps the separation of true neighbouring defects.

Counting application. There are situations where 4-connectivity is the better choice. If the customer's grading rule counts separate pinholes per metre, merging two holes that happen to touch diagonally would reduce the count and could change the grade. In that case, 4-connectivity or m-connectivity gives a count that follows the physical definition more closely, while the thread-break detector, which needs continuity, uses 8-connectivity. Different classes may therefore use different connectivity.

Section summary	Statement
Given	Thin diagonal defects cross the pixel grid.
Finding	A six-pixel diagonal becomes six fragments under 4-connectivity.
Decision	Avoid 4-connectivity for threads; use linking if it is kept.

Neighbourhood sketch (1 = foreground pixel):

```
4-neighbour:      N
                W P E
                  S

8-neighbour:    NW N NE
                 W P E
                SW S SE
```

OpenCV example: `n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)`

4. 8-connectivity in fabric masks

Eight-connectivity treats corner-touching pixels as connected. It is useful for irregular contours and diagonal structures, including broken yarns that cross the pixel grid diagonally. It often gives a more continuous representation of a small hole or thread defect.

The risk is over-merging. Two distinct diagonal defects close together may be labelled as one component, affecting defect count, bounding box and area. A local threshold that creates a diagonal chain of noise pixels can also connect a false region to a true defect.

The risk can be controlled by cleaning the mask before labelling, using size and shape filters after labelling, and separating merged components when geometry or distance criteria indicate multiple regions. Eight-connectivity is a strong practical default for defect candidate extraction, but it is not sufficient by itself.

Merging example. Consider two small defects, each three pixels wide, whose centres are at x = 10 and x = 16 pixels. A chain of diagonal noise pixels between them makes 8-connectivity label them as one component with an area of about 7–8 pixels and a centroid near x = 13. The centroid is three pixels from each true defect, i.e. about $3 \times 0.49 \approx 1.5$ mm, which is as large as the minimum defect itself. A marker aimed at the merged centroid could miss both defects, so merged components must be tested and split.

Control	How it helps
Opening with a small kernel before labelling	Breaks one-pixel noise bridges
Area and aspect-ratio limits after labelling	Flag components too large for one defect
Distance-transform or watershed split	Separates two blobs joined by a thin neck
Reporting each sub-region separately	Ensures the marker receives correct positions

Why 8-connectivity remains the default. The over-merging risk is controllable by post-labelling checks, whereas fragmentation of thin diagonal defects loses information that cannot be recovered. A false merge produces a larger mark or a mark between two defects, which can usually be reviewed; a lost defect is a missed defect. Because the missed-defect rate is the stricter requirement, the conservative choice is the one that keeps thin structures intact.

Testing the choice. The connectivity decision is easy to test: generate a set of labelled masks of diagonal threads, jagged holes and closely spaced spots, run the labelling with 4 and with 8 connectivity, and compare the number of components and areas with the known truth. A table of counts for each connectivity gives numerical evidence that the chosen method keeps threads in one piece without merging separate spots, and the table can be filed with the acceptance results.

Neighbourhood sketch (1 = foreground pixel):

```
4-neighbour:      N
                W P E
                  S

8-neighbour:    NW N NE
                 W P E
                SW S SE
```

OpenCV example: `n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)`

5. Mixed connectivity

Mixed connectivity defines diagonal connections carefully to avoid ambiguity. It is especially useful in digital topology, where the connectivity of foreground and background should not produce contradictory interpretations of a narrow passage. It can provide a compromise between 4- and 8-connectivity.

The cost is additional conceptual and implementation complexity. The rule for allowing a diagonal link must be implemented consistently, and engineers must understand why two pixels were joined or kept separate. A connectivity method that cannot be explained or reproduced is difficult to maintain in a production quality system.

For a factory implementation, mixed connectivity may be selected when testing reveals systematic overmerging under 8-connectivity. The choice should be driven by measured component statistics and defectlevel recall rather than by theoretical preference alone.

Aspect	8-connectivity	m-connectivity

Rule	Any of 8 neighbours	4-neighbours, diagonal only if both shared 4-neighbours are background
Redundant loops	Possible	Avoided
Implementation in OpenCV	Built in (connectivity = 8)	Not built in; needs custom code or post-processing
Explainability	Simple	Needs a clear written rule
Speed	Fast	Slower if implemented in plain loops

Practical note about libraries. OpenCV's connected-component function accepts connectivity 4 or 8 only. Mixed connectivity therefore needs additional code, for example a pass that removes redundant diagonal pixels (thinning) before labelling, or a custom union-find labeller. For a factory system running within a 0.5 s deadline, the extra code must be benchmarked, and it should be used only if the measured over-merging with 8-connectivity is a real problem.

When mixed connectivity is worthwhile. The method is most attractive when the fabric masks contain thin outlines that touch themselves diagonally, for instance the boundary of a small hole drawn with a onepixel line. m-connectivity produces cleaner boundary topology, so perimeter and hole-count measurements become more reliable. For compact blobs such as stains, the difference from 8-connectivity is usually negligible.

Summary of the three rules. In short: 4-connectivity separates, 8-connectivity joins, and m-connectivity joins without creating ambiguity. The textile application mostly needs joining because defects such as threads and hole outlines are thin and oblique, so 8-connectivity with later geometric checks is the practical choice, with m-connectivity kept as an option if boundary topology becomes important for hole counting or perimeter measurement.

Neighbourhood sketch (1 = foreground pixel):

```
4-neighbour:      N
                W P E
                  S

8-neighbour:    NW N NE
                 W P E
                SW S SE
```

OpenCV example: `n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)`

6. Effect on region measurements

Connected components produce a label image and region statistics such as area, bounding box and centroid. If one physical defect is split, its individual areas are too small and its count may be inflated. If two defects merge, the measured area and bounding box are too large and the marker may be placed between the true defect locations.

The defect coordinate must be transformed from pixels to millimetres using the calibrated pixel scale. If the scale is s millimetres per pixel, a centroid at (x,y) maps approximately to (x·s, y·s), subject to lens distortion and perspective correction. For a moving web, the along-web coordinate also needs the encoder position at acquisition.

Component size filtering must be based on the required minimum defect size. A 1.5 mm defect should not be removed by a minimum-area threshold chosen only to suppress noise. Use known-size calibration targets and real defect examples to choose thresholds.

Worked coordinate example. Let the pixel scale be s = 0.49 mm/pixel in both directions. A component has 12 pixels, so its area is $12 \times 0.49^2 \approx 2.9$ mm². Its centroid is at x = 1,224 px and y = 200 px in a frame whose first row corresponds to an encoder reading of 12,000 mm. The cross-web position is $1{,}224 \times 0.49 \approx 600$ mm from the left edge of the field, and the along-web position is $12{,}000 + 200 \times 0.49 = 12{,}098$ mm. Because the marker is 300 mm downstream, it must fire when the encoder reaches about 12,398 mm. Any labelling error that shifts the centroid by three pixels moves the target by about 1.5 mm.

Labelling error	Effect on measurement	Consequence
Defect split into pieces	Several small areas, high count	Small pieces filtered as noise or multiple marks
Two defects merged	One large area and mid-point centroid	Marker aimed between defects
Threshold too high	Region too small	Underestimated size and severity
Threshold too low	Region too large	Overestimated severity; wasteful rejection

Sub-pixel considerations. The centroid is computed as the mean of pixel coordinates inside the component and therefore has sub-pixel precision for a symmetric blob. Weighting each pixel by its contrast can improve the position of faint stains. The position error budget should include the calibration residual, the centroid error, the encoder resolution and the marker mechanical tolerance, and the sum should be compared with the marking tolerance agreed with the customer.

Size filters. The minimum-area threshold should be derived from physical size. For a 1.5 mm defect at 0.49 mm per pixel, a circular defect of diameter 1.5 mm has an area of $\pi \times 0.75^2 \approx 1.77$ mm², or about 7 pixels. A filter that removes every component below 10 pixels would discard it. A safer rule is to keep components above roughly half the expected area, here about 3–4 pixels, and to rely on contrast, shape and location tests to reject noise.

Neighbourhood sketch (1 = foreground pixel):

```
4-neighbour:      N
                W P E
                  S

8-neighbour:    NW N NE
                 W P E
                SW S SE
```

OpenCV example: n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)

7. Morphology and component separation

Morphological opening can remove isolated noise, while closing can fill small gaps in a mask. Both operations can change topology: opening may break a thin thread, and closing may join two nearby defects. Kernel dimensions should be selected in millimetres and converted to pixels after calibration, rather than chosen arbitrarily in pixels.

A good sequence is illumination correction, thresholding, small-noise suppression, connectivity labelling, then geometric validation. The detector should preserve a copy of the pre-morphology mask so that engineers can inspect whether a missing defect was erased during cleanup.

A class-specific strategy may use 8-connectivity for thread-like structures and a more conservative separation rule for compact spots. A unified system can extract with 8-connectivity and then apply orientation, gap and distance logic to split suspicious merged regions.

Kernel size in millimetres. The structuring element should be specified physically. At 0.49 mm per pixel, a 3 × 3 kernel spans about 1.5 mm, a 5 × 5 kernel about 2.5 mm. An opening with a 3 × 3 kernel removes every structure thinner than three pixels, so a one-pixel-wide thread break about 0.5 mm wide would be erased completely. This is the reason closing and opening kernels must be chosen from the smallest defect that must survive, not from a general rule of thumb.

Operation	Effect on mask	Danger for textile defects
Opening (erode then dilate)	Removes small isolated noise	Deletes thin threads and 1-pixel features
Closing (dilate then erode)	Fills small gaps and holes	Joins nearby defects
Dilation	Grows regions	Increases merged area
Erosion	Shrinks regions	Splits thin links and removes small defects
Skeletonisation	Reduces to a centre line	Useful for thread length measurement

Recommended procedure. Keep the unprocessed threshold mask, apply a minimal opening with the smallest kernel that removes isolated noise, label with 8-connectivity, apply geometric tests, and store both masks for audit. If a known test defect disappears after the opening step, the kernel is too large for this fabric. This test, repeated for every fabric recipe, avoids silent loss of small defects.

Quality control of the mask. The mask sequence should be tested with a set of synthetic defects of known size, such as 1.5 mm discs and 0.5 mm wide lines. After each step the software or the engineer counts how many remain. A step that removes any of these is adjusted or removed. Keeping this test with the software means that a later change of kernel or threshold is immediately checked against the smallest defect requirement.

Neighbourhood sketch (1 = foreground pixel):

4-neighbour: N
W P E

S

8-neighbour: NW N NE
W P E
SW S SE

OpenCV example: n, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)

TASK 5 – Image Acquisition Analysis

1. Rolling-shutter camera principle

A rolling-shutter sensor exposes or reads different image rows at different times. It can be inexpensive and provide high pixel counts, but moving fabric may shift between rows during acquisition. This can distort the apparent shape of a hole or make a straight thread look slanted. The severity depends on line timing, exposure, readout direction and fabric speed.

At 0.6 m/s, motion during exposure is $d = v \cdot t$. For an exposure of 1 ms, $d = 0.6$ mm; at 2 ms, $d = 1.2$ mm; at 5 ms, $d = 3$ mm. These distances are significant relative to a 1.5 mm defect. Short exposure reduces blur but requires sufficient illumination and may increase image noise.

Rolling shutter may be acceptable for stationary or slow-moving scenes, but for continuous textile web inspection it introduces a risk that is difficult to control under changing light. It should only be chosen after a motion test proves that distortion remains below the defect-scale tolerance.

Skew calculation. Suppose a rolling-shutter sensor with 2,048 rows reads one row every 20 µs (an assumed value). The last row is then read $2{,}048 \times 20\ \mu s \approx 41$ ms after the first. During this time the fabric moves $600\ mm/s \times 0.041\ s \approx 24.6$ mm. A vertical thread imaged across the frame would therefore be sheared by about 25 mm across the field, and a circular hole would be stretched into an oblique ellipse. Even if the effect is corrected geometrically, any fabric flutter or speed variation during readout remains uncorrected.

Rolling-shutter effect	Cause	Consequence for inspection
Shear of straight threads	Rows exposed at different times	False thread-break or weave-irregularity alarms
Shape distortion of holes	Motion during readout	Wrong size and aspect ratio
Position error along the web	Row time offset	Marker coordinate error of several millimetres
Flicker bands	LED or mains light modulation sampled row by row	False colour variation

Mitigation if a rolling shutter must be used. A global strobe that lights the scene only while all rows are exposed removes most skew, because the motion is frozen by the light pulse instead of by the sensor. The strobe pulse must be shorter than about 250 µs for 0.15 mm blur at 0.6 m/s, and all rows must be integrating during that pulse. This arrangement needs a sensor that supports a common exposure window, which should be confirmed in the data sheet before purchase.

Where rolling shutter is acceptable. A rolling shutter is acceptable for stationary inspection of cut samples, for slow-moving webs where motion during readout is a small fraction of the pixel size, and for operator viewing or documentation images. It is not suited to measuring width or position on a web moving at 0.6 m/s, because those measurements require the whole frame to represent one instant. Keeping it for non-critical views can reduce cost without risking the main inspection.

Exposure	Motion at 0.6 m/s	Assessment
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

2. Global-shutter camera principle

A global-shutter sensor exposes all pixels at approximately the same time. This avoids row-to-row temporal skew and is suitable for moving fabric. A short exposure can freeze small holes and thread discontinuities, especially when paired with a pulsed light source. Global shutter is generally a good choice for area images when the full field of view can be covered at sufficient resolution.

The sensor still has finite exposure time, so motion blur remains. At 0.6 m/s, a 100 µs exposure produces 0.06 mm of movement, while a 500 µs exposure produces 0.3 mm. The exposure should be short enough that blur is a small fraction of the 1.5 mm defect dimension. Bright, uniform illumination makes this practical.

Global shutter is easier to integrate than a line-scan system and can work with the available CPU. The main concern is whether the selected camera resolution and frame rate cover the required width and defect detail without exceeding storage or processing limits.

Frame coverage calculation. With a field height of 400 mm along the web and a web speed of 600 mm/s, one frame is valid for 400/600 = 0.667 s of travel. If consecutive frames must overlap by at least 10 % (40 mm) to avoid gaps and to permit stitching, the fabric may advance at most 360 mm between frames, so the frame period must be at most 360/600 = 0.6 s, i.e. a minimum frame rate of about 1.67 frames/s. For safety against dropped frames, a design rate of 3–5 frames/s gives larger overlap. These values are coverage minima; the processing rate must be at least as high.

Parameter	Typical requirement	Reason
Exposure time	≤ 250 µs (0.15 mm blur)	Freeze motion at 0.6 m/s
Trigger	Encoder or hardware trigger	Constant spatial spacing between frames
Frame overlap	≥ 10 % of field height	No unobserved strips
Interface	Gigabit Ethernet or USB3 / CoaXPress class	Sustained transfer of 5 MP frames
Lighting	Strobed LED, synchronised	Enough photons in a short exposure

Limitation of area imaging at fine pitch. A global-shutter area camera covering 1,200 mm with 4,096 pixels gives about 0.29 mm/pixel. This is adequate for the 1.5 mm defect in the present order, but it cannot scale to the finer sampling needed later in Task 10. For that reason an area camera is described as a good prototype or fallback, while the production recommendation remains a line-scan system.

Buffering and transfer. A 5 MP frame at 8 bits is 5.01 MB, and at 5 frames/s the data rate is about 25 MB/s, which is comfortable for a Gigabit Ethernet camera. If the plant selects 12 bits or a higher frame

rate, the interface must be rechecked: 12-bit data stored in 16-bit words doubles the rate. A sustained test of several hours, with a count of lost frames, is needed because short tests may hide occasional bus or driver problems.

Section summary	**Statement**
Given	0.6 m/s web; 400 mm field height.
Finding	Exposure up to 250 µs keeps blur at 0.15 mm; frame period up to 0.6 s gives coverage.
Decision	Global shutter is a valid fallback or prototype.

Exposure	**Motion at 0.6 m/s**	**Assessment**
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

TASK 5 – Image Acquisition Analysis (continued)

3. Line-scan camera principle

A line-scan camera captures one line of pixels at a time as the fabric moves beneath the camera. Successive lines are combined into a two-dimensional image using the web motion or an encoder. It is widely suited to continuous web inspection because it can provide a large cross-web pixel count and continuous coverage without repeatedly capturing a large area frame.

The line rate must match the web speed and desired along-web pixel pitch. If the target along-web pitch is 0.5 mm and speed is 600 mm/s, the required rate is 600/0.5 = 1,200 lines/s. For 0.25 mm pitch, it is 2,400 lines/s. Encoder synchronisation prevents scale distortion when conveyor speed varies.

Line-scan systems need careful mechanical alignment, stable web motion, line-rate configuration and more integration effort. However, for a wide, continuous fabric web, they offer strong geometric consistency and are a compelling industrial option.

Along-web pitch	Line rate at 0.6 m/s	Line rate at 1.2 m/s
1.0 mm	600 lines/s	1,200 lines/s
0.5 mm	1,200 lines/s	2,400 lines/s
0.25 mm	2,400 lines/s	4,800 lines/s
0.08 mm	7,500 lines/s	15,000 lines/s

Data-rate example. A line-scan camera with 4,096 pixels across the web at 8 bits and 2,400 lines/s produces 4,096 × 2,400 ≈ 9.8 MB/s, which is small compared with the 150 MB/s of the area-camera example. Even a 16,384-pixel camera at 2,400 lines/s needs about 39 MB/s. This is the efficiency advantage of line scan: every pixel is useful and no overlap or stitching is needed. The data rate rises proportionally with both pixel count and line rate, so Task 10's 15,000-pixel requirement at 15,000 lines/s must be checked carefully.

Synchronisation requirement. The encoder pulses define the line trigger. If the web speed changes by 10 % but the line rate stays fixed, the along-web pitch changes by 10 % and circular holes appear as ellipses, so measurement and marking positions are wrong. Using the encoder as the master clock keeps the pixel aspect ratio constant for any speed. Mechanical alignment is equally important: the line of pixels must be perpendicular to the direction of travel, otherwise straight yarns appear skewed.

Alignment procedure. To align a line-scan camera, an operator places a test chart with a printed square or circle on the web and views the assembled image while the web moves. If the square appears as a rectangle, the line rate or encoder scale is wrong; if it appears slanted, the camera is rotated relative to the web. Adjusting the camera mount and the encoder scaling until the chart appears undistorted yields a geometrically correct image without relying on software correction.

Section summary	Statement
Given	Continuous web at 0.6 m/s.
Finding	1,200–2,400 lines/s give 0.5–0.25 mm along-web pitch at about 10 MB/s.
Decision	Line scan with encoder trigger is the preferred architecture.

Exposure	Motion at 0.6 m/s	Assessment
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

4. Motion blur calculation

Motion blur in millimetres is approximately $b = v \cdot t$, where v is the fabric speed and t is exposure duration. With v = 0.6 m/s = 600 mm/s, a 0.1 ms exposure gives $b = 600 \times 0.0001 = 0.06$ mm. A 1 ms exposure gives 0.6 mm, and a 2.5 ms exposure gives 1.5 mm.

Because the smallest defect is 1.5 mm, an exposure that creates 1.5 mm blur could erase or smear the entire defect. A sensible design target is to keep motion blur well below one pixel and preferably below 10–20% of the minimum defect size. For example, a 0.15 mm blur target implies $t \leq 0.15/600 = 0.00025$ s = 250 µs.

This calculation assumes constant speed and translation along the web. Real performance also depends on sensor timing, lens sharpness, fabric vibration and illumination. Verify blur using a moving target rather than relying only on the nominal exposure setting.

Exposure	Blur at 0.6 m/s	Blur as fraction of 1.5 mm defect	Assessment
50 µs	0.03 mm	2 %	Excellent
100 µs	0.06 mm	4 %	Very good
250 µs	0.15 mm	10 %	Acceptable target
500 µs	0.30 mm	20 %	Marginal
1 ms	0.60 mm	40 %	Poor
2.5 ms	1.50 mm	100 %	Defect can be erased

Relationship to pixel size. At 0.49 mm/pixel (the medium resolution of Task 6), a 0.15 mm blur is about 0.3 pixel, so it is well below the pixel size. At 0.29 mm/pixel, 0.15 mm is about half a pixel, still acceptable. The goal is to keep blur below one pixel and below 10–20 % of the defect size, whichever is stricter. Contrast is reduced by blur in proportion to the fraction of the feature width smeared, so a 5–8 level stain on a narrow feature is affected more than a high-contrast hole.

Exposure and light level. Shortening the exposure from 2.5 ms to 250 µs reduces collected light tenfold. To keep the same signal, the lamp irradiance must rise tenfold or the lens aperture must open, with a tradeoff in depth of field. Strobed LED illumination provides high peak power without continuous heating. The image brightness must be verified at the shortest exposure, because noise at low signal can again hide the 5–8 level stain contrast.

Verification test. The calculated blur should be checked with a printed pattern of black and white bars of known width mounted on the moving web. The sharpness of the bar edges is measured at rest and at 0.6 m/s using the same exposure; the increase of the edge transition width is the real motion blur. Agreement

with the calculated 0.15 mm gives confidence in the exposure and trigger settings, and a larger value indicates timing jitter or vibration.

Section summary	Statement
Given	b = v × t with v = 600 mm/s.
Finding	250 µs gives 0.15 mm, 10 % of the defect; 2.5 ms erases it.
Decision	Specify exposure of 250 µs or less and verify with a moving target.

Exposure	Motion at 0.6 m/s	Assessment
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

5. Comparison of acquisition options

Rolling shutter: lower cost and easy area imaging, but row skew can occur. Global shutter: good for moving objects and easier area-camera integration, with exposure blur still requiring control. Line scan: best aligned with continuous web inspection and high cross-web resolution, but needs encoder synchronisation and more integration work.

Image quality should be compared using the same fabric sample, speed, lens, lighting and defect set. Measurements should include sharpness, defect visibility, geometric distortion, line coverage, dropped frames, latency and the number of false alarms. The selected camera should also be compatible with the PC interface and the storage bandwidth.

For this plant, a line-scan camera is preferred for the continuous web if it fits the budget and integration capacity. A global-shutter area camera is a reasonable fallback for a prototype or a more limited inspection zone. A rolling-shutter camera is not recommended as the primary sensor for the smallest moving defects.

Criterion	Rolling-shutter area	Global-shutter area	Line scan
Motion distortion	High risk	Low (blur only)	Low, if synchronised
Pixels across the web	Limited by sensor width	Limited by sensor width	Can be 8k–16k
Continuous coverage	Needs overlap and stitching	Needs overlap and stitching	Inherent
Lighting need	Strobe advisable	Strobe for short exposure	Bright line illumination
Integration effort	Low	Low to medium	Medium to high

Relative cost for the same coverage	Lowest	Medium	Medium to high
Fit for this web	Not recommended	Fallback or prototype	Preferred

Test method. A fair comparison mounts each camera at the same height, uses the same lens focal length scaled to the field, the same lamp and the same fabric samples bearing known defects, and records images at several web speeds. The metrics are the sharpness of a printed test chart, the apparent shape of a circular hole, the measured width error, the fraction of known defects visible and the number of dropped frames over a sustained run. This evidence, not the catalogue data alone, supports the final purchase decision.

Cost should be compared on a complete-system basis, including lens, lighting, frame grabber or interface card, encoder interface and software licence. A line-scan camera is more costly than an area camera, but the cost per useful pixel along a wide web can be lower because no pixels are wasted on overlap.

Decision summary. For the plant, the ranking by suitability is line scan first, global-shutter area camera second and rolling-shutter area camera third. The ranking is based on motion distortion, cross-web coverage and the ability to hold 1.5 mm defects at several pixels. The costs and integration effort run in the opposite order, which is why the final choice involves a trade-off that should be documented with the quotations received, not decided by technical preference alone.

Section summary	**Statement**
Given	Three sensor types compared under equal conditions.
Finding	Line scan is best aligned; rolling shutter is not suitable.
Decision	Choose line scan, with global-shutter area camera as fallback.

Exposure	**Motion at 0.6 m/s**	**Assessment**
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

6. Lighting and sensor synchronization

Controlled illumination is essential because ambient light varies sixfold from 500 to 3,000 lux. An enclosed hood and diffuse LED lighting reduce spatial gradients and changing daylight. A short strobe can supply enough photons during a brief exposure. For oil stains, diffuse bright-field lighting and crosspolarization can reduce glare; dark-field lighting may emphasize surface discontinuities.

For line scan, the encoder should determine line acquisition timing. For an area camera, the trigger should synchronize frame capture with web position. Exposure, strobe and camera trigger must be configured together. A frame captured without a valid trigger or with saturated highlights should be marked as unreliable, not classified as defect-free.

The lighting arrangement must be evaluated across fabric colours and finishes. A setup that works on white cotton may fail on dark polyester blends or glossy material. Store calibration references and check them periodically for dirt, LED ageing and mechanical shifts.

Lighting arrangement	Strength	Weakness	Best defect
Diffuse bright-field	Uniform brightness, low shadow	Low contrast for relief defects	Colour variation, stains
Cross-polarised	Reduces specular glare	Needs more light	Oil stains on glossy fibres
Dark-field / low angle	Emphasises raised fibres and holes	Strong shadows, uneven	Holes, thread breaks
Back-lighting	Strong hole contrast	Only for thin or open fabrics	Holes

Quantifying the light problem. Ambient illumination varies from 500 to 3,000 lux, a ratio of 6:1. With no control, a pixel that reads 150 grey levels at 500 lux would try to read about 900 at 3,000 lux and would saturate the 8-bit range at 255. If an enclosure limits the ambient contribution to 5 % of the controlled illumination at 3,000 lux, the ambient swing between 500 and 3,000 lux is only about 5 % × (1 − 1/6) ≈ 4 % of the controlled level, which flat-field correction can handle. The 5 % figure is a design target to be checked by measurement with a lux meter inside the enclosure.

Synchronisation faults. Typical faults are a strobe that fires before the exposure begins, a trigger that is missed under bus load, and an encoder that loses pulses. The software should monitor the number of pulses between frames, the frame brightness and the strobe-current feedback, and should mark the corresponding fabric as uninspected when any of them is outside limits.

Strobe timing example. For an area camera with a 250 µs exposure, the strobe must start at or slightly before the exposure window opens and end after it closes, for example a 300 µs pulse starting 25 µs early. A mismatch of even 100 µs can leave part of the exposure unlit, producing a darker frame. The timing is verified by capturing frames while changing the delay by small steps and plotting the mean brightness against delay, choosing the middle of the flat region.

Section summary	Statement
Given	500–3,000 lux ambient range.
Finding	An enclosure with constant-current LED lighting limits drift to a few per cent.
Decision	Trigger camera and light from the encoder; flag bad frames.

Exposure	Motion at 0.6 m/s	Assessment
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

7. Marker timing and position

The marker is 0.3 m downstream. At 0.6 m/s, the nominal travel time is 0.3/0.6 = 0.5 s. This exactly equals the stated maximum decision time, leaving no margin if the system waits until the deadline to process an image. Therefore the actual design should finish earlier and schedule the mark from encoder distance.

If the defect is detected at along-web position y and the marker is at distance D, the target position is y + D along the direction of travel. The controller should track cumulative encoder pulses and activate the marker when the measured travel distance reaches D. A fixed timer is vulnerable to speed changes, slip and pauses.

The application should test the complete chain from image exposure to physical mark. Report median, 95th percentile and worst observed latency, plus marking-position error in millimetres. If a decision misses its deadline, the system should raise a fault or use a defined safe response.

Contribution to delay	Planning value	Notes
Exposure and readout	≤ 50 ms	Interface dependent
Processing	≤ 250 ms	Includes worst-case frames
Communication to controller	≤20ms	Fieldbus or Ethernet; measure
Actuator response (valve or pen)	≤30ms	Check data sheet
Total	≤ 350 ms	Leaves 150 ms margin within 0.5 s

Position error from latency. Any delay that is not compensated moves the mark downstream. A 30 ms uncompensated actuator delay at 0.6 m/s causes 600 × 0.03 = 18 mm of error. Using the encoder, the controller can subtract the known actuator delay by firing the command 18 mm early, which is more accurate than a fixed time offset. The remaining error is the variation of the delay, called jitter; a jitter of ±5 ms corresponds to ±3 mm, which is a useful bound to compare with the marking tolerance agreed with the customer.

Acceptance test. Place calibrated defects of known position on a test fabric, run the line at normal and at slightly varied speeds, and measure the distance between each defect and its mark with a ruler. Report the mean, standard deviation and worst error, together with the number of missed or duplicate marks. Compare these with the tolerance (for example ±10 mm) and repeat after any recalibration or mechanical change.

Safe default behaviour. If the system is unsure whether a defect is real, a possible policy is to mark it as “check” using a different colour or pattern, so that the rework station can examine it. This preserves the missed-defect rate while limiting the effect of false alarms. The policy must be agreed with the customer in advance because a marking that is too aggressive wastes fabric and one that is too lenient violates the 2 % limit.

Section summary	Statement
Given	0.3 m marker offset; 0.5 s travel.
Finding	Fixed timers fail when speed changes; the encoder follows true travel.
Decision	Fire from encoder distance and subtract actuator delay.

Exposure	Motion at 0.6 m/s	Assessment
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

8. Recommended acquisition design

The recommended acquisition is an industrial line-scan camera, encoder-synchronised, with controlled diffuse LED lighting and an enclosure. Select the cross-web pixel count to cover 1,200 mm with enough pixels for a 1.5 mm defect, and choose the line rate based on the required along-web sampling pitch at 0.6 m/s. Use a calibrated lens and flat-field correction.

If cost or integration constraints make line scan impractical, use a global-shutter area camera with short exposure, high-intensity strobe and a field of view that does not sacrifice the required defect sampling. Avoid using a low-cost rolling-shutter sensor without measured evidence that motion distortion is acceptable.

Final recommendation: line scan is the most suitable production architecture for continuous fabric, but final purchase depends on vendor specifications and a bench test. Confirm resolution, frame/line rate, interface bandwidth, lens coverage, lighting and CPU processing time before procurement.

Element	Recommended specification	Verification
Sensor	Line scan with ≥ 4,096 pixels across the 1,200 mm view	Pixel pitch and defect span measured on a chart
Line rate	≥ 1,200 lines/s for 0.5 mm; up to 2,400 lines/s for 0.25 mm pitch	Encoder pulse count versus lines
Exposure	≤ 250 µs equivalent	Blur test with moving target
Lighting	Diffuse LED line light inside an enclosure	Lux and uniformity map
Calibration	Flat-field and grid calibration	Residual error report
Interface	Frame grabber or high-speed link, tested for sustained throughput	Dropped-frame count over a long run

Procurement checklist. Before purchase, request from each vendor the pixel size, the number of pixels, the maximum line rate, the bit depth, the interface bandwidth, the lens mount, and the sensitivity at short exposure. Ask for sample images of a textile with a comparable defect. A vendor who cannot provide such evidence is a risk for the 5–8 level stain requirement, which depends directly on sensor noise and sensitivity.

Fall-back plan. If the line-scan system exceeds the budget of □1,50,000, the prototype configuration is a global-shutter area camera with a 4,096-pixel-wide view, strobed LED lighting and the same encoder trigger. This fallback must be clearly labelled as meeting the 1.5 mm requirement with less margin and it must pass the same acceptance tests before production use.

Documentation for procurement. The specification sent to suppliers should state the field width, working distance, minimum defect size, maximum web speed, required pixel pitch, bit depth, line rate, interface, trigger source, lighting requirement and the acceptance tests, including the moving-target blur test. Writing the requirements in measurable terms allows each supplier's offer to be compared directly and later checked at delivery against the same numbers.

Section summary	**Statement**
Given	Budget □1,50,000; 1.5 mm defects; 1,200 mm field.
Finding	Line scan with 4,096 pixels or more meets sampling at manageable data rate.
Decision	Confirm vendor data and bench tests before purchase.

Exposure	**Motion at 0.6 m/s**	**Assessment**
0.1 ms	0.06 mm	Good motion-freezing candidate
1 ms	0.6 mm	Large relative to 1.5 mm defect
2.5 ms	1.5 mm	May smear the full defect

Formula: motion blur (mm) = fabric speed (mm/s) × exposure time (s).

TASK 6 – Sampling Analysis

1. Pixel counts and spatial scale

The three candidate resolutions are 1280×1024, 2448×2048 and 4096×3000. Their pixel counts are 1,310,720; 5,013,504; and 12,288,000 respectively. If image width spans the full 1,200 mm field of view, the cross-web sampling interval is FOV width divided by image width.

Low resolution: 1200/1280 = 0.9375 mm/pixel. Medium resolution: 1200/2448 ≈ 0.4902 mm/pixel. High resolution: 1200/4096 ≈ 0.2930 mm/pixel. A 1.5 mm defect spans approximately 1.6, 3.1 and 5.1 pixels across the image width, respectively.

A feature represented by fewer than two pixels is fragile, and reliable classification commonly needs several pixels across the feature. The low-resolution option is inadequate for dependable detection of the 1.5 mm defect. Medium resolution is a practical minimum candidate, while high resolution provides more margin but increases processing and data volume.

Option	Pixels (W × H)	Total pixels	Pitch across 1,200 mm	Pixels across 1.5 mm
Low	1280 × 1024	1,310,720	0.9375 mm	1.60
Medium	2448 × 2048	5,013,504	0.4902 mm	3.06
High	4096 × 3000	12,288,000	0.2930 mm	5.12

Along-web pitch in the area-camera case. If the same cameras view 400 mm along the web, the vertical pitch is 400/1024 = 0.39 mm for the low option, 400/2048 = 0.195 mm for the medium option and 400/3000 = 0.133 mm for the high option. The medium and high options are therefore finer along the web than across it. For a defect at 1.5 mm, the weaker direction is the cross-web direction, so it controls the design. This calculation shows that the quoted pixel counts should always be examined separately along each axis.

Reading the table. A feature spanning fewer than two pixels (the low option) cannot be reliably separated from noise. Three pixels (medium) allow detection of a high-contrast hole but give limited boundary information. Five pixels (high) gives enough pixels for a stable measurement of size and shape. The recommended minimum for the 1.5 mm defect is therefore the medium option, with the high option preferred if the processing budget allows.

Common mistake. A frequent mistake is to divide the sensor width in pixels by the field of view in the wrong direction or to use the fabric width (1,000 mm) instead of the field width (1,200 mm). The scale applies to the object plane seen by the camera, so the 1,200 mm figure is correct for the sampling pitch, while the 1,000 mm fabric width is used when converting the measured edge positions into a fabric width. Keeping units and planes explicit prevents a 20 % error.

Section summary	Statement	
Given	Three candidate resolutions across 1,200 mm.	
Finding	Pitch 0.94, 0.49 and 0.29 mm; defect spans 1.6, 3.1 and 5.1 pixels.	
Decision	Reject low resolution; medium is the practical minimum.	
Parameter	**Calculation / observation**	**Design implication**

Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

2. Nyquist and practical defect detection

Nyquist sampling theory says that a periodic signal requires at least two samples per cycle, but defect detection is not solved merely by satisfying Nyquist. A small irregular hole or broken thread needs enough pixels to preserve its shape, contrast and boundary. Five pixels across a defect is a useful engineering target for initial design, not a universal guarantee.

At 0.4902 mm/pixel, a 1.5 mm defect occupies about 3.06 pixels across the width. This may detect a highcontrast defect but is less robust for a low-contrast oil stain or a thread discontinuity. At 0.2930 mm/pixel, it occupies about 5.12 pixels, improving shape and boundary representation.

The final choice must consider both cross-web and along-web sampling. With line scan, line pitch is controlled by line rate and conveyor speed; with area imaging, vertical pixel pitch depends on the fieldof-view height. A system is only as good as its coarser sampling direction.

Applying the Nyquist criterion. A periodic pattern with spatial period p is only represented if the sampling interval $\Delta \leq p/2$. For a hole of diameter 1.5 mm, regarded as half a period, the equivalent period is about 3 mm, so Nyquist asks for $\Delta \leq 1.5$ mm. All three options pass that minimum, which shows that Nyquist is a necessary but not a sufficient criterion for defect inspection: it ensures the presence of a signal but not its shape, contrast or position accuracy.

Pitch	**Pixels across 1.5 mm**	**Expected capability (design expectation)**
0.9375 mm	1.6	Marginal; detection of high-contrast holes only, unreliable otherwise
0.4902 mm	3.1	Detects holes and clear thread breaks; stains less reliable
0.2930 mm	5.1	Better edges and shape; faint stains still depend on contrast and noise

Sampling and the point-spread function. The lens and sensor blur every point into a small spot. If this spot is larger than the pixel, adding more pixels does not add detail. For a design with 0.29 mm pixels, the lens should resolve a line pair of about 0.6 mm (two pixels), which can be checked with a resolution test chart across the full field. Corners usually resolve worse than the centre, so the test must include the edges where the fabric border and width measurement take place.

Safety factor. Engineering practice for machine-vision inspection often asks for three to five pixels across the smallest defect and at least two pixels across the narrowest feature that must be resolved, such as a

thin thread. These factors cover the blur of the lens, the unknown alignment between the defect and the pixel grid, and the noise. The factor chosen should be confirmed by a test in which defects are moved by subpixel steps and detection is repeated.

Section summary	Statement
Given	Nyquist needs two samples per period.
Finding	Defect inspection needs about three to five pixels across the defect.
Decision	Aim for 0.29–0.49 mm pitch and verify by sub-pixel shift tests.

Parameter	Calculation / observation	Design implication
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

3. Data and storage calculations

An 8-bit grayscale image requires one byte per pixel; an 8-bit RGB image requires three bytes per pixel. For low resolution, grayscale is 1.31 MB/frame and RGB is 3.93 MB/frame. For medium resolution, grayscale is 5.01 MB/frame and RGB is 15.04 MB/frame. For high resolution, grayscale is 12.29 MB/frame and RGB is 36.86 MB/frame, using decimal MB.

At 10 frames/s, medium RGB data is about 150 MB/s before compression and protocol overhead. Continuous storage for seven days would be extremely large if every full frame were stored uncompressed at this rate. The retention policy therefore needs a defined acquisition rate, lossless or suitable compression, event clips, and a storage-capacity calculation based on actual throughput.

The 2 TB disk is a hard resource constraint. Store a full-resolution image or lossless crop for every relevant inspection event and retain compact metadata for all frames. If policy requires every fruit or fabric segment image to be kept for seven days, verify that the chosen frame rate and compression meet the requirement before deployment.

Option	Grayscale frame	RGB frame	RGB at 10 frames/s	Per day (24 h)
Low (1,310,720 px)	1.31 MB	3.93 MB	39.3 MB/s	3.4 TB
Medium (5,013,504 px)	5.01 MB	15.04 MB	150 MB/s	13.0 TB
High (12,288,000 px)	12.29 MB	36.86 MB	369 MB/s	31.8 TB

Compare with the 2 TB disk. A 2 TB disk (2,000,000 MB) holds about 2,000,000 / 15.04 ≈ 133,000 uncompressed medium-resolution RGB frames, i.e. less than four hours at 10 frames/s. A seven-day retention of continuous raw RGB images is therefore impossible. If lossless compression gives a ratio of

2:1 (an assumption that varies with the fabric texture), the time is still under eight hours. The policy must therefore store only defect events and a thin sample of normal frames, or reduce the rate.

Example of a feasible retention plan. Suppose one defect is detected every 20 s on average and a fullresolution grayscale crop of 512 × 512 pixels (0.26 MB) is stored for each, with a 5.01 MB overview every 30 s. Per day this is 4,320 crops × 0.26 MB ≈ 1.1 GB and 2,880 overviews × 5.01 MB ≈ 14.4 GB, about 15.5 GB per day, or roughly 109 GB for seven days, which fits easily inside 2 TB. The assumed event rate is illustrative; the real rate depends on fabric quality, and a margin of at least a factor of two should be kept.

Units. Data sizes in this report use decimal megabytes (1 MB = 10^6 bytes) because camera and disk manufacturers do the same. Operating systems often display binary units (MiB, 2^{20} bytes), and 5.01 MB is about 4.78 MiB. Mixing the two causes an error of about 5 % per prefix level, which matters when checking whether seven days of data fit on a 2 TB disk, where the usable capacity is also smaller than the label value.

Parameter	Calculation / observation	Design implication
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

4. Resolution comparison

Low resolution has the lowest bandwidth and processing cost but represents the smallest defect with only about 1.6 pixels. Medium resolution provides about 3.1 pixels per defect and is a reasonable compromise for a CPU-based prototype. High resolution provides about 5.1 pixels per defect and better boundary detail, but its pixel count is roughly 2.45 times the medium image and can increase processing latency.

Processing time does not always scale exactly with pixel count, because algorithms differ and can use region-of-interest or downsampled views. However, higher resolution generally increases memory traffic and the cost of filtering, thresholding, feature extraction and model inference. Measure all candidates on the actual 4-core PC.

The decision should be based on defect-level recall, false alarms, width accuracy, latency and storage. A resolution that looks sharp to a person is not automatically sufficient for reliable automated inspection.

Criterion	Low	Medium	High
Pixels across 1.5 mm	1.6	3.1	5.1
Relative pixel count	1.0×	3.8×	9.4×
Relative RGB data per frame	1.0×	3.8×	9.4×
Width error for a 2-pixel boundary error	1.9 mm	0.98 mm	0.59 mm

CPU load	Lowest	Moderate	Highest
Overall suitability	Reject	Practical minimum	Preferred if affordable

How the ratios are obtained. The medium option has 5,013,504 / 1,310,720 = 3.83 times the pixels of the low option, and the high option has 12,288,000 / 1,310,720 = 9.38 times. The high option has 12,288,000 / 5,013,504 = 2.45 times the medium. These ratios give the first-order scaling of memory traffic and of simple per-pixel operations. Algorithms with a fixed number of steps per region scale less steeply, which is why region-based processing is attractive.

Measure rather than guess. Run the actual preprocessing and detection code on recorded images at the three resolutions, using the same 4-core CPU. Record the mean and worst-case time per frame, and the sensitivity for each defect class. If the high option achieves 2 % lower missed defects with an extra 0.1 s of processing and still meets the latency budget, it is justified; if not, the medium option is more sensible. The numbers decide, not the pixel count.

Practical choice. Given the CPU limit and the need for 1.5 mm detection, the medium option is the minimum viable choice and the high option the preferred one if its latency can be tolerated. A trial that records the same fabric with both and runs the same detector would show whether the extra pixels reduce misses on the thread breaks and faint stains, which are the classes most sensitive to resolution.

Parameter	Calculation / observation	Design implication
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

5. Resolution versus width accuracy

Width measurement within ±2 mm requires calibration and stable edge detection. If the width spans 1,000 mm and the cross-web scale is 0.4902 mm/pixel, a two-pixel error corresponds to about 0.98 mm, which is below the ±2 mm limit. However, perspective distortion, lens distortion and uncertain edge positions can add further error.

At 0.9375 mm/pixel, two pixels correspond to 1.875 mm, leaving little margin for optical and calibration errors. At 0.2930 mm/pixel, two pixels correspond to 0.586 mm, giving more room for other error sources. The width accuracy should be verified using a traceable reference ruler or calibrated fabric-width target.

Width accuracy and defect detection are related but distinct requirements. A system can measure overall width correctly while missing a small low-contrast stain. Both requirements must be validated separately.

Resolution	Pitch	2-pixel edge error	Share of ±2 mm budget
Low	0.9375 mm	1.88 mm	94 %

Medium	0.4902 mm	0.98 mm	49 %
High	0.2930 mm	0.59 mm	29 %

Error budget. Width is measured from two edges, so the independent errors of the two edges combine. Take the medium option with a 1-pixel (0.49 mm) uncertainty at each edge: the root-sum-square is $\sqrt{(0.49^2 + 0.49^2)} \approx 0.69$ mm. Add a calibration residual of 0.5 mm and a lens-distortion residual of 0.5 mm: $\sqrt{(0.69^2 + 0.5^2 + 0.5^2)} \approx 0.99$ mm, still within ±2 mm. At the low option, the same method gives $\sqrt{(0.94^2 + 0.94^2 + 0.5^2 + 0.5^2)} \approx 1.5$ mm, leaving little margin for fabric flutter. These are budget estimates and must be confirmed on a calibrated target.

Sub-pixel edge detection. The edge of the fabric can be located to a fraction of a pixel by fitting a profile to the intensity transition, which reduces the effective error below one pixel. However, sub-pixel accuracy needs a sharp, stable edge and a contrasting background, such as a dark belt under light fabric. If the web edge is curled, frayed or fluttering, the apparent edge changes with time and no resolution can fix this.

Practical edge detection. A reliable width measurement depends on contrast between the fabric edge and the background. A matt dark background under light fabric gives a strong edge, while white fabric on a light belt does not. The background must therefore be part of the design, and its colour should be tested with the darkest and lightest fabric in the product range. Edge position is estimated as the maximum of the intensity gradient along several rows, and the median over rows rejects local fraying.

Section summary	**Statement**
Given	Width tolerance ±2 mm.
Finding	Two-pixel edge error is 1.88, 0.98 and 0.59 mm for the three options.
Decision	Use medium or high resolution with calibration and sub-pixel edges.

Parameter	**Calculation / observation**	**Design implication**
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

6. Area-camera frame rate and line-scan rate

For area imaging, if the along-web field of view is 400 mm and the fabric moves at 600 mm/s, the view covers about 400/600 = 0.667 s of fabric travel. To avoid gaps, frame overlap and trigger strategy must be considered; the minimum frame rate for contiguous coverage depends on the actual field orientation and acquisition scheme.

For line scan, line rate is determined by desired along-web pixel pitch. At 0.6 m/s, 0.5 mm pitch requires 1,200 lines/s; 0.25 mm pitch requires 2,400 lines/s. These are moderate rates for industrial line-scan cameras, but the pixel width and bit depth determine total data throughput.

Choose the sampling architecture before finalizing the camera. An area camera may be simpler for a demonstration, while line scan can make more efficient use of pixels for continuous web inspection. In either case, test dropped frames, encoder mismatch and data transfer under sustained operation.

Case	Parameter	Calculation	Result
Area camera	Frame period for no gap with 10 % overlap	(0.9 × 400 mm) / 600 mm/s	0.6 s (1.67 frames/s)
Area camera	Frame period for 50 % overlap	(0.5 × 400 mm) / 600 mm/s	0.33 s (3 frames/s)
Line scan	Line rate for 0.5 mm pitch	600 / 0.5	1,200 lines/s
Line scan	Line rate for 0.25 mm pitch	600 / 0.25	2,400 lines/s
Line scan	Data rate, 4,096 px × 2,400 lines/s × 1 byte	4,096 × 2,400	9.8 MB/s

Matching along-web and cross-web pitch. For isotropic sampling, the along-web pitch should match the cross-web pitch. With 4,096 pixels across 1,200 mm (0.293 mm), the line rate should be 600 / 0.293 ≈ 2,048 lines/s. With 2,448 pixels (0.49 mm), 600 / 0.49 ≈ 1,224 lines/s is sufficient. Over-sampling along the web wastes data, while under-sampling blurs defects along the direction of motion.

Hidden costs. Higher line rates reduce the exposure available for each line. A line period of 1/2,400 s is 417 µs, so the exposure cannot exceed this and in practice is shorter. The lighting must provide enough photons within that time, and the resulting noise has to be checked against the 5–8 level stain contrast.

Choice of architecture. The calculation shows that both architectures can meet the sampling needs of the present order. The line-scan system provides continuous coverage and a modest data rate, whereas the area camera uses off-the-shelf components and simple optics but produces larger data volumes and needs overlap handling. The decision therefore depends on budget, local support and integration skills, as well as on the technical figures.

Parameter	Calculation / observation	Design implication
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin
Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

7. Recommendation under CPU limits

The balanced recommendation is medium-to-high sampling with a line-scan or global-shutter sensor, but the processing design should avoid sending every full-resolution pixel through expensive algorithms. Use fast preprocessing, candidate-region extraction and a compact classifier or class-specific rules. Preserve high-resolution crops around candidate defects for audit.

A useful two-stage strategy is to scan at sufficient resolution, then perform detailed analysis only on candidate regions. This can reduce CPU load while retaining defect detail. The candidate detector must be sensitive enough not to discard weak oil stains before the detailed stage.

Final choice: do not use 1280-pixel width as the production resolution for a 1.5 mm minimum defect. Medium resolution is the lowest practical candidate; high resolution offers a stronger margin if the data rate and latency are manageable. The final selection requires measured performance on the actual hardware.

Stage	Processing region	Resolution	Purpose
Fast screening	Whole frame or line stream	Full, with simple operations	Find candidate regions
Detailed analysis	Candidate crops only	Full, with feature extraction	Classify and measure
Audit storage	Defect crops and thumbnails	Full for crops	Evidence for 7 days

Why two stages help. Suppose screening uses an operation costing 10 ns per pixel on the full 5.01 MP frame, i.e. about 50 ms. If 20 candidate regions of 128 × 128 pixels (about 0.33 MP in total, 6.5 % of the frame) are passed to a more expensive stage costing 200 ns per pixel, that stage needs about 65 ms. The same expensive stage over the full frame would need about one second, which would break the deadline. These costs are illustrative; they show why the work should be concentrated where it is needed.

Safeguard against missed candidates. The screening threshold should be set conservatively, so that faint stains are still passed on, and then verified with a test in which every labelled defect must appear in the candidate list before the detailed stage is applied. Candidate recall is measured separately from final classification, and the overall missed-defect rate is the product of the two stages' miss rates. Therefore a 1 % miss at screening and a 1 % miss at classification already approach the 2 % limit.

Section summary	Statement
Given	4-core CPU, 0.5 s deadline.
Finding	Two-stage processing keeps the expensive work on a small part of the frame.
Decision	Medium-to-high sampling with candidate-region analysis.

Parameter	Calculation / observation	Design implication
Low resolution	1200/1280 = 0.9375 mm/px; 1.5 mm ≈ 1.6 px	Insufficient margin

Medium resolution	1200/2448 ≈ 0.4902 mm/px; 1.5 mm ≈ 3.1 px	Practical minimum candidate
High resolution	1200/4096 ≈ 0.2930 mm/px; 1.5 mm ≈ 5.1 px	Better defect detail; higher data load

Formula: spatial pitch (mm/pixel) = field of view (mm) ÷ pixel count across the field.

TASK 7 – Quantization Analysis

1. Quantization levels

An n-bit quantizer represents 2^n discrete intensity levels. Eight-bit data has 256 levels, twelve-bit data has 4,096 levels, four-bit data has 16 levels, five-bit data has 32 levels, and six-bit data has 64 levels. More levels allow smaller intensity differences to be represented, but only if the sensor signal-to-noise ratio and illumination stability support them.

The stated oil-stain contrast is only 5–8 grey levels in an 8-bit representation. Reducing bit depth can erase or weaken this difference. If 8-bit values are reduced to 4 bits by simple right shift, each reduced level covers 16 original levels; a 5–8 level difference can map to the same quantized value.

Quantization should therefore be selected based on low-contrast defect visibility, not only on storage. Higher bit depth does not fix blur, poor focus or uneven lighting, but it can preserve subtle intensity differences that would otherwise be lost.

Bits	Levels	Step on 8-bit scale	RMS quantization noise (step/√12)	Ideal SNR (6.02n + 1.76 dB)
4	16	16	4.6 levels	25.8 dB
5	32	8	2.3 levels	31.9 dB
6	64	4	1.15 levels	37.9 dB
8	256	1	0.29 level	49.9 dB
12	4,096	1/16	0.018 level	74.0 dB

Survival of a small contrast. If a stain differs from the fabric by d grey levels (8-bit scale) and the quantizer step is q > d, the two values fall in different bins only when a bin boundary lies between them, which happens with probability about d/q for a random alignment. For d = 5–8 and q = 16 (4-bit), the probability is 31–50 %, so roughly half of the stains vanish completely. For q = 8 (5-bit), it is 63–100 %, and for q = 4 (6-bit) the difference always survives as at least one level. These probabilities show that bit depth is a first-order variable for faint stains.

Quantization noise compared with the signal. At 4 bits the rms quantization noise, about 4.6 levels, is nearly as large as the 5–8 level signal, so the contrast-to-noise ratio cannot exceed about 1–2. At 8 bits the quantization noise is only 0.29 level and is normally smaller than the sensor's own noise. This explains why 8 bits is usually sufficient if the sensor and lighting are good, while 4 bits is not.

Summary rule. Each added bit halves the step size and, in an ideal quantizer, improves the signal-tonoise ratio by about 6 dB. For faint stains the key quantity is the step size relative to the stain contrast: the step should be well below the contrast, for example below half of it, so the stain is represented by at least two levels. For a 5-level stain this means a step of at most 2–3 levels, which is why 6 bits is a marginal minimum and 8 bits is preferred.

Section summary	Statement
Given	n bits give 2^n levels.

Finding	A 4-bit step of 16 is larger than the 5–8 level stain contrast.
Decision	Never use 4-bit data for the primary detector.

Parameter	**Calculation / observation**	**Design implication**
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

2. Eight-bit quantization

Eight-bit quantization is widely supported and computationally efficient. It uses 256 levels and one byte per pixel, which keeps memory bandwidth low and is convenient for OpenCV processing. Many standard image operations, file formats and pretrained models expect 8-bit inputs.

Its weakness is limited dynamic range when the defect contrast is only a few levels. If sensor noise, lighting variation or compression artefacts are comparable to the 5–8 level stain contrast, the defect can be hard to separate from normal fabric texture. Careful exposure and local normalization are required.

Eight-bit is an acceptable processing format when the camera provides stable illumination and the sensor has adequate signal quality. If the camera supports 12-bit acquisition, it may be useful to retain the raw high-bit-depth image and create an 8-bit normalized image for the CPU detector.

Aspect	8-bit data	Consequence
Levels	256	Step of 1; quantization noise 0.29 level
Memory per pixel	1 byte	5.01 MB per 2448 × 2048 frame
Compatibility	Native for OpenCV and common models	No conversion needed
Dynamic range	About 48 dB	Bright reflections may clip while dark areas are noisy
File format	PNG or lossless TIFF	Avoid JPEG for low-contrast defects

Is 8-bit really the limit? Suppose the sensor's own noise is 2 grey levels rms. Adding quantization noise of 0.29 level gives $\sqrt{(2^2 + 0.29^2)} = 2.02$ levels, a negligible change. In that case moving to 12 bits would not improve the contrast-to-noise ratio of a 6-level stain, because the sensor noise, not the quantizer, limits it. 12 bits becomes useful only when the sensor noise is below about one 8-bit level, or when the scene has a wide dynamic range that must be mapped without clipping.

Compression warning. Lossy image compression is a second quantization stage. A JPEG quantizes frequency coefficients and can flatten a 5–8 level stain, particularly at low quality settings, while introducing block artefacts that look like texture changes. The detector should therefore operate on uncompressed data from the camera, and archives of defect images should use lossless compression unless tests prove that detection results do not change.

Gain and offset. With 8-bit data, using the full range matters. If the fabric occupies only grey levels 100–140, 60 % of the range is unused and the stain contrast stays small relative to the step. Adjusting the exposure or lamp so that the fabric sits at around mid-to-upper range (but below saturation) maximises the use of levels. A histogram of the clean fabric should be a regular check in commissioning, with limits for mean and for the fraction of saturated pixels.

Section summary	Statement
Given	256 levels, 1 byte per pixel.
Finding	Quantization noise 0.29 level is usually below sensor noise.

TASK 7 – Quantization Analysis (continued)

Decision	Use 8 bits when sensor noise and lighting are good; avoid JPEG.

Parameter	**Calculation / observation**	**Design implication**
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

3. Twelve-bit quantization

Twelve-bit acquisition offers 4,096 levels and can preserve subtle intensity differences over a wider dynamic range. This is useful when fabric areas contain both bright reflections and faint oil stains, provided that the sensor noise is low enough to exploit the extra levels. Industrial cameras may deliver 12-bit samples in 16-bit containers.

The storage and memory cost can be about twice that of 8-bit single-channel data when stored in 16-bit words. Some OpenCV operations require conversion or scaling, and poorly chosen normalization can compress useful contrast. Twelve-bit acquisition should not be treated as an automatic accuracy improvement.

The recommended strategy is to acquire at 10–12 bits if the selected sensor supports it, then perform a calibrated conversion to the processing range. Keep raw frames for diagnostic samples and use an 8-bit normalized view for operations that require 8-bit input.

Storage option	Bytes per pixel	Frame size (2448 × 2048)	Remarks
12-bit packed	1.5	7.52 MB	Compact; needs unpacking
12-bit in 16-bit word	2	10.03 MB	Simple and common
8-bit normalised copy	1	5.01 MB	For OpenCV routines expecting 8 bits
32-bit float feature map	4	20.05 MB	Only for selected intermediate maps

Mapping 12 bits to 8 bits. A simple right shift by four bits discards the low bits and is equivalent to 8-bit acquisition. A better approach is a windowed linear stretch: choose the interval of raw values that contains the fabric (for instance 1,800–3,000 counts), map it to 0–255 and clip the rest. The window narrows the range, so each 8-bit level then represents about 4.7 raw counts instead of 16, preserving more of the stain contrast. The window limits must be taken from the flat-field reference, not set arbitrarily.

Cost of 12-bit data. Twice the memory traffic and, because a 128-bit vector register holds 8 sixteen-bit values instead of 16 eight-bit values, about half the number of pixels processed per instruction for simple operations. The consequences for the 0.5 s limit are small for a region-based pipeline but can be significant for whole-frame filtering. Benchmark both options on the target CPU before deciding.

When to adopt it. Twelve-bit acquisition should be adopted if the chosen camera offers it without a large cost or bandwidth penalty, if the sensor noise is below one 8-bit level, and if tests show that faint stains are detected better with the higher depth. Otherwise it adds storage and computation without benefit. A simple comparison, imaging the same 5–8 level stains at 8 and 12 bits, shows within a day whether the extra depth is useful for the mill’s fabrics.

Section summary	Statement
Given	4,096 levels, usually in 16-bit words.
Finding	Benefit appears only when sensor noise is below one 8-bit level.

TASK 7 – Quantization Analysis (continued)

Decision	Acquire at 10–12 bits if available; use a windowed 8-bit copy.

Parameter	**Calculation / observation**	**Design implication**
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

4. Reduced 4–6 bit quantization

Reduced quantization lowers memory and may speed simple operations, but it removes subtle levels. Four-bit data has only 16 levels, five-bit has 32, and six-bit has 64. When the stain difference is 5–8 levels on an 8-bit scale, reduction to 4 bits can collapse the stain and background into the same value.

Six-bit quantization may retain some differences after careful mapping, but its behaviour depends on the quantization step and where the values fall relative to bin boundaries. A contrast difference that is visible in one part of the range can disappear in another. This is especially risky for a production detector that must work across multiple fabric colours and light conditions.

TASK 7 – Quantization Analysis (continued)

Reduced bit depth may be suitable for preview images or archival thumbnails, not as the primary input for low-contrast defect detection. Any reduction should be validated with paired images and reported with per-class recall.

Pair of 8-bit values	4-bit (value >> 4)	5-bit (value >> 3)	6-bit (value >> 2)
150 and 144 (6 apart)	9 and 9 – lost	18 and 18 – lost	37 and 36 – kept, 1 level
150 and 143 (7 apart)	9 and 8 – kept	18 and 17 – kept	37 and 35 – kept
152 and 146 (6 apart)	9 and 9 – lost	19 and 18 – kept	38 and 36 – kept
130 and 123 (7 apart)	8 and 7 – kept	16 and 15 – kept	32 and 30 – kept

What the table shows. The same 6–7 level stain contrast is preserved or destroyed depending only on where the grey values fall relative to the bin edges. The first row is the numerical example used in the earlier tasks: the fabric at 150 and the stain at 144 map to the same 4-bit and 5-bit level, so the stain disappears. In a real image, local brightness varies across the fabric, so a stain would be visible in some places and invisible in others, giving an inconsistent detector that is difficult to tune and to validate.

Where reduced depth can be used. Four- to six-bit versions are acceptable for operator thumbnails, for coarse overview maps showing where defects occurred along a roll, and for fast pre-screening of large defects such as a hole or major colour change. They must not feed the detector for faint stains, and any pre-screening stage must be verified so that it does not reject frames that would later have contained a stain.

Dithering. Noise or dithering can sometimes preserve fine differences through coarse quantization on average, because pixels near a bin edge flip between levels in proportion to their true value. Averaging over a region then restores part of the contrast. However, this works only when the noise is comparable to the step and the region is large, so it is not a reliable way to detect small stains in real time and should not be used to justify reducing the bit depth.

Section summary	Statement	
Given	Right-shift mapping of 8-bit values.	
Finding	150 and 144 merge at 4 and 5 bits and separate only at 6 bits.	
Decision	Restrict reduced depth to previews and overviews.	
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

Parameter	Calculation / observation	Design implication

5. Quantization and contrast

For a simple uniform quantizer, the quantization step is approximately the input range divided by the number of levels. If an 8-bit range 0–255 is reduced to 4 bits, the step is about 16. A 5–8 level difference is smaller than one quantization step and can be lost. At 6 bits, the step is about 4, so the difference may survive as one or two levels but remains vulnerable to noise.

Contrast enhancement can make a difference easier to view, but it cannot recreate information that was lost during quantization. If the raw sensor output is quantized too coarsely, later histogram equalization only stretches the remaining levels and may amplify noise or banding.

TASK 7 – Quantization Analysis (continued)

Parameter	Calculation / observation	Design implication

4-bit	2^4 = 16 levels	5–8 grey-level contrast can

The preferred sequence is stable illumination, high-quality acquisition, flat-field correction and then appropriate scaling. Do not reduce bit depth before the stage that detects subtle oil stains.

Bits	Step (levels)	6-level stain survives?	Banding after contrast stretch
4	16	Only about 38 % of alignments	Severe
5	8	About 75 % of alignments	Visible
6	4	Always as 1–2 levels	Mild
8	1	Always as 6 levels	None
12	1/16	Always, with fine gradation	None

Why contrast stretching cannot rescue the data. Suppose an image after 4-bit quantization uses only 6 distinct levels in the fabric region. Histogram equalisation can spread them over 0–255, but they remain 6 separate values with gaps of about 40 levels between them. The output looks high-contrast but contains no more information than before: a stain that mapped into the same bin as the fabric is still identical to it, and noise is amplified at the boundaries. Contrast enhancement is therefore a display and normalisation aid, not a source of lost information.

Order of operations. The reliable order is: acquire at the highest practical depth, correct illumination with flat-field division in floating point or 16-bit, compute local contrast, and only then, if required for a library function, convert to 8 bits with a window that preserves the stain range. Doing the conversion before flat-field correction compresses the very differences the correction is intended to expose.

Display versus analysis. Contrast stretching is useful for showing the operator where a stain is, because the human eye cannot see 5–8 levels on a standard display. The stretched image should be generated only for display from the full-depth data, and labelled as enhanced. The detector itself should work with the original numerical values, so that the display tools cannot change the detection result.

Section summary	Statement
Given	Contrast stretching works on remaining levels.
Finding	It cannot recreate information lost in quantization.
Decision	Preserve depth before enhancement; stretch only for display.

		disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

TASK 7 – Quantization Analysis (continued)

6. Memory and processing comparison

For a single 2448×2048 frame, 8-bit grayscale is approximately 5.01 MB. A 16-bit container for 12-bit samples is approximately 10.03 MB. RGB at 8 bits per channel is approximately 15.04 MB. These are uncompressed decimal sizes; file compression and metadata add or reduce the final storage requirement.

A higher bit depth can increase memory traffic and may require scaling before some operations, but thresholding and feature extraction can still be practical on a 4-core CPU. The system should benchmark the complete pipeline rather than assuming that 12-bit data doubles processing time.

Use full bit depth only where it adds information. A useful design retains the raw 12-bit image for audit or selected samples, processes a normalized 8-bit representation for common operations, and avoids needless conversion copies.

Format	Size per frame	Relative memory traffic	Processing implication
8-bit grey	5.01 MB	1×	Fastest; 16 pixels per 128-bit vector
12-bit packed grey	7.52 MB	1.5×	Unpacking cost
12-bit in 16-bit word	10.03 MB	2×	8 pixels per 128-bit vector
8-bit RGB	15.04 MB	3×	Three channels to process
16-bit RGB	30.08 MB	6×	Rarely justified

Bandwidth estimate. At 10 frames/s the 8-bit grey stream is about 50 MB/s and the 16-bit grey stream about 100 MB/s, which are small compared with the memory bandwidth of a modern 4-core PC (tens of GB/s). The real concern is not raw bandwidth but the number of passes made over the image. If the pipeline makes ten passes (convert, filter, subtract, threshold, morphology and so on), a 16-bit image moves about 100 MB per frame, still modest. Careful buffer reuse and region-based processing matter more than the choice between 8 and 12 bits.

Disk implication. Doubling the storage of raw frames halves the number of frames that fit on the 2 TB disk. If raw 12-bit images are kept only for defect crops and for periodic calibration samples, the extra storage is small. Keeping every full 12-bit frame is neither necessary nor possible within the seven-day, 2 TB limit, as shown in Task 6.

Practical conclusion. The extra memory and processing needed for 12-bit data are modest compared with the gain for faint stain detection when the sensor supports it. The practical constraint is storage of raw data, not processing. The design therefore processes in the format with the best contrast preservation and stores only the representation needed for evidence, maintaining the balance between accuracy and the 2 TB disk limit.

Section summary	Statement
Given	2448 × 2048 frame.
Finding	8-bit grey 5.01 MB; 12-bit in 16-bit 10.03 MB; RGB 15.04 MB.
Decision	Keep high depth for crops; process a normalised copy.

Parameter	Calculation / observation	Design implication
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

TASK 7 – Quantization Analysis (continued)

7. Experiment plan

Capture the same defect-free fabric and oil-stained samples at the same exposure using 8-bit, 12-bit and reduced-bit representations. Create matched crops and compare stain-to-background contrast, signal-tonoise ratio, detector sensitivity, false alarms and processing time. Use multiple stain intensities and fabric colours.

The test should include the faintest available stains because average contrast can hide the difficult cases. Report the number of true stains, detected stains, missed stains and false detections for each bit depth. If using synthetic data, label it clearly and do not present it as a measured factory result.

Section summary	Statement

Given	Same raw images requantized to each depth.

Ablation testing should isolate quantization from lighting and threshold changes. If a 12-bit sensor is used with a different exposure or illumination, any accuracy improvement cannot be attributed to bit depth alone.

Factor	Levels tested	Measured quantity
Bit depth	4, 5, 6, 8 and 12 bits (12-bit as reference)	Stain-to-background contrast, SNR
Stain strength	Faintest, typical and strongest available	Detection rate
Fabric	Light and dark; cotton and blend	Detection rate and false alarms
Lighting	500, 1,500 and 3,000 lux ambient with enclosure	Stability of contrast
Processing	Same detector and thresholds for all bit depths	Processing time per frame

Procedure. Capture each sample once at 12 bits, then create the lower-depth versions numerically by requantizing the same raw image. This guarantees that every bit-depth comparison uses identical optics, light and noise, so the difference is attributable to quantization only. Run the same detector with thresholds recalibrated on clean fabric for each depth, and record true detections, misses and false alarms. Use at least 50–100 stains of each strength where possible.

Analysis. Because the same images are used for every depth, a paired comparison such as McNemar's test on detected/missed outcomes is appropriate. Report the contrast-to-noise ratio of each stain at each depth, a table of detection rates, and a short list of images where a stain was lost. The minimum bit depth that keeps the detection rate within the agreed tolerance of the 12-bit reference becomes the production choice. Simulated requantization does not model sensor behaviour at different native depths, so the final check should use the actual camera.

Documentation. The experiment record should include the camera model and settings, exposure, lamp current, the list of samples with their stain type and age, the processing parameters, the software version and the raw images. The results are presented as a table of detection rate against bit depth with confidence intervals, and a short statement of the conclusion. This documentation allows the test to be repeated if the camera or fabric changes.

Finding	Paired comparison isolates the effect of bit depth.
Decision	Choose the minimum depth that matches the 12-bit reference.

Parameter	Calculation / observation	Design implication
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

TASK 7 – Quantization Analysis (continued)

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

8. Recommendation

The recommended acquisition is 10–12 bit if the camera supports it with acceptable data rate, with calibrated scaling to 8-bit or floating-point features for processing. Do not use 4-bit quantization for the primary detector. Six-bit data may be useful for previews, but not without evidence that the stain contrast is retained.

The production design should prioritize stable light and a low-noise sensor. More bits cannot compensate for saturation, motion blur or poor focus. The raw acquisition should be retained where feasible so that thresholds can be reviewed if a missed stain is reported.

Final choice: preserve high-bit-depth acquisition where affordable and process a normalized representation. Validate the choice using real 5–8 level oil-stain examples and verify that the complete pipeline remains within the 0.5-second decision limit.

Situation	Recommended depth	Reason
Low-noise sensor, stable light, faint stains	10–12 bit acquisition, normalised to 8 bit	Preserves subtle contrast
Sensor noise already above 2 levels	8 bit	Extra bits give little benefit
Operator preview or overview	6 bit or lossless 8 bit	Only for display
Archive of defect images	Lossless 8 or 12 bit crops	Evidence for investigation
Primary detector for oil stains	Never below 8 bit	5–8 level contrast

Decision rule. Choose the lowest depth that keeps the faintest required stain at a contrast-to-noise ratio of at least about 3–5 (a design guideline) and does not reduce detection sensitivity relative to the 12-bit reference by more than the agreed tolerance. The guideline is not a standard; the plant should confirm it with its own samples. If the camera supports only 8 bits, the design must instead rely on better illumination and noise control.

Link to the other tasks. Bit depth interacts with exposure (Task 5), because short exposures reduce signal and increase relative noise; with sampling (Task 6), because smaller pixels collect less light; and with storage (Task 9). The recommended front end is therefore judged as a whole: the best bit depth is useless if blur, glare or insufficient pixels remove the defect before the sensor quantizes it.

Limit of the conclusion. The recommendation applies to the stains and fabrics tested. If the mill introduces a new fabric colour or finish, the contrast of oil stains may change, and the bit-depth decision should be revisited by repeating the experiment on samples of the new fabric. Keeping the raw high-depth images for a sample of production helps, because requantizing stored data is a quick way to re-evaluate the depth question later.

Section summary	Statement
Given	Faint 5–8 level stains; stable lighting.
Finding	10–12 bit acquisition with calibrated normalisation is preferred.

TASK 7 – Quantization Analysis (continued)

Decision	Validate on real stains within the 0.5 s limit.

Parameter	**Calculation / observation**	**Design implication**
4-bit	2^4 = 16 levels	5–8 grey-level contrast can disappear
8-bit	2^8 = 256 levels	Efficient baseline; stable lighting needed
12-bit	2^12 = 4096 levels	Better tonal precision; more data

Formula: number of discrete levels = 2^n, where n is the number of bits per sample.

TASK 8 – Image Formation Analysis

1. Perspective image formation

A camera forms an image by projecting three-dimensional points through the lens onto the sensor. In a pinhole approximation, image coordinates depend on focal length and the point's distance from the camera. If the fabric plane is not parallel to the sensor or the camera is not perpendicular to the web, the apparent scale varies across the image.

The camera is mounted 700 mm above the fabric. A 1,200 × 400 mm field of view must be covered. A wide field of view requires a suitable focal length and sensor size. The exact lens cannot be chosen from camera height alone; sensor dimensions, distortion, working distance and required field of view are also needed.

Uncalibrated perspective imaging is simple but can produce systematic width errors. For a ±2 mm width requirement, calibration and mechanical alignment are necessary. The system should use a reference target placed in the fabric plane to determine the pixel-to-millimetre mapping.

Pinhole relations. For a point at lateral position X and distance Z from the lens, the image coordinate is $x = f X / Z$. The magnification is $m \approx f/(Z - f)$. With $f = 7.7$ mm and $Z = 700$ mm, $m = 7.7/692.3 \approx 0.0111$, so 1 mm on the fabric becomes 0.0111 mm on the sensor, and a 13.2 mm sensor covers $13.2/0.0111 \approx 1{,}190$ mm. The same sensor therefore matches the 1,200 mm field. Because x depends on Z, any change in distance changes the scale: a 1 % increase in Z causes a 1 % shrinkage of the image.

Symbol	Meaning	Value in this system
Z	Working distance (camera to fabric)	700 mm
FOV	Required field of view	1,200 × 400 mm
f	Focal length	About 7.7 mm for a 13.2 mm wide sensor
m	Magnification	About 0.011
s	Pixel scale on the fabric	FOV width / pixel count, e.g. 0.49 mm/pixel

Effect of camera tilt. If the camera is tilted by 1° about the along-web axis, the distance to the fabric changes by about ±600 mm × sin 1° ≈ ±10 mm across the 1,200 mm field, which is ±1.5 % of the working distance. At a fabric edge 500 mm from the centre, a 1.5 % scale difference corresponds to an error of the order of 7 mm if left uncorrected. This is a first-order estimate for illustration, but it shows that a 1° mounting error is large compared with the ±2 mm width tolerance and must be removed by mechanical alignment and homography calibration.

Measuring the working distance. The working distance in the calculations is measured from the lens to the fabric plane, not from the camera body or housing. A laser distance meter, or a mechanical gauge at the mounting, gives it to within a millimetre. Since the scale changes in proportion to the distance, a 7 mm error on a 700 mm working distance (1 %) produces a 1 % scale error and therefore up to 10 mm error across a 1,000 mm width, which would be far above the tolerance.

Section summary	Statement
Given	Z = 700 mm; FOV 1,200 × 400 mm.

TASK 8 – Image Formation Analysis (continued)

Finding	Scale depends on distance; a 1° tilt gives errors of several millimetres.
Decision	Align mechanically and map with a homography.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

2. Lens and field-of-view calculation

For a rectilinear lens, the approximate field of view is FOV ≈ sensor dimension × working distance / focal length when angles are modest. Rearranging gives focal length ≈ sensor dimension × working distance / FOV. The calculation should be performed independently for horizontal and vertical dimensions, using the camera sensor's active dimensions.

For example, if the sensor active width is 13.2 mm, the working distance is 700 mm and the required horizontal field is 1,200 mm, the approximate focal length is 13.2×700/1200 ≈ 7.7 mm. This is only a firstorder estimate; the actual lens and sensor format must be checked against manufacturer field-of-view data and distortion specifications.

The camera should be positioned square to the fabric and rigidly mounted. A wide-angle lens may introduce barrel distortion, while a long lens may make the required field of view impossible at the available height. Final selection should be based on calibration images and edge-to-edge sharpness.

Sensor width	Focal length for 1,200 mm at 700 mm	Vertical field (sensor height × 700 / f)	Comment
6.4 mm (1/2 inch)	≈ 3.7 mm	≈ 0.9 m (4.8 mm height)	Very wide-angle; strong distortion
8.8 mm (2/3 inch)	≈ 5.1 mm	≈ 0.9 m (6.6 mm height)	Common machine-vision format
13.2 mm (1 inch)	≈ 7.7 mm	≈ 0.8 m (8.8 mm height)	Matches the worked example
Line scan, 4,096 × 7 µm = 28.7 mm	≈ 16.7 mm	Single line	Lens image circle must cover 28.7 mm

Checking the result. Each focal length is f = sensor width × 700 / 1,200. The check for the 13.2 mm sensor is: 13.2 × 700 / 1,200 = 7.7 mm. In practice, the nearest standard lens (for example 8 mm) is chosen and the working distance adjusted slightly to obtain the exact field. A lens that is too wide in angle reduces resolution at the edges and increases distortion, whereas a longer lens would need a greater camera height than the available 700 mm.

Depth of field. An approximate depth of field is DOF ≈ 2 N c / m², where N is the f-number, c the acceptable blur circle (taken as one pixel, 5.5 µm) and m the magnification. For N = 5.6 and m = 0.0111, DOF ≈ 2 × 5.6 × 0.0055 / (0.0111)² ≈ 500 mm, so a few millimetres of fabric flutter is well within focus. The estimate assumes a diffraction-free, ideal lens and should be verified; stopping down too far increases diffraction blur and demands more light.

Section summary	Statement
Given	f = sensor width × 700 / 1,200.
Finding	A 13.2 mm sensor needs about 7.7 mm; a 28.7 mm line sensor about 16.7 mm.
Decision	Select the nearest lens and confirm distortion and resolution.

Geometric relationship:

TASK 8 – Image Formation Analysis (continued)

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

3. Geometric calibration

Camera calibration estimates intrinsic parameters such as focal length in pixels, principal point and lens distortion coefficients. A checkerboard or calibrated grid is captured at multiple positions and orientations. The estimated parameters allow radial and tangential distortion to be corrected before measuring fabric width or defect location.

After lens calibration, a planar homography can map the fabric plane to a top-down coordinate system. This is useful when the camera is slightly tilted or the web plane is not perfectly aligned. A calibration target with known dimensions should be placed at the same height as the fabric to avoid height-related scale errors.

Calibration must be checked periodically. Camera movement, lens focus changes, vibration or a change in working distance can invalidate the mapping. The system should store calibration date, parameters, target image and residual error as part of the audit record.

Step	Action	Output / acceptance
1	Print a rigid checkerboard with known square size (for example 25 mm) and mount it flat in the fabric plane	Target flatness and dimension certificate
2	Capture 15–20 images with different positions and tilts	Coverage of the whole field including corners
3	Detect corners(for example cv2.findChessboardCorners) and solve for intrinsics and distortion	Camera matrix, distortion coefficients
4	Check reprojection error	Target below about 0.3 pixel (≈ 0.15 mm at 0.49 mm/pixel)
5	Compute a homography from four or more known points on the fabric plane	Pixel-to-millimetre map in the web plane
6	Verify on an independent ruler or target	Measured position errors in mm

Calibration in a line-scan system. A line-scan camera sees only one line, so a checkerboard is imaged by moving it past the camera under encoder control, which builds a two-dimensional image. The crossweb scale comes from the pixel pitch and lens magnification, and the along-web scale from the encoder, so the pixel aspect ratio must be tested explicitly: a circular target must appear as a circle in the assembled image. A skew angle between the sensor line and the direction of travel shows up as a slanted rectangle and is removed by mechanical adjustment before software correction.

Recalibration schedule. Calibration should be repeated after any mechanical impact, lens or camera replacement, change of working distance, and at a regular interval such as monthly. A quick daily check uses a fixed reference mark within the field: if the measured position or width of the mark changes by more than a set limit (for example 0.5 mm), the system warns that recalibration is needed.

Calibration records. Each calibration generates a record that includes the date, operator, target description, camera matrix, distortion coefficients, reprojection error and the homography. These files are stored with the software version so that images recorded in the past can be reprocessed with the calibration

that was valid at the time. A calibration that is overwritten without a record makes it impossible to analyse old measurements.

Section summary	Statement
Given	Checkerboard images across the field.
Finding	Reprojection error below about 0.3 pixel corresponds to 0.15 mm.
Decision	Calibrate, verify with a ruler and repeat periodically.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

4. Illumination effects

Lighting varies from 500 to 3,000 lux. A factor-of-six change in illuminance can shift pixel intensities and alter thresholds, particularly for stains with only 5–8 grey levels of contrast. A bright reflection can saturate a region, while a shadow can resemble a dark stain or hole.

Controlled illumination uses fixed LED output and an enclosure to reduce daylight variation. Diffuse lighting can provide uniform fabric illumination, while cross-polarization can reduce specular reflections from some finishes. Directional or dark-field lighting can emphasize raised fibres, holes or surface texture, but it may create shadows that complicate colour measurements.

The chosen lighting should be evaluated on light and dark fabric, cotton and cotton-polyester blends, matte and glossy regions, and every target defect. A flat-field reference can correct smooth illumination variation, but it cannot recover clipped highlights or photons that were never captured.

Illumination issue	Effect on the image	Control
Ambient 500–3,000 lux	Six-fold brightness change	Enclosure; controlled LED light dominates
Mains light flicker (100 Hz for 50 Hz supply)	Brightness bands in short exposures	Constant-current LED driver; avoid fluorescent ambient
Specular highlights	Clipped regions without colour	Cross-polarisation, diffusers, adjusted angle
Shadows from raised yarns	False dark regions	Diffuse or symmetric lighting
LED ageing and heating	Gradual brightness drift	Periodic flat-field and lamp-current monitoring
Non-uniform lamp	Brightness gradient	Flat-field correction using a white reference

Why flicker matters. Indian mains frequency is 50 Hz, so lamps powered directly from the supply fluctuate at 100 Hz, a period of 10 ms. A 250 µs exposure samples only about 2.5 % of the cycle and can fall at a bright or dark part of the cycle, changing the brightness from frame to frame. A DC-driven LED at constant current avoids this. If the lamp is strobed, the strobe must be timed by the camera trigger and not by the mains.

Verification. Illumination should be mapped by imaging a uniform grey card and computing the mean brightness in a grid across the field. A non-uniformity of more than about 10 % is difficult to correct digitally without increasing noise at the dim edge, so lamp layout and diffusers should be adjusted until the map is acceptable. The map must be repeated at the lowest and highest ambient light and for each fabric colour in the product range.

Colour temperature. LEDs of different batches have slightly different colour temperatures. Mixing lamps from different batches in one array can produce a colour gradient across the field, which the system may report as colour variation. Using lamps from one batch, checking the colour of a neutral target across the field, and applying a per-channel flat-field correction minimises this effect and keeps colour measurement stable when a lamp is replaced.

TASK 8 – Image Formation Analysis (continued)

Section summary	Statement
Given	Light varies sixfold; mains flicker is 100 Hz.
Finding	Uncontrolled light and flicker change thresholds and brightness.
Decision	Use enclosed constant-current LED lighting with a brightness map.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

5. Standard versus calibrated imaging

Standard perspective imaging without calibration has the lowest setup effort but leaves lens distortion and perspective effects uncorrected. Calibrated imaging corrects geometric mapping and improves width and coordinate accuracy. Controlled-illumination imaging adds stable brightness and contrast, which improves the reliability of defect detection as well as measurement.

The three approaches should be compared using a ruler or calibrated grid across the field, plus actual fabric samples with known defect positions. Record width error at the left, centre and right of the field, as well as the displacement between a known defect coordinate and the predicted coordinate.

The preferred design combines calibration with controlled illumination. Calibration alone cannot solve changing brightness, and lighting alone cannot remove geometric distortion. Both are required for dependable defect marking.

Approach	Geometric error at edge (illustrative)	Brightness stability	Setup effort	Expected width result
Standard imaging (no calibration, ambient light)	Up to about 5 mm for 1 % barrel distortion at 500 mm	Poor	Low	Likely to exceed ±2 mm
Calibrated imaging, ambient light	About 0.15–0.3 mm	Poor	Medium	Good width, unreliable stains
Calibrated plus controlled illumination	About 0.15–0.3 mm	Good	High	Meets width and supports faint stains

Origin of the 5 mm figure. A barrel distortion of 1 % means a point at radius r in the undistorted image appears displaced by about 1 % of r. For a fabric edge at r = 500 mm from the centre, this is 0.01 × 500 = 5 mm, which alone is more than twice the width tolerance. Wide-angle lenses commonly show distortion of this order, so uncalibrated imaging cannot be accepted for a ±2 mm measurement. The 1 % value is an assumption for illustration; the real figure comes from the calibration of the lens used.

Experiment. Measure a calibrated ruler laid across the field at the left, centre and right, for each of the three approaches, with three repeats. Compute the mean and maximum absolute error. Then place defects at known coordinates and compare predicted and true positions. The three-row comparison turns the qualitative ranking into evidence and shows which improvement (calibration or lighting) contributes the larger gain for each requirement.

Cost-benefit. Calibration adds engineering time and requires a good target, but its cost is small compared with the camera and lighting. The controlled lighting costs more, yet it is the main contributor to reliable detection of faint stains. Because the two measures fix different errors, omitting either would reduce performance, and the budget should include both as essential items.

Section summary	Statement
Given	Three imaging approaches compared.
Finding	Calibration fixes geometry; controlled light fixes brightness.

TASK 8 – Image Formation Analysis (continued)

Decision	Adopt calibration plus controlled illumination.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

6. Width measurement accuracy

The fabric is 1,000 mm wide, while the camera field is 1,200 mm across. The extra 200 mm provides approximately 100 mm of margin on each side if the fabric is centred. The detector should locate both fabric edges, reject background regions, and calculate the width using the calibrated mapping.

If the pixel scale is 0.49 mm/pixel, a two-pixel boundary error corresponds to about 0.98 mm. However, width error also includes calibration residual, lens distortion, edge threshold uncertainty and fabric flutter. A budget of ±2 mm should be allocated among these sources rather than assuming the pixel scale alone guarantees compliance.

Use a traceable width target and measure at multiple positions and times. Report mean absolute error, maximum absolute error and repeatability. The line should alarm if the fabric edge leaves the calibrated field or if the edge confidence is too low.

Error source	Estimated contribution (1σ, mm)	How it is reduced
Edge location (two edges, medium resolution)	0.69	Sub-pixel edge fitting, higher resolution
Calibration residual	0.50	Better target, more images, validation
Lens distortion residual	0.30	Calibration with a distortion model
Fabric flutter and edge curl	0.50	Guides, roller tension, fixed edge background
Root-sum-square total	$\sqrt{(0.69^2 + 0.50^2 + 0.30^2 + 0.50^2)} \approx 1.0$	Compare with ±2 mm tolerance

Using the budget. The contributions are independent estimates, so they combine as the root of the sum of squares, giving about 1.0 mm (1σ). A ±2 mm tolerance corresponds to roughly two standard deviations (95 %), so the design has a small margin, not a large one. If a measured test shows a larger standard deviation, the largest contributor should be improved first; the table shows that edge location and flutter dominate. The values are assumed for planning and must be replaced by measured ones.

Reporting. The width log should store the measured width, a quality flag, the left and right edge positions and the timestamp for every section. A rolling mean and the standard deviation can be shown to the operator, and an alarm raised if the width leaves the specified band or if the edge cannot be detected, which usually indicates a folded edge, a shadow or a camera problem.

Validation. Validation uses at least three reference widths, for example by placing a calibrated bar or a strip of known width within the web area, and several lateral positions. For each, 30 or more repeat measurements are made. The mean error shows bias (correctable by calibration) and the standard deviation shows precision. The specification is considered met if the mean plus three standard deviations stays within ±2 mm, or a more relaxed statistical criterion agreed with the customer.

Section summary	Statement
Given	Fabric 1,000 mm in a 1,200 mm field.

TASK 8 – Image Formation Analysis (continued)

Finding	Combined error is about 1.0 mm (1σ) in the planning budget.
Decision	Measure on a traceable target and alarm on weak edges.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

7. Image formation error sources

Important error sources include lens distortion, perspective, defocus, vibration, fabric height variation, uneven illumination, sensor noise, saturation and motion blur. Each can affect defect detection differently. Blur weakens a small hole boundary; uneven illumination changes thresholds; geometric error shifts the reported marker position; saturation can hide colour information.

The front-end should use a rigid mount, suitable lens aperture, controlled exposure, stable LED lighting, calibration and periodic checks. The image-processing pipeline should flag low-quality frames using focus, brightness, saturation and motion metrics before classification.

A defect detection result should carry an image-quality status. This allows the system to distinguish 'no defect found' from 'image not reliable enough to decide', preventing poor acquisition from silently generating a false clean result.

Error source	Typical effect	Detection metric	Mitigation
Defocus	Soft edges, smaller apparent contrast	Variance of the Laplacian	Fixed focus, rigid mount, focus check
Motion blur	Smeared small defects	Edge width along the web	Short exposure, encoder timing
Lens distortion	Position and width error	Reprojection error	Calibration
Uneven illumination	False stains	Brightness map	Flat-field correction
Saturation	Clipped colour and edges	Fraction of pixels at maximum	Exposure control, polarisation
Vibration	Shifts between frames	Registration shift	Isolation rigid mounts, frame
Sensor noise	Hides faint stains	Standard deviation on clean fabric	Cooling, more light, averaging

Image-quality gate. Before a frame is accepted for classification, the software computes a few cheap metrics: mean brightness within limits, fraction of saturated pixels below a small value (for example 1 %), focus metric above a limit, and no sign of dropped lines. These limits are tuned on good images from the line. A frame that fails any test is flagged “quality poor”, the section is marked for manual inspection, and a warning is logged. This prevents the system from reporting “no defect” where it effectively saw nothing.

Maintenance. Most of the listed sources change slowly: dust on the lens, lamp ageing, loosened mounts. A weekly cleaning and calibration-target check, plus trend plots of the quality metrics, allow these to be caught before the missed-defect rate is affected.

Priority order. Because resources are limited, the sources are ranked by effect on the requirements: illumination uniformity and stability, motion blur and focus affect detection of small and faint defects;

TASK 8 – Image Formation Analysis (continued)

calibration and perspective affect width and position; vibration and noise affect both. A reasonable order of improvement is lighting and enclosure first, then exposure and focus, then calibration, and finally finetuning. This order follows the likely gain per unit cost.

Section summary	Statement
Given	Seven physical error sources.
Finding	Each has a cheap quality metric and a mitigation.
Decision	Gate every frame by quality before classification.

Geometric relationship:

Approximate scale: mm/pixel = field-of-view width (mm) / image width (pixels).

Planar correction: image pixel coordinates → calibrated fabric-plane coordinates → marker coordinates.

Formula: approximate focal length = sensor dimension × working distance ÷ desired field of view.

TASK 9 – Implementation and Performance Evaluation

1. System architecture

The complete system contains camera and lens, controlled illumination, acquisition interface, CPU processing unit, local storage, network interface, encoder and marking mechanism. The camera captures fabric continuously. The processing unit corrects and analyzes each image, estimates defect class and coordinates, and schedules a marking event. Storage preserves the image and metadata for seven days.

A block diagram is: Fabric web + encoder → Camera/lighting → Frame acquisition → Calibration and preprocessing → Defect candidate extraction → Classification and coordinate conversion → Deadline/quality check → Marker controller. In parallel, frame, mask, labels and timestamps are sent to the audit store over the local network.

The architecture should isolate the real-time path from slower archive tasks. Image writing should use a queue and must not block the marker-control loop. If storage is full or the camera stops delivering frames, the system should raise a fault and follow a predefined safe operating procedure.

Module	Function	Interface	Timing constraint
Camera and lens	Capture lines or frames	Gigabit Ethernet, USB3 or CoaXPress class link	Triggered by encoder pulses
Lighting controller	Constant or strobed LED	Trigger line from camera	Pulse no longer than exposure
Encoder	Web position	Quadrature input to controller	Position known at every frame
Acquisition thread	Fill a ring buffer of frames	Camera driver	No dropped frames at the sustained rate
Processing workers	Preprocess, detect, classify	Shared memory	Inside the latency budget
Marker controller (PLC or microcontroller)	Fire at target position	Ethernet or fieldbus	Command before the fabric arrives
Audit storage	Save crops and metadata	SSD or HDD queue	Never block the realtime path

Encoder resolution example. If a 1,000 pulse-per-revolution encoder drives a 100 mm diameter measuring wheel (circumference 314.16 mm), each pulse corresponds to 0.314 mm of web travel, or 0.0785 mm per count with 4× quadrature decoding. The 300 mm marker offset is then 300 / 0.314 ≈ 955 pulses (3,820 counts). The resolution is far finer than the likely marker tolerance, so the encoder is not the limiting element, but wheel slip and mounting must be checked because the pulse count measures the wheel, not the fabric.

Thread plan on four cores. One reasonable arrangement dedicates a core to acquisition and buffering, two cores to processing workers (each handling a different frame or region set), and the fourth core to the control thread, communications and archiving. Real-time critical work is given higher priority than

TASK 8 – Image Formation Analysis (continued)

archiving, and the disk writer uses a bounded queue so that a slow disk causes dropped archive images rather than a late marker command. This plan is a starting point to be tested under full load.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

2. Budget configuration

The budget configuration uses a low-cost camera, basic lighting and CPU-based OpenCV. Its advantages are low initial cost and simple installation. Its risks include insufficient spatial resolution, rolling-shutter distortion, unstable lighting and false alarms from fabric texture. It may be useful for a proof of concept but should not be selected solely because it is cheapest.

A budget estimate must include camera, lens, lighting, mounting, trigger/encoder interface, cables, enclosure, PC integration, storage and commissioning. A low camera price does not represent the total installed system cost. All cost figures in this report are planning estimates and must be replaced by current vendor quotations before procurement.

The budget configuration should be tested on small holes, broken threads, faint oil stains and weave irregularities. If the system cannot reliably represent a 1.5 mm defect or keep false alarms manageable, the configuration should be rejected regardless of its price.

Item (planning allocation, not a quotation)	**Amount (□)**	**Comment**
Low-cost camera and lens	20,000	Small sensor; rolling shutter likely
Basic LED lighting	10,000	Little or no enclosure
Encoder and interface	10,000	Needed to trigger and to locate defects
Mounting and cabling	8,000	Rigid mount still required
Software and commissioning	15,000	Mostly engineering time
Total	63,000	Leaves 87,000 unused of the 1,50,000 budget

Technical limits of this configuration. A low-cost camera with 1280 × 1024 pixels over a 1,200 mm width has a pitch of 0.9375 mm, giving 1.6 pixels across a 1.5 mm defect (Task 6), and a rolling shutter introduces skew (Task 5). Because these are physical limits, the unspent 87,000 does not indicate good value; it indicates that money should be moved to the camera and lighting, where it improves the measured outcome. This is the main reason the configuration is acceptable for a demonstration but not for production.

Acceptance gate. A quick test on the budget system with known 1.5 mm holes and faint stains gives an early answer. If fewer than the required fraction of test defects are found, or the false-alarm rate exceeds what operators will accept, the configuration is rejected without further development, avoiding wasted software effort on hardware that cannot see the defect.

Possible role. A low-cost system can serve as a pilot to collect images from the real line, even if its detection performance is limited. The images and operator feedback guide the choice of the final lens, resolution and lighting. The pilot should be run for a defined period, and its results should be summarised as a recommendation for the balanced configuration, so that its purpose is learning rather than permanent installation.

Section summary	**Statement**
Given	Low-cost parts and □63,000 planning total.

TASK 9 – Implementation and Performance Evaluation (continued)

Finding	Resolution and shutter limits reject it for production.
Decision	Use only as a pilot for image collection.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

3. Balanced configuration

The balanced configuration uses a suitable industrial global-shutter or line-scan camera, controlled lighting, a calibrated lens and optimized CPU processing. It balances sampling, reliability, cost and ease of maintenance. The camera should have enough resolution to represent the minimum defect with multiple pixels, and the lighting should remain stable across fabric types.

The software should use reusable preprocessing buffers, region-based feature extraction and class-specific checks. A compact classifier can be considered after a baseline rule-based detector is established. Realtime processing and image archival should be separated so that disk activity does not delay the marking decision.

This configuration is the recommended starting point if its quoted components fit within □1,50,000. The final purchase decision depends on measured defect recall, width error, latency and storage. No accuracy or latency number should be claimed until the chosen hardware is actually tested.

Item (planning allocation, not a quotation)	Amount (□)	Specification target
Industrial camera and lens	45,000	Global shutter or line scan, ≥ 4,096 pixels across the field, ≥ 8 bit
LED lighting and enclosure	30,000	Diffuse, constant-current, shielded from ambient light
Encoder and trigger/marker I/O	20,000	Encoder, input/output module, relay or valve driver
Mounting, cabling, calibration targets	15,000	Rigid frame, checkerboard and width target
Software development and commissioning	25,000	OpenCV-based pipeline, tests, documentation
Contingency	15,000	About 10 % of the budget
Total	1,50,000	Equals the stated budget

Feasibility remark. The amounts are illustrative planning splits and have not been confirmed by suppliers. Industrial 4k line-scan cameras and their lenses can cost more than the camera allocation shown, in which case the balanced option would use a 5-megapixel global-shutter area camera with strobed lighting, accepting a smaller margin on the 1.5 mm defect, or negotiate additional budget. The report should therefore state clearly which of the two sub-options was costed.

Why it is the recommended starting point. This configuration reaches about 0.29–0.49 mm per pixel, controls the light and uses the existing 4-core computer. It leaves room for a second-stage upgrade, for example a UV channel, without discarding the first investment, since the enclosure, encoder, mounting and software architecture remain valid. Performance claims remain provisional until the acceptance test of section 8.

Software structure. The software is organised as independent modules for acquisition, preprocessing, detection, coordinate mapping, marker communication and logging, connected by queues. Each module can be tested on recorded data without the camera, and the whole system can be replayed for regression

TASK 9 – Implementation and Performance Evaluation (continued)

testing after a change. Configuration files hold thresholds and calibration so that engineers can adjust them without changing code, and every change is version controlled.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

4. Premium configuration

The premium configuration may use a high-resolution industrial line-scan camera, advanced illumination and higher-performance processing hardware. It offers greater sampling margin and may support more complex analysis. It also increases acquisition bandwidth, integration complexity, cost and the need for specialized support.

A premium camera is not automatically more accurate. If illumination is poor or the classifier is unsuitable, extra pixels can merely create larger files and slower processing. The premium option should be evaluated on whether it improves detection of the smallest defects and reduces false alarms enough to justify its cost.

Because the project budget is fixed, premium equipment may be infeasible unless a vendor offers a package within the limit. The report should explicitly separate the technically strongest configuration from the configuration that is financially feasible.

Aspect	Balanced	Premium	Comment
Pixels across the web	4,096	8,192–16,384	Premium offers 0.15–0.07 mm pitch
Line rate	About 2,000 lines/s	5,000–15,000 lines/s	Needs a fast interface
Data rate (8bit, mono)	About 8 MB/s	40–245 MB/s	Premium needs frame grabber and fast storage
Lighting	Diffuse LED	High-power line light, possibly multi-angle	Heat management needed
Computer	Existing 4-core PC	Probably more cores or an accelerator	Outside the stated budget
Cost	Within □1,50,000 (target)	Likely above □1,50,000 (to be quoted)	Compare on installed cost

Where premium equipment pays. The premium system makes sense when the minimum defect is smaller than the balanced system can resolve, as in the export order of Task 10, or when low false-alarm rates are worth more than the extra cost. For the present 1.5 mm requirement, the additional resolution would give about 5–10 pixels across the defect, which is more than necessary and would only increase data and processing time unless the extra pixels measurably improve recall.

Evaluating value. A fair comparison estimates the annual saving from fewer escaped defects and fewer false rejections. For example, if the plant ships a lot that is later rejected, the loss includes the fabric, freight and reputation. The premium system should be chosen only if the measured reduction in such losses exceeds its extra cost over the expected life of the equipment. Without plant cost data, this report treats the question as open.

Risk of over-specification. A premium configuration may bring unneeded cost and complexity. An engineer should compare the specification of the camera with the actual need: if the system is to detect 1.5 mm defects, a camera that resolves 0.07 mm adds little unless the export order is accepted. A modular

TASK 9 – Implementation and Performance Evaluation (continued)

design that can be upgraded later is often a better use of money than purchasing the highest specification at the start.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

5. Defect detection and coordinates

Holes can be detected from local contrast, edge structure and a region interior that differs from the fabric. Broken threads require directional texture and continuity analysis. Oil stains require local brightness and colour comparison because the contrast is small. Weaving irregularities can be detected using local texture statistics or deviations from the expected periodic weave. Colour variations require calibrated colour features.

After segmentation, each candidate region receives a bounding box, centroid, area and class. Pixel coordinates are converted to millimetres using calibration. The along-web position is combined with encoder distance. The marker controller receives roll ID, defect ID, cross-web coordinate, along-web coordinate, class and required action.

The system should retain confidence and quality flags. If a region is ambiguous, the record should say 'review' rather than 'clean'. A coordinate error can cause the marking mechanism to miss a correctly detected defect, so location accuracy must be measured independently from classification accuracy.

Field in defect record	Example	Source
Roll ID	R-2026-1031-07	Operator input or barcode
Defect ID	D-000342	Software counter
Class and confidence	Oil stain, 0.86	Classifier
Cross-web position x (mm)	600.0	Calibrated centroid
Along-web position y (mm)	12,098	Encoder count at frame plus offset
Bounding box (mm)	12.0 × 8.5	Region statistics
Image-quality flag	OK	Quality gate
Action	Mark for rework	Severity rule

Marker target. The marker command is generated when the encoder reaches y + D, where D = 300 mm is the camera-to-marker offset along the web. In the example the target is 12,098 + 300 = 12,398 mm. The cross-web coordinate selects which marker nozzle or position to use if a multi-position marker is fitted. If the marker spans the full width, only the along-web target matters, but the stored cross-web position remains useful for later rework.

Independent measurement of position accuracy. Coordinate accuracy depends on calibration, centroid precision, encoder reading and mechanical position of the marker. Each is tested separately: calibration against a ruler, the centroid against labelled defect positions, the encoder against a measured length of fabric, and the marker by measuring the distance between the intended and actual mark. Only the sum of all these errors in millimetres can be compared with the marking tolerance.

Duplicate suppression. A defect that lies on a boundary between two frames or lines can be detected twice. The software should merge detections whose positions are within a tolerance in the along-web direction and which have compatible class, and then send one marker command. The tolerance is chosen from the position accuracy of the system, for example 5 mm. Without this step, the marker could fire twice and waste fabric or confuse the rework team.

System flow:

TASK 9 – Implementation and Performance Evaluation (continued)

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

6. Timing and marker communication

The marker is 0.3 m downstream and fabric speed is 0.6 m/s. Nominal travel time is t = d/v = 0.3/0.6 = 0.5 s. Since the stated maximum decision time is also 0.5 s, the design has no spare margin if processing finishes exactly at the deadline. The real system should complete processing earlier and use encoder position to schedule the mark.

A fixed 0.5-second delay assumes constant speed. An encoder-based system is more robust because it tracks the actual fabric travel. The controller should validate that each defect event is received once, associated with the correct roll, and activated at the intended position. Network delay and actuator response should be measured.

The test report should include average, 95th percentile and maximum end-to-end latency; dropped-frame count; missed marker count; and position error. A successful software benchmark alone does not prove that the physical marking mechanism works reliably.

Event	Planning time after exposure (ms)	Notes
Exposure and readout complete	50	Frame in memory
Preprocessing complete	130	Flat-field, denoise, representations
Candidates and features complete	240	Worst-case frame
Classification and coordinates complete	300	Marker target computed
Command received by controller	320	Network and driver latency
Actuator responds	350	Valve or pen response
Margin to 500 ms	150	Absorbs jitter and rare slow frames

Encoder-based firing. At 0.6 m/s the fabric moves 0.6 mm every millisecond. If the controller fired using a fixed 0.5 s timer and the web slowed to 0.55 m/s, the mark would be placed 0.05 × 500 = 25 mm too early, while at 0.65 m/s it would be 25 mm too late. Because the encoder counts actual movement, it removes this error. The controller should also compensate for the known actuator delay by subtracting speed × delay (for 30 ms at 0.6 m/s, 18 mm) from the firing distance.

Fault handling. If a decision arrives after the target position has passed, the event must not be silently dropped. The system should log it, show an alarm to the operator, and use a safe fallback such as marking the entire section a further fixed distance downstream or flagging the roll for manual re-inspection. Similarly, a communication timeout with the controller should raise an alarm, since marking silently stops.

Measurement method. To measure latency, the software writes a timestamp at the moment of exposure and another when the command leaves the PC, and the PLC records the time of arrival and of actuator output using a common clock or a hardware signal captured on an oscilloscope. Comparing the two gives

TASK 9 – Implementation and Performance Evaluation (continued)

the network and actuator contributions. Measurements are taken at normal and maximum loads, with archiving enabled, and are repeated over a long period to capture rare slow events.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

7. Seven-day retention and storage

The storage requirement depends on frame rate, image dimensions, bit depth and compression. For example, 5.01 megapixel RGB at 8 bits/channel is about 15.04 MB per raw frame. At 10 frames/s this is about 150 MB/s, or approximately 12.96 TB per day uncompressed. This exceeds a 2 TB drive by a large margin.

The actual system must therefore define whether it stores every line/frame, stitched web images, compressed images, or event crops. If every image must be retained for seven days, calculate the acquisition cadence and compression ratio that fit the 2 TB limit. A rolling retention policy must delete or archive data only after the required seven-day period and should alert when capacity is near the limit.

Lossy compression may alter small low-contrast defects. Test the selected codec on real defects and compare detection results with the original. Prefer lossless compression or storing full-resolution crops for defect regions while keeping a lower-bandwidth overview, if this complies with the buyer's retention requirement.

Storage policy	Data per day	Seven days	Fits in 2 TB?
Raw RGB frames, 2448 × 2048 at 10 frames/s	≈ 13.0 TB	≈ 90.7 TB	No
Raw grey frames, same size and rate	≈ 4.3 TB	≈ 30.3 TB	No
Line scan 4,096 px × 2,048 lines/s, grey, uncompressed	≈ 0.72 TB	≈ 5.1 TB	No
Same, with 3:1 lossless compression (assumed)	≈ 0.24 TB	≈ 1.7 TB	Barely; no headroom
Defect crops plus periodic overviews (example plan)	≈ 15.5 GB	≈ 0.11 TB	Yes, large headroom

How the numbers arise. The 2448 × 2048 RGB frame is 15.04 MB, and at 10 frames/s the rate is 150 MB/s, which over 86,400 s gives 12.96 TB. A grey frame is 5.01 MB, giving 4.33 TB/day. The line-scan stream is 4,096 × 2,048 = 8.39 MB/s, or 725 GB per day, which for seven days is 5.07 TB; dividing by an assumed 3:1 compression gives 1.69 TB. Reserving 20 % of the disk for the operating system and safety margin leaves only 1.6 TB, so full continuous retention would require at least 57:1 compression for the raw RGB case, which is impractical for lossless methods.

Recommended policy. Retain full-resolution crops for every detected or uncertain event, a lower-rate overview of normal sections, and complete metadata for every frame. Delete data older than seven days with a scheduled job, but never before, and raise a warning at about 80 % capacity. If the customer requires every metre to be imaged and retained, the plant should discuss either an external storage array or a lower retention frame rate, as the 2 TB drive cannot hold the full stream.

Monitoring. The storage service should record the daily growth in gigabytes, the number of files, the oldest retained timestamp and the free space. A simple linear projection shows when the disk would fill, which gives early warning of changes such as a higher defect rate. Deleting should be automated and logged, and the log must show that no file less than seven days old was removed.

TASK 9 – Implementation and Performance Evaluation (continued)

Section summary	Statement
Given	2 TB disk; 7-day retention.
Finding	Continuous raw RGB needs about 90.7 TB; event storage needs about 0.11 TB.
Decision	Store crops, overviews and metadata; monitor capacity.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

8. Performance evaluation and final configuration

The three configurations should be evaluated on the same held-out fabric set, including every defect class, different fabric colours, light levels, and clean regions. Metrics should include per-class sensitivity, precision, false alarms per metre, width error, processing latency, storage use, dropped frames and marker position error. The missed-defect target is at most 2%, so sensitivity must be high and its uncertainty must be considered.

A balanced configuration is recommended because it offers a realistic chance of fitting the budget while meeting sampling and latency requirements. The premium option may exceed the budget; the budget option may lack sufficient image quality. These are engineering expectations until vendor quotes and measured test results are available.

Commission only after the balanced system passes an agreed acceptance test. Include a representative production run, independent ground-truth review, fault injection for missing frames and storage errors, and a documented rollback plan. The final report should label estimates, simulations and measured values separately.

Metric	Target	Method	Evidence to file
Per-class sensitivity	Missed-defect rate ≤ 2 %	Hold-out rolls with independent labels	Confusion matrix, list of misses
Precision / false alarms	Operator-acceptable (set by plant)	Clean-fabric runs	False alarms per metre
Width error	Within ±2 mm	Calibrated width target at several positions	Mean, maximum, repeatability
Latency	Worst case below 0.5 s with margin	Timestamps under full load	Median, 95th percentile, maximum
Marker position error	Within agreed tolerance	Marked test defects measured by ruler	Error table
Storage	Seven days within 2 TB	Measure actual daily growth	Disk usage log
Robustness	No silent failure	Fault injection (cover lens, unplug camera, fill disk)	Fault log

Sample size and uncertainty. To claim a missed-defect rate of at most 2 % with 95 % confidence, a test with no misses needs about 150 defects per class (the rule of three, 3/150 = 2 %). If the test finds 2 misses in 150, the observed rate is 1.3 % but the upper confidence limit is above 2 %, so more samples or an improved detector are needed. The plan must therefore include enough rare-defect samples, which may require collecting defects over several weeks or creating realistic defects deliberately on test fabric.

Final configuration. The balanced configuration is carried forward: a calibrated, enclosure-lit camera system with encoder-triggered acquisition, a CPU pipeline using RGB, HSV and grayscale representations with 8-connectivity labelling, an event-based audit store, and an encoder-timed marker. Each assumption

TASK 9 – Implementation and Performance Evaluation (continued)

in this chapter (costs, timings, compression ratios) must be replaced by a quotation or measurement before the system is commissioned.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

Marker travel time at baseline speed: 0.3 m ÷ 0.6 m/s = 0.5 s. Use encoder distance rather than a fixed delay.

TASK 10 – Front-End Adaptation Challenge

1. Changed requirements

The new order requires detection of 0.4 mm defects, identification of stains visible under ultraviolet illumination, conveyor speed of 1.2 m/s, twice the current production volume and a decision within 0.3 s. The existing PC and □1,50,000 budget remain unchanged. These requirements change the optical resolution, spectral sensitivity, acquisition rate, data rate and processing deadline.

The original 1.5 mm defect assumption is no longer valid. A camera selected to show a 1.5 mm defect across only three pixels may represent a 0.4 mm defect with less than one pixel. Visible RGB imaging may not reveal a stain that responds primarily under UV. The doubled speed also increases motion blur for the same exposure time and requires faster acquisition.

The unchanged budget and PC make the problem a constrained redesign rather than a simple software update. The team must identify which requirements can be achieved through component selection and optimization, and which require a budget or hardware exception.

Requirement	Original	Export order	Ratio / effect
Smallest defect	1.5 mm	0.4 mm	3.75× smaller; pitch must be 3.75× finer
Stain visibility	Visible light, 5–8 levels	Visible under UV	New spectral channel
Web speed	0.6 m/s	1.2 m/s	2× blur and 2× line rate for the same pitch
Production volume	1×	2×	Twice the fabric area per hour to inspect
Decision time	0.5 s	0.3 s	Latency budget cut by 40 %
Marker (if distance unchanged)	0.3 m, 0.5 s travel	0.3 m, 0.25 s travel	Travel time shorter than decision time
PC and budget	4-core, no GPU; □1,50,000	Unchanged	No extra computing or money

Compound effect. If the pixel pitch must shrink by 3.75 times across the web and the speed doubles, the line rate rises by 2 × 3.75 = 7.5 times for the same type of line-scan system, and the data rate by 3.75 × 7.5 ≈ 28 times (pixels per line times lines per second). The line period also shrinks: it is 1/15,000 = 66.7 µs at 0.08 mm pitch and 1.2 m/s, compared with 833 µs for a 0.5 mm pitch at 0.6 m/s, i.e. 12.5 times shorter. The new order is therefore not an incremental modification but a change of several orders of difficulty at once, and each requirement must be tested for feasibility separately before a solution is promised.

TASK 10 – Front-End Adaptation Challenge (continued)

Order of analysis. The sections that follow take the requirements in the order resolution, blur, spectral channel, then compare three adaptation strategies and give a feasibility conclusion. The marker-distance conflict is treated explicitly at the end, because it concerns the plant layout, not the camera.

Customer communication. Because the export order may not be feasible within the original budget and equipment, the plant should communicate with the customer early. A clear table showing which requirements the present system meets, which would be met by proposed changes and which remain uncertain allows the customer to decide whether to accept a phased delivery, adjust the specification or provide extra funding.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm □ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target □ exposure ≤ 66.7 μs.

2. Required spatial resolution

A defect should span multiple pixels for reliable shape and contrast analysis. Using a five-pixel target for a 0.4 mm defect gives a pixel pitch of 0.4/5 = 0.08 mm/pixel. Across a 1,200 mm field, the required crossweb sampling is 1200/0.08 = 15,000 pixels. This is far beyond a 4,096-pixel area-camera width.

Three pixels across the defect would require 0.133 mm/pixel and about 9,000 pixels across the field. Even that may be inadequate for low-contrast stains and irregular defects. A line-scan camera can allocate many pixels across the web more efficiently, but it still requires high data bandwidth and suitable optics.

The calculation shows why keeping the old camera and changing only software is not credible for the new smallest-defect requirement. A narrow region of interest could reduce width coverage, but it would violate the requirement to inspect the full fabric width unless multiple cameras or scans are used.

Pixels across a 0.4 mm defect	Pixel pitch	Pixels across 1,200 mm	Data rate (8-bit, mono) at 1.2 m/s
2	0.200 mm	6,000	6,000 × 6,000 lines/s = 36 MB/s
3	0.133 mm	9,000	9,000 × 9,000 lines/s = 81 MB/s
4	0.100 mm	12,000	12,000 × 12,000 lines/s = 144 MB/s
5	0.080 mm	15,000	15,000 × 15,000 lines/s = 225 MB/s

Interpreting the numbers. The line rate equals speed divided by pitch: 1,200 mm/s ÷ 0.08 mm = 15,000 lines/s, and the data rate is pixels per line times lines per second. Five pixels across the defect gives 225 MB/s for a single monochrome channel, about 19.4 TB per day if recorded continuously, far above the 2 TB disk and heavy for a 4-core CPU if every pixel were processed by expensive algorithms. Colour (RGB) and a UV channel multiply this by three or four. Sensors of 8k–16k pixels exist but must be confirmed with suppliers for line rate, sensitivity and price.

Options if one camera is not enough. Several cameras can each view part of the width, with overlap and stitching, which allows more pixels in total and may use lower-cost sensors but increases calibration effort. Alternatively, a reduced requirement (such as three pixels across the defect, 9,000 pixels) may be negotiated with the customer when tests show that 0.4 mm defects can still be found reliably. Any such compromise must be tested on real 0.4 mm defects and recorded as a documented deviation, not assumed.

Sensor selection note. A sensor with enough pixels is only part of the solution. The lens must resolve 0.08 mm features over a 1,200 mm width, which means the optical resolution at the object must be about 6 line pairs per millimetre. Very wide fields at this resolution require high-quality lenses and a longer working distance, which may be difficult at 700 mm. The camera height and lens design would therefore need to be reviewed together with the sensor choice.

Section summary	Statement
Given	0.4 mm defect, five pixels across.

TASK 10 – Front-End Adaptation Challenge (continued)

Finding	0.08 mm pitch needs 15,000 pixels over 1,200 mm.
Decision	A single ordinary area camera is not enough.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm □ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target □ exposure ≤ 66.7 μs.

3. Motion blur at increased speed

At 1.2 m/s, fabric speed is 1,200 mm/s. Motion blur is $b = v \cdot t$. At 100 µs exposure, blur is 0.12 mm; at 250 µs, blur is 0.30 mm; at 500 µs, blur is 0.60 mm. A 0.4 mm defect can be severely degraded by a 0.3–0.6 mm blur.

To keep blur below 0.08 mm, exposure should satisfy $t \leq 0.08/1200 = 0.0000667$ s, or about 67 µs. This requires strong controlled illumination, a sensitive sensor and stable optics. The exposure target is an engineering estimate; actual blur also depends on sensor timing, fabric vibration and the direction of movement.

A short exposure alone is insufficient if the sensor noise becomes comparable to the UV stain contrast. The lighting and sensor must be tested together. An encoder-synchronized line-scan camera can help preserve geometric scale, but its line rate must be high enough for the required along-web sampling.

Exposure	Blur at 1.2 m/s	Fraction of a 0.4 mm defect	Verdict
25 µs	0.03 mm	7.5 %	Excellent
50 µs	0.06 mm	15 %	Good
66.7 µs	0.08 mm	20 % (one 0.08 mm pixel)	Target limit
100 µs	0.12 mm	30 %	Poor
250 µs	0.30 mm	75 %	Defect largely smeared
500 µs	0.60 mm	150 %	Defect lost

Light requirement. At 15,000 lines/s the line period is 1/15,000 = 66.7 µs, so the exposure cannot exceed this value anyway. The previous design could use up to 250 µs; the signal collected by a sensor pixel of the same size and the same lens aperture is proportional to exposure time, so it falls by 250 / 66.7 ≈ 3.75 times. The scene radiance, i.e. the LED output at the fabric, must therefore rise by about 3.75 times, or a larger aperture, a more sensitive sensor or time-delay integration (TDI) must be used to recover the signal. A 5–8 level stain on a noisier image may no longer be detectable.

TDI as a possible solution. A TDI line-scan sensor exposes the same strip of fabric on several rows in succession, adding the signals in synchronism with the motion. With N stages the signal increases by N and the noise by about $\sqrt{N}$, so N = 4 would double the signal-to-noise ratio, while keeping the effective exposure per row short. TDI requires precise synchronism with the web, a very rigid mount and, usually, higher cost, so it is a candidate to evaluate in Strategy B or C rather than a guaranteed fix.

Mechanical contribution. At 1.2 m/s vibration of the web or camera mount becomes more significant. A vibration amplitude of 0.05 mm during a 67 µs exposure would add little blur on its own, but a lowfrequency flutter of the fabric can change the distance to the lens and the scale. Web guides, tension control and a flat inspection roller reduce this, and a displacement measurement of the fabric height during commissioning would show whether further stabilisation is needed.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

TASK 10 – Front-End Adaptation Challenge (continued)

New-order calculations: five pixels across 0.4 mm □ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target □ exposure ≤ 66.7 μs.

4. UV illumination and spectral channel

If a stain is visible under ultraviolet illumination but not in ordinary visible light, an RGB camera under white light cannot reliably identify it. A UV-sensitive sensor or suitable UV illumination and optical filter may be required. The optical system must transmit the relevant UV wavelength; many ordinary lenses and sensor cover glass attenuate ultraviolet light.

A practical design may use a UV channel in addition to visible imaging. The UV image can be registered with the visible image so that the system can associate the stain with a physical fabric coordinate. Registration errors must be small compared with the defect size and marking tolerance.

UV illumination raises safety and integration concerns. The wavelength, intensity, exposure, enclosure and operator access must be reviewed. The detector should be validated on actual target stains, because not every stain fluoresces or absorbs UV in the same way.

Design question	Typical answer	Check
Which wavelength?	UV-A near 365 nm is commonly used for fluorescence of many oils and finishes	Test actual stained samples
Sensor sensitivity	Standard silicon sensors respond weakly below about 400 nm unless UV-enhanced	Quantum efficiency curve from supplier
Lens and filters	Ordinary glass lenses absorb much UV; need UV-grade optics and a filter that blocks reflected excitation light	Transmission data
Imaging mode	Fluorescence (UV light in, visible light out) or reflectance	Which mode shows the stain
Safety	UV exposure to eyes and skin must be controlled	Enclosure and interlocks; applicable photobiological safety standard

Registration between channels. If a visible line sensor and a UV line sensor view the web at positions 20 mm apart along the direction of travel, the visible data arrive 20 / 0.08 = 250 lines before the UV data at 0.08 mm pitch (a delay of 16.7 ms at 1.2 m/s). The software must delay one stream by exactly this amount, determined from the encoder, and a mechanical error of 0.08 mm in the sensor spacing already corresponds to one line. Cross-web misalignment and different magnifications must also be corrected by calibration so that a UV stain overlays the visible fabric structure.

Validation of the UV claim. Not every stain responds to UV. The plant should assemble a library of stained fabric samples (oils, lubricants, finishing agents) and image them under UV and visible light with the proposed camera and filters. Only the stain types that appear in the UV images with sufficient contrast can be claimed as detectable. The rest remain outside the guaranteed performance and must be stated as such in the offer to the customer.

Detection logic. With a UV channel, the detector can combine information: a region that is bright in UV but not visible in colour is a likely oil stain, while a region visible in both channels may be a dye or fibre difference. The rules must be derived from the sample library. For the first prototype, a simple difference

TASK 10 – Front-End Adaptation Challenge (continued)

between the UV image and a clean-fabric reference after flat-field correction gives a quick measure of feasibility.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm ⍰ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target ⍰ exposure ≤ 66.7 μs.

5. Strategy A: retain existing system

Retaining the existing camera, lighting and PC with only software changes is the lowest-cost option. Software can crop regions of interest, optimize image processing, reuse buffers and reduce unnecessary operations. These changes may reduce latency, but they cannot recover spatial information that the sensor never captured or spectral information that was not measured.

At 0.4 mm, the original spatial sampling is likely inadequate if it was designed around a 1.5 mm defect. The existing visible channel also cannot reliably detect a stain that is only apparent under UV. Faster conveyor speed increases blur and acquisition requirements. Therefore Strategy A cannot credibly satisfy all new requirements.

Strategy A may be used temporarily for non-critical visible defects while the upgraded system is prepared, but it should not be represented as compliant with the new export order. The remaining limitations must be documented and communicated to production management.

Requirement	Status with software-only changes	Reason
0.4 mm defects	Not met	At 0.49 mm/pixel, 0.4 mm is 0.8 pixel; even 0.29 mm/pixel gives only 1.4 pixels
UV-visible stains	Not met	No UV illumination or sensitivity
1.2 m/s	Not met	A 250 µs exposure gives 0.30 mm blur, 75 % of the defect
0.3 s decision	Possible with processing only optimisation	But travel time to the marker is 0.25 s if unchanged
Double volume	CPU load doubles	Needs profiling
Budget	Met	No new equipment

What software can still do. Software can reduce processing time by using region-of-interest processing, vectorised code, reuse of buffers and multi-threading. It can also improve detection on the existing visible images, for instance by better local background modelling for faint stains. These improvements are worth making in any case, as they free time for later stages, but they cannot bring back details that the sensor did not record.

Recommended use. Strategy A should be used only as an interim measure for orders that still follow the old specification, and for baseline recording while the new front end is evaluated. It must not be offered as compliant with the export order. A clear statement of the gaps (defect size, UV and speed) protects the company from accepting a contract it cannot fulfil.

Residual value. Even if the existing system cannot meet the new order, its recorded images and detection software are valuable. They provide baseline false-alarm statistics for the current fabrics, a labelled dataset and a working marker interface. The new front end can reuse the encoder, the marker controller and the audit store, so the investment in Strategy A is not wasted when the plant moves on to B or C.

TASK 10 – Front-End Adaptation Challenge (continued)

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm ⍰ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target ⍰ exposure ≤ 66.7 μs.

6. Strategy B: modify selected components

Modifying selected components could replace the camera with a higher-resolution global-shutter or linescan sensor, add short-pulse illumination, install UV optics or a second spectral channel, and optimize software on the existing PC. This is more feasible than retaining everything, but the cost and data-rate implications are substantial.

The 15,000-pixel cross-web estimate for five pixels across a 0.4 mm defect suggests that a single ordinary area camera may be unsuitable. A high-resolution line-scan camera could address the width sampling, but its line rate and output bandwidth may exceed what the existing PC can process at full resolution. A UV channel adds data and calibration work.

Strategy B is appropriate if vendor testing demonstrates a combination of camera, lighting and interface that fits the unchanged budget and CPU. Otherwise it becomes a partial upgrade that improves some requirements but leaves the smallest-defect or UV requirement unmet.

Component change	Benefit	Risk / cost
Higher-resolution line-scan camera (8k–16k)	Meets sampling for 0.4 mm	Cost, interface bandwidth, CPU load
Pulsed or high-power LED lighting	Short exposure with enough signal	Heat, synchronisation
UV-grade lens, filter and UV source	Reveals UV stains	Cost, safety, registration
Faster encoder interface and trigger	Accurate line timing at 15,000 lines/s	Must verify with the controller
Optimised software on existing PC	Stays within the computer limit	May still be too slow at full resolution

Where the PC becomes the bottleneck. A single monochrome stream of 15,000 × 15,000 pixels per second (225 MB/s) already exceeds the capability of Gigabit Ethernet (about 125 MB/s) and needs a faster link and frame grabber. Even if received, a 4-core CPU would have to apply screening operations to 225 million pixels per second; if each pixel needs only 10 CPU cycles at 3 GHz on four cores, the budget is about 1.2 billion pixel-operations per second, so 225 million pixels with 10 cycles consume about a fifth of the CPU. This leaves little for colour, UV and classification, so the processing must be strictly regionbased.

Conditions for success. Strategy B is acceptable if a vendor demonstration, using the proposed camera, lighting and the plant's own fabric, shows 0.4 mm defect detection with the required recall at 1.2 m/s, and if the quoted prices fit the budget. If the vendor proof is positive, the cost is limited to the front end. If only some requirements are met, the offer to the customer must be limited to those.

Staged approach. A staged implementation first upgrades the camera and lighting for the visible channel at the new resolution and speed, tests detection of 0.4 mm defects, and only then adds the UV channel if the first stage works. This reduces the risk and allows the budget to be spent in steps. Each stage has its own acceptance test, so the plant knows at every stage what it can promise to the customer.

TASK 10 – Front-End Adaptation Challenge (continued)

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm ⍰ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target ⍰ exposure ≤ 66.7 μs.

7. Strategy C: redesign architecture

A redesigned architecture could use high-resolution RGB plus UV line-scan sensing in an enclosed station, encoder-synchronized acquisition, region-based processing and a dedicated real-time path. The design should separate acquisition and buffering from CPU analysis, and process candidate regions in parallel where possible.

This architecture is technically best aligned with a continuous wide fabric web and the new spectral requirement. However, the unchanged PC and budget may make full compliance unrealistic. Highresolution dual-channel data can overwhelm the PC, storage and network unless the acquisition and processing plan is carefully engineered.

A redesign is the correct engineering recommendation if the new order is mandatory, but the company must validate feasibility before promising performance. If the fixed budget prevents a compliant sensor or UV arrangement, management must negotiate additional resources or a reduced inspection scope.

Element	Redesigned architecture	Reason
Sensors	RGB and UV line-scan channels in one enclosed station	Spectral coverage and registration
Lighting	Pulsed, synchronised visible and UV sources	Short exposure; safe UV enclosure
Acquisition	Frame grabber with onboard buffering and optional preprocessing	Offload the CPU
Processing	Two stages: screening on the whole stream, detailed analysis on candidate crops	Meets 0.3 s with a 4-core CPU
Control	Dedicated real-time marker controller with encoder-based scheduling	Deterministic timing
Storage	Event-based retention, compressed crops	Fits 2 TB

Data rate of the full design. With 15,000 pixels per line, 15,000 lines/s and four 8-bit channels (R, G, B and UV), the stream is 15,000 × 15,000 × 4 = 900 MB/s. At this rate, a 2 TB disk would fill in about 2,222 s (37 minutes) if everything were stored. The stream cannot go through an unmodified 4-core PC for full analysis; either the grabber must preprocess (flat-field and screening) in hardware, or the analysis must be limited to candidate regions and a reduced-resolution channel. This is the most demanding but also the most technically complete solution.

Programme risks. Integration of two spectral channels, a high-speed grabber and a real-time controller needs specialised skills and takes longer than the other options. The company should plan a staged project with an early prototype on one camera and the existing PC, so that the highest risks (UV contrast, blur at 1.2 m/s and CPU load) are tested before the full budget is committed.

Why a separate real-time path. Because the decision window is only 0.3 s, a general-purpose operating system on the main PC may be unable to guarantee timing when other tasks are active. Putting the marker scheduling in a dedicated controller with hardware inputs for encoder counts ensures deterministic firing,

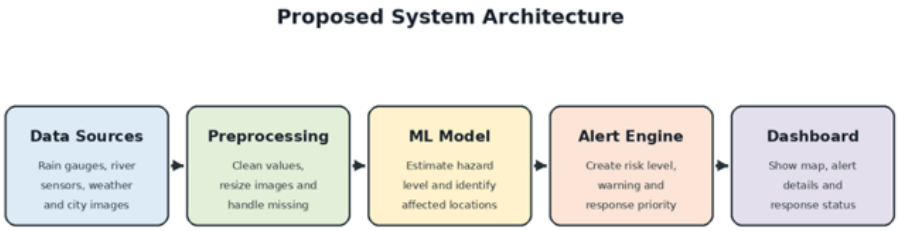
TASK 10 – Front-End Adaptation Challenge (continued)

while the PC provides the detection result, with a time stamp, in advance of the target position. The architecture then tolerates PC delays up to a defined limit.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm □ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target □ exposure ≤ 66.7 μs.



8. Comparison of adaptation strategies

Strategy A has the lowest cost and least disruption but cannot fix missing spatial or spectral information. Strategy B can address selected limitations with a camera/lighting upgrade but may remain constrained by the PC and bandwidth. Strategy C changes the acquisition architecture and is most capable, but carries the greatest integration and budget risk.

A comparison table should score each option against 0.4 mm sampling, UV stain detection, 1.2 m/s motion, 0.3 s decision, full-width coverage, 2 TB storage, maintenance and cost. The score must be based on sensor specifications, test measurements and realistic data-rate calculations, not on assumptions that software optimization can solve every physical limitation.

The preferred strategy is C if the export requirements cannot be relaxed. If the fixed budget is absolute, Strategy B should be evaluated through a vendor proof-of-concept, but the factory must not claim full compliance until it has measured the new defect classes and latency.

Requirement	Strategy A	Strategy B	Strategy C
0.4 mm sampling	Not met	Met if 8k–16k camera is affordable	Met
UV stain detection	Not met	Partly (UV channel added)	Met
1.2 m/s blur control	Not met	Met with strobe/TDI	Met
0.3 s decision	Uncertain	Possible with region-based processing	Designed for it
Full-width coverage	Yes (old sampling)	Yes	Yes
2 TB retention	Met	Met with event storage	Met with event storage
Cost within □1,50,000	Met	Doubtful	Unlikely
Integration risk	Low	Medium	High

Reading the matrix. No single strategy satisfies every row: A fails the physical requirements, C satisfies the technical requirements but exceeds the budget, and B is the compromise whose outcome depends on supplier data. The selection is therefore conditional, and the conditions are stated: if the export order is

TASK 10 – Front-End Adaptation Challenge (continued)

mandatory and the budget fixed, B must be tested first; if B fails the proof-of-concept, management has to choose between more money (C) and a reduced specification.

Next steps. (1) Ask suppliers for line-scan cameras with 8k–16k pixels, TDI options, UV-enhanced sensitivity and complete package prices. (2) Image stained samples under UV. (3) Run a bench blur test at 1.2 m/s. (4) Benchmark the region-based pipeline on the existing PC using recorded high-resolution data. (5) Confirm the marker-distance and decision-time conflict with the line engineer. The decision to proceed is made after these five results are available.

Decision under uncertainty. The technical information needed for the final decision is not yet available, so the recommendation is expressed as conditions. If the proof-of-concept shows that the 0.4 mm and UV requirements can be met at 1.2 m/s with the existing PC and budget, Strategy B is adopted. If not, the plant presents the customer with the minimum extra cost for Strategy C or a revised specification. This avoids committing to a result that the data do not support.

System flow:

Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm ⍰ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target ⍰ exposure ≤ 66.7 µs.

9. Updated architecture and feasibility conclusion

Updated flow: Fabric + encoder → enclosed RGB/UV line-scan camera → synchronized visible/UV illumination → high-speed acquisition buffer → geometric and channel registration → fast candidate detector → class-specific defect analysis → confidence/deadline check → encoder-based marker controller → compressed audit storage. The data path should prioritize timely marking while an asynchronous path writes audit images.

The 0.3-second decision time is stricter than the old 0.5-second requirement. At 1.2 m/s, the marker distance must be confirmed; if it remains 0.3 m, travel time is 0.3/1.2 = 0.25 s, which is less than the decision limit. The system would then need a decision earlier than the physical travel window or the marker location must be changed. This is a key feasibility conflict to resolve.

Final decision: redesign the acquisition front end, but state clearly that full compliance may not be achievable under an unchanged PC and budget without a successful vendor prototype. Preserve the existing system for baseline inspection during transition, add UV and high-resolution sensing only after a measured proof-of-concept, and request additional resources if the 0.4 mm, UV, 1.2 m/s and 0.3 s requirements cannot all be demonstrated.

System flow:

> Fabric + encoder → Camera / controlled light → Acquisition buffer → Calibration & preprocessing → Candidate regions → Feature extraction / classifier → Coordinate mapping → Deadline check → Marker controller → Audit storage

New-order calculations: five pixels across 0.4 mm □ 0.08 mm/pixel; 1200/0.08 = 15,000 pixels across the field. At 1.2 m/s, 0.08 mm blur target □ exposure ≤ 66.7 μs.